



**GREEN
CLIMATE
FUND**

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9 June 2025

Consideration of funding proposals – Addendum VII

Funding proposal package for FP265

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Climate-resilient landscapes for sustainable livelihoods in northern Ghana";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

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Funding Proposal

Project/Programme title:	Climate-resilient landscapes for sustainable livelihoods in northern Ghana
Country(ies):	Republic of Ghana
Accredited Entity:	UNEP
Date of first submission:	2018/06/25
Date of current submission	2025/04/14
Version number	<u>[V.000]</u>



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Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

“FP-[Accredited Entity Short Name]-[Country/Region]-[YYYY/MM/DD]”

Acronyms and abbreviations

AE	Accredited Entity
AFAWA	Program on Affirmative Finance Action for Women in Africa: Financing Climate Resilience Agricultural Practices in Ghana
AO	Administrative Officer
BMUB	Federal Ministry for the Environment, Nature Conservation, Building and Nuclear
CCAP	Community Climate Action Plan
CCAG	Climate Change Action Group
CC-DARE	Climate Change and Development — Adapting by Reducing Vulnerability
CCV	Climate Change Vulnerability
CIG	Common Interest Groups
CRA	Climate-resilient Agriculture
CSIR	Council for Scientific and Industrial Research
DA	District Assembly
DAP	District Adaptation Plan
DACF	District Assembly Common Fund
DCS	Directorate of Crop Services
DEMC	District Environmental Management Committee
DMTDP	District Medium-Term Development Plan
DoA	Department of Agriculture
DPCU	District Planning and Coordination Unit
EbA	Ecosystem-based Adaptation
EE	Executing Entity
EIA	Environmental Impact Assessment
EMC	Environmental Management Committee
ERP	Enterprise Resource Planning
ESES	Environmental, Social and Economic Sustainability
ESERN	Environmental, Social and Economic Review Note
ESMF	Environmental and Social Management Framework
ESS	Environmental and Social Safeguard
EPA	Environmental Protection Agency
FAA	Funded Activity Agreement
FAO	Food and Agriculture Organization of the United Nations
FBO	Farmer-based Organisation
FDI	Fire Danger Index
FM	Financial Management
FPO	Financial and Procurement Officer
GAP	Gender Action Plan
GCM	Global Climate Model
GDP	Gross Domestic Product
GEF	Global Environment Facility
GEMP	Ghana Environmental Management Project
GMet	Ghana Meteorological Agency
GoG	Government of Ghana
ICA	Internal Cooperation Agreement
INDC	Intended Nationally Determined Contributions
IP	Impact Pathway
IPSAS	International Public Sector Accounting Standards
ITCZ	Intertropical Convergence Zone
LFI	Local Financial Institution
LSC	Local Steering Committee
M&E	Monitoring and Evaluation
MESTI	Ministry of Environment, Science, Technology and Innovation
MLGCRA	Ministry of Local Government, Chieftaincy and Religious Affairs
MoFA	Ministry of Food and Agriculture
MoFEP	Ministry of Finance and Economic Planning
MMDA	Metropolitan, Municipal and District Assemblies



MSME	Micro Small Medium-sized Enterprises
MTR	Mid-Term Review
NAP	National Action Programme
NACAP	National Anti-Corruption Action Plan
NCAFSAP	National Climate-Resilient Agriculture and Food Security Action Plan
NCCAS	National Climate Change Adaptation Strategy
NCCC	National Climate Change Committee
NCCP	National Climate Change Policy
NDA	National Designated Authority
NDC	Nationally Determined Contribution
NGO	Non-governmental Organisation
NPV	Net Present Value
NER	North East Region
NTFP	Non-timber Forest Products
PCA	Project Cooperation Agreement
PET	Potential evapotranspiration
PHL	Post-harvest loss
PM	Project Manager
PMU	Project Management Unit
PSC	Project Steering Committee
REMC	Regional Environmental Management Committee
RMTDP	Regional Medium-Term Development Plan
RPCU	Regional Planning and Coordinating Unit
SADA	Savannah Accelerated Development Authority
SCI-SLM	Stimulating Community Initiatives in Sustainable Land Management project
SDG	Sustainable Development Goals
SLWM	Sustainable Land and Water Management
SLWMP	Sustainable Land and Water Management Project
SPEI	Standardized Precipitation Evapotranspiration Index
TAC	Technical Advisory Committee
TE	Terminal Evaluation
UER	Upper East Region
UNDP	United Nations Development Programme
UNEP	United Nations Environment Programme
UNFCCC	United Nations Framework Convention on Climate Change
UNSDP	United Nations Sustainable Development Partnership
USAID	United States Agency for International Development
UWR	Upper West Region
VSLA	Village Savings and Loans Associations
WBS	Work Breakdown Structure
WBSE	Work Breakdown Structure Elements
WRI	World Resources Institute

A. PROJECT/PROGRAMME SUMMARY			
A.1. Project or programme	Project	A.2. Public or private sector	Public
A.3. Request for Proposals (RFP)	<u>Not applicable</u>		
A.4. Result area(s)		GCF contribution	Co-financers' contribution¹
	Mitigation total	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Energy generation and access	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Low-emission transport	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Buildings, cities, industries and appliances	<u>Enter number</u> %	<u>Enter number</u> %
	☐ Forestry and land use	<u>Enter number</u> %	<u>Enter number</u> %
	Adaptation total	<u>Enter number</u> %	<u>Enter number</u> %
	☒ Most vulnerable people and communities	Enter number18.26%	23.53 %
	☒ Health and well-being, and food and water security	33.21 %	32.8 %
	☒ Infrastructure and built environment	21.78 %	18.29 %
	☒ Ecosystems and ecosystem services	26.75 %	25.38 %
A.5. Expected mitigation outcome <i>(Core indicator 1: GHG emissions reduced, avoided or removed / sequestered)</i>	N/A	A.6. Expected adaptation outcome <i>(Core indicator 2: direct and indirect beneficiaries reached)</i>	2,861,674 beneficiaries
			691,125 direct beneficiaries (52% women)
			2.2% of the total population
A.7. Total financing (GCF + co-finance²)	70,198,822 USD	A.9. Project size	Medium (Upto USD 250 million)
A.8. Total GCF funding requested	63,211,141 USD		

¹ Co-financer's contribution means the financial resources required, whether Public Finance or Private Finance, in addition to the GCF contribution (i.e. GCF financial resources requested by the Accredited Entity) to implement the project or programme described in the funding proposal.

² Refer to the Policy of Co-financing of the GCF.

A.10. Financial instrument(s) requested for the GCF funding	☒ Grant 63,211,141 USD ☐ Equity <u>Enter number</u> ☐ Loan <u>Enter number</u> ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>		
A.11. Implementation period	7 years	A.12. Total lifespan	28 years
A.13. Expected date of AE internal approval	6/8/2018	A.14. ESS category	<i>Refer to the AE's safeguard policy and GCF ESS Standards to assess your FP category.</i> B
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☐ No ☒
A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?	Yes ☒ No ☐
A.19. Complementarity and coherence	Yes ☒ No ☐		
A.20. Executing Entity information	Government of Ghana through the Ghana Environmental Protection Agency (EPA) and Ghana Meteorological Agency (GMet).		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			
The agriculture sector in northern Ghana — which supports ~80% of the total workforce — is dominated by smallholder farmers practising mostly rainfed, subsistence farming, using mostly traditional agricultural practices and minimum technology. The long duration of the dry season experienced in northern Ghana results in prolonged periods of food shortages. High temperatures in northern Ghana already contribute to reduced water-retention capacity and frequent drying of water bodies — such as wells, dams, and streams — that serve as sources of water for both livestock and household use, including for dry season gardening. Increased temperatures linked to climate change exacerbate the problem through the expansion of ranges of the weeds, pests and fungi that cause food spoilage. Conversely, extreme rainfall events often result in severe flooding across many areas of northern Ghana, destroying crops, taking lives and damaging assets and public infrastructures, as well as reducing storage capacity of reservoirs in northern Ghana, causing them to dry-up quicker in the dry season as well as increase the risk of dam collapse, leading to downstream flooding. The proposed project objective is to enhance the climate resilience of vulnerable smallholder farming communities in northern Ghana by improving food security and enhancing the agro-based rural economy. The project strategy is three-fold comprising the improved climate data and early warnings made available to facilitate proactive drought and flood			

management, the adoption of climate-resilient agriculture and water storage to enable dry season farming, and investments in landscape restoration (such as riverbank restoration, agroforestry, reforestation and fire management) to improve soil health and integrity, physical and land-based water retention and protect physical assets from flooding.

The project will work at three levels: community level on planning and implementation systems, at the regional level on strengthening weather forecast capability that will serve the Northern Sector, complementing investments in forecasting capability in the South of the country as well as national level in establishing a national action plan for drought and flood hazard management. The community level integrated ecosystem-based adaptation approach will inform the District level planning and budgeting systems, through building capacity and awareness of District planning officers to scale up the project approach. Combined with associated community training, extension services and awareness-raising, this will increase opportunities for knowledge and technology exchange between communities, and thereby promote autonomous upscaling of these interventions.

The proposed project will be implemented in the North East, Upper East and Upper West Regions of northern Ghana, in eight districts in northern Ghana that have been specifically chosen because of their high vulnerability to climate change impacts. Across the eight districts, the project will provide agricultural and livelihood support for 120 communities over seven years as well early warning advisories for the entire population in the 8 Districts.

The EPA is the main executing entity, that will work in partnership with Ministry of Food Security and Agriculture together with the Ghana Meteorological services and Water Resources Commission to deliver an integrated, ecosystem-based adaptation strategy that will transition the smallholder farming communities of northern Ghana more climate-resilient, productive and sustainable livelihoods.

B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Context³

Approximately 80% of the 4.16 million people living in northern Ghana are smallholder farmers depending on traditional, small-scale, rainfed agricultural systems to generate household incomes and maintain food security. Smallholder farming households in northern Ghana, comprising of Upper West Region (UWR), Upper East Region (UER), and North East Region (NER), frequently experience severe food insecurity for between three to six months annually during the dry season. They are highly vulnerable to the degradation of natural ecosystems and changes in climate. The main drivers of degradation are inter alia: i) deforestation for wood and charcoal production and clearing for agricultural activities; ii) wildfires to clear natural vegetation for agriculture; iii) overgrazing by cattle; and iv) soil erosion and fertility loss because of unsustainable farming practices.

The results of the climate change vulnerability assessment (see Section 2 of Annex 2 — Feasibility Assessment) demonstrate that these administrative Regions of northern Ghana, which have high levels of poverty, are the most climate vulnerable in the country (Figure 14 – Annex 2 Feasibility Assessment), and therefore, are a priority target for climate change adaptation interventions.

Climate analysis⁴

Current trends

Most of the rainfall in northern Ghana occurs between June and October; this is also the period with the largest variation in total monthly precipitation volume. Historical trends show a substantial change in rainfall patterns with a reduction in the amount of rainfall during the early wet season (~-8 mm April, ~-8 mm May, ~-17 mm June ~-9 mm July, ~-12 mm August in the 2010 decade) and an increase during the late wet season (~+5 mm September and ~+17 mm October in the 2010 decade). In northern Ghana, rainfall patterns have become increasingly erratic with the start of the wet season becoming more difficult to predict. In the past, the wet season started in April and ended around late September or early October. However, in recent times, the wet season starts in June or July with extremely heavy rainfall in September or October, indicative of a shortening of the historical rainfall season. Similarly, the number of rainfall days (days with 0.1 mm or more) has decreased by 5.3 days per year in the peak precipitation months April to September from the 2000s, with a decreasing trend of 2.6 days per decade. The overall trend has been prolonged dry seasons, an increased frequency and intensity of drought conditions, and a shorter rainy season with increased occurrence of extreme rainfall events with severe consequences for smallholder farmers.

Climate change projections

Temperature: Under RCP 8.5, the annual average temperature of Upper East Region is predicted to increase from 28.2°C to 33.1°C (an increase of 4.9°C) by 2085, higher than the national average of 4.5°C. Annual minimum and maximum temperatures are also expected to increase in all the regions of Ghana, with northern Ghana again expected to experience the largest changes. The annual maximum temperature of the Upper East Region is expected to increase



Figure 1: Ghana's 16 Administrative Regions, including the project Regions North East, Upper East and Upper West

³ For more detail on information in this section and a full set of references to support the arguments, please consult the Feasibility Assessment (Annex 2).

⁴ Please refer to Section 4 of Annex 2: Feasibility Study for a detailed analysis of observed and predicted climate change in northern Ghana.

from 34.1°C to an extreme 38.8°C by 2085 under RCP 8.5. In summary, maximum and minimum temperature will increase throughout the year, and heatwaves will become more frequent.

Potential evapotranspiration: GCMs suggest that potential evapotranspiration (PET)⁵ will increase across Ghana throughout the 21st century under both RCP scenarios. This increase will be most apparent in northern Ghana where PET will increase by 8.5% by 2085. The predicted increase in PET will be marginally greater than the predicted minor increase in annual rainfall, resulting in a decrease in the annual moisture index⁶ in most regions of the country.

Rainfall: Ensemble climate projections suggest that in northern Ghana the change in rainfall to include a reduction in early wet season rainfall (April to June) and increase in late wet season rainfall (July to September). The projections, under RCP 8.5 scenario estimate an increase of 5mm/month in April and July/August, increasing per decade towards the end of the century. By contrast, the wet season onset and cessation months of March and October are estimated to experience a decrease in rainfall per decade of 10mm/month. For both RCP 4.5 and RCP 8.5, there will be a reduced occurrence of low-volume rainfall (<11mm) events and an increased occurrence of high-volume rainfall (11–21mm) events. In summary, the rainfall profile in northern Ghana will change where the wet season will be shorter and rain will be more likely to occur less frequently but more intensely during high volume events.

Extreme events: Extreme rainfall events are likely to become more frequent over time, with events that historically occurred 1 in every ~100 years predicted to occur 1 in ~73 years and 1 in every ~80 years under RCP 4.5 and RCP 8.5 scenarios, respectively. Furthermore, extreme rainfall events are predicted to become more intense over time, with 1 in 10-year events projected to be ~8.3% more intense (under both RCP 4.5 and RCP 8.5 scenarios).

The increased occurrence of higher rainfall events is associated with an increase in flood risk, while the overall reduction in the number of rainfall events is associated with more frequent dry spells. The number of dry days per month is predicted to increase above the historical average throughout the year under both the RCP 4.5 and RCP 8.5 scenarios, with the largest increases occurring in the dry season months of November to February. Noteworthy is the increased frequency of dry days in the wet season onset and cessation months of March (increasing by 2.8–5.6 days per month) and October (increasing by 1.4–1.7 days per month). An increasing number of dry days per month will increase the risk of droughts occurring in northern Ghana.

Climate change impacts⁷

The *Guina savanna* and *Sudan savanna* agroecological zones of northern Ghana, on which the livelihoods of smallholder farmers rely, are already under considerable stress because of *inter alia*: i) unfavourable climate conditions and trends; ii) extensive land and environmental degradation related to increasing population growth rates; and iii) unsustainable farming methods. The effects of future climate change are expected to exacerbate the current problems in these systems through the negative impact pathways (See Annex 2 – Section 5 - **Error! Reference source not found.**

Impact on food systems: Climate change is impacting dry season food production. Few smallholder farmers have access to irrigation infrastructure and as a result are highly dependent on rain for their livelihoods. As described above, the long duration of the dry season experienced in northern Ghana results in prolonged periods of food shortages. Dry season gardening can supplement food production outside of the major growing season, but can only be practiced in areas with a nearby water source, where water is available even during the driest months of the year. However, climate change is exacerbating water shortages during the dry season. High temperatures in northern Ghana already contribute to reduced water-retention capacity and frequent drying of water bodies — such as wells, dams, and streams — that serve as sources of water for both livestock and household use, including for dry season gardening.

Food security during the dry season is further impacted by the high rate of post-harvest losses (PHLs). Post-harvest losses in staple foods in Ghana such as maize, cassava, yam and rice amount to over 35%, 34%, 24% and 7% of yields, respectively. Increased temperatures linked to climate change exacerbate the problem through the expansion of ranges of the weeds, pests and fungi that cause food spoilage — as well as an increase in their multiplication rates. In addition to causing food spoilage, an increased rate of fungal growth results in: i) reduced seed viability; ii) discolouration; iii) caking; iv) mouldy smells; and v) the production of mycotoxins in stored food.

⁵ Defined as the amount of evaporation that would occur if a sufficient water source were available.

⁶ Annual moisture index = Mean annual rainfall / Potential evapotranspiration.

⁷ Please see Section 5 of Annex 2: Feasibility Study for a detailed assessment of climate change impacts in northern Ghana.

Impact on water resources: Decreasing wet-season length, coupled with rising temperatures, is increasing the frequency in which surface water bodies such as dams and streams are drying up. It is estimated that changing rainfall patterns in Ghana will lead to a general reduction in annual river flow by 15–20% for the year 2020 and 30–40% for 2050⁸. The groundwater system in Ghana is a dynamic system that mostly recharges through direct rainfall infiltration, especially in the northern regions. The shallow aquifer ground water resources are already classified moderately vulnerable to flood and drought impacts. Increased temperatures, drought occurrence and potential evapotranspiration will reduce the resource recharge.

Inversely, extreme rainfall events often result in severe flooding across many areas of northern Ghana, destroying crops, taking lives and damaging assets and public infrastructures. Numerous such extreme events have been observed over the past 60 years (Figure 4.7 of Annex 2: Feasibility Study) resulting in human deaths and displacement of 16 million people. Floods have become increasingly frequent over the past 30 years and is a significant risk to agriculture in flood-prone areas in northern Ghana.

Fluvial flooding results in river overbank flow, and there expose crops and livestock to flood damage and loss, especially for those farmers farming close to the river due to a lack of irrigation systems, and in severe cases also affect houses and roads. Pluvial floods, also called flash floods, occur after heavy rainfall, often only in short time periods. Pluvial floods play a role in all districts. Flash floods result in crops and fertile soil being washed away. Breaches of small reservoir dams cause flood risk in all districts, especially when dams are poorly constructed, maintained or treated.

The problem of reduced water availability in the dry season is further exacerbated by the increasing frequency and intensity of extreme rainfall events which lead to more eroded material and pollutants being deposited into water bodies, further decreasing water quality and reducing the reservoir storage capacity. Reduction in storage capacity of reservoirs in northern Ghana, in turn cause them to dry-up quicker in the dry season as well as increase the risk of dam collapse, leading to downstream flooding.

Impact on local infrastructure and livelihoods: Extreme climate events may undermine the effectiveness of existing rural infrastructure and the impacts on rural livelihoods and agricultural systems will thus also depend on their climate-resilience and compatibility with changing climatic patterns. Extreme climate events have already caused important economic losses. According to EPA, 2007 floods in northern regions affected and displaced more than 300,000 people, damaged or destroyed almost 26,000 houses, collapsed 13 bridges, destroyed 1,060 km of feeder roads, contaminated 634 drinking water facilities, amongst other damages to social infrastructure.

Social infrastructure, including community buildings, are also vulnerable to the impacts of climate change. Recurrent floods, heavy downpours and long dry spells are the ways in which climatic change can affect social and community infrastructure. Houses, school buildings, water and sanitation facilities are infrastructures that are both sensitive and vulnerable to climate change.

Impact on ecosystems: Uncontrolled bushfires in northern Ghana are reducing the capacity of agroecosystems to provide critical ecosystem services such as NTFPs, forage, soil stabilisation and flood mitigation. Bushfire impact agriculture through several pathways, including: i) causing extensive damage to crops and fruit trees; ii) reducing the fertility of farmlands by removing soil organic matter; and iii) causing vegetation loss in fallowed/grazing lands and woodland. The frequency and intensity of uncontrolled bushfires in northern Ghana is increasing. While the increase is largely driven by rapid population growth and the break-down of socio-cultural norms that used to regulate land in northern Ghana, climate change is exacerbating the problem. The extent of the impact of uncontrolled bushfires in northern Ghana are estimated as follows:

- **Forest loss:** Research indicates that approximately 30% of Ghana's reserved forest area is destroyed by fire each year⁹.
- **Burned areas:** Between 2001 and 2021, the Guinea-savanna and Forest-savanna mosaic zones experienced significant fire activity. For instance, in the Guinea-savanna zone, December recorded an average burned area of 107.539 km² with 59,288 fires detected¹⁰.

⁸ Republic of Ghana. Ghana National Drought Plan. Available at https://knowledge.unccd.int/sites/default/files/inline-files/1%20FINAL_NDP_Ghana.pdf.

⁹ Available at: https://weadapt.org/knowledge-base/nature-based-solutions/bush-fires-for-natural-resources-and-environmental-management-in-ghana-a-review/?utm_source=chatgpt.com

¹⁰ Available at: https://fireecology.springeropen.com/articles/10.1186/s42408-023-00198-z?utm_source=chatgpt.com

- **Fire frequency:** The overall fire frequency was estimated at 60% in the Guinea-savanna and 43% in the Forest-savanna mosaic zones¹¹.

Scaling up good adaptation practice¹²

This project draw on past and ongoing programmes and projects which have generated positive adaptation results for beneficiaries in Ghana, the key one being the sustainable land and water management (SLWM) project to enhance soil and land productivity and watershed services for climate resilience in northern Ghana (evaluation report at Annex 18).

B.2 (a). Theory of change narrative and diagram (max. 1500 words, approximately 3 pages plus diagram)

Problem statement

The agriculture sector in northern Ghana — which supports ~80% of the total workforce — is dominated by smallholder farmers practising mostly subsistence farming. Given that smallholder farms in northern Ghana are almost entirely rainfed — using mostly traditional agricultural practices and minimum technology — agricultural production is strongly dependent on the availability of natural resources, making them highly vulnerable to changes in their environment and the climate. The agroecological systems which underpin natural resource-based livelihoods, are notably already under considerable stress because of *inter alia*: i) unfavourable climate conditions; ii) extensive environmental degradation related to high population growth rates, and traditional, unsustainable farming methods and iii) dam spills in Tamne reservoir and Bagre Dam in Burkina Faso. These baseline challenges are exacerbated by climate change, including: i) decreases in the duration of the wet season; ii) increases in rainfall variability, with less frequent but more intense rainfall events; iii) increases in the number of dry days; iv) increases in maximum and minimum temperatures, with more frequent heatwaves; v) increased evapotranspiration; and vi) a reduced annual moisture index.

The impacts of these changes to climate on smallholder farmers include:

- increased frequency and severity of floods — particularly in areas near rivers — that will lead to soil erosion, major crop losses and damage to agricultural infrastructure, including post-harvest storage and irrigation/water facilities, and transport infrastructure and small dams being breached;
- increased frequency and intensity of droughts leading to, among other things: i) increased frequency in the drying-up of surface water bodies such as dams and streams; and subsequent stress on water systems for both agriculture and human use; ii) a decline in agricultural productivity as the ability of small-scale farmers to engage in dry season gardening is reduced;
- shortened agricultural production periods resulting in decreased crop yields and income as the number of achievable wet season cropping cycles and harvests of non-timber forest products (NTFPs) from agroecosystems are reduced;
- pushing common northern Ghanaian crop and livestock varieties beyond their optimum thermal limits and expanding the ranges of weeds and pests that prefer warmer climates as previously cooler areas experience warmer temperatures;
- increased soil erosion and loss of soil nutrients leading to decreased crop yields and increased pollution and eutrophication of water bodies; and
- more frequent and severe uncontrolled bushfires resulting in the destruction of crop fields — particularly of the dry mature crop — and reduced capacity of agroecosystems to provide critical ecosystem services such as NTFPs, forage, soil stabilisation and flood mitigation.
- Increased incidence of climate change induced disasters on the built environment and settlements, ultimately resulting in disruptions to normal human activities and economic losses in terms of asset damages.

These impacts, when combined with the non-climate drivers of environmental degradation and dam spillages, have a substantial impact on the livelihoods of smallholder farmers and their households in northern Ghana, with subsequent impacts on food security and income generation of the country's most vulnerable people. Without effective adaptation action supported by access to timely climate information and analysis, northern Ghanaian smallholder farmers will likely remain vulnerable to the current and future effects of climate change, with serious repercussions for the northern Ghanaian economy, ultimately increasing the north-south divide in Ghana.

An analysis of current drought and flood early warning services in Ghana indicates that the majority have incomplete coverage and limited channels for information transfer and dissemination. Several studies — including a gap analysis of drought and flood risk management and early warnings in Ghana, a capacity needs assessment of the Ghana

¹¹ Ibid.

¹² Please see Annex 2 - Section 10 for information on past and ongoing projects and Section 11 on adaptation best practices and lessons learned.

Meteorological Agency (GMet) , a Gender Assessment and Action Plan (Annex 8), and the assessments under the UNEP CLIMWARN project — have also revealed several technological barriers to building the resilience in the agricultural sector to the impacts of climate change, namely relating to information access. Stakeholder consultations (see Annex 7h: Stakeholder Engagement Plan) revealed that if women were provided improved technologies and financial services, they would likely invest in the land, leading to increased production — and potential increased income — which benefits not only themselves, but their families and communities too. Further to this, drought and flood information management and early warning systems have been identified as gaps by stakeholder consultations during the project development process (Annex 7h).

Preferred adaptation solution

The preferred adaptation solution would see a reduction in the vulnerability to climate change of smallholder farmers in northern Ghana through the implementation and adoption of climate-resilient agriculture, ecosystem-based adaptation approaches, alternative climate-resilient livelihoods and locally led investments in climate resilient rural infrastructure. These three suites of climate change adaptation interventions will complement one another to increase the resilience of smallholder farming communities to climate change impacts. Climate-resilient Agriculture (CRA) is an integrative approach that assists with the reorientation and transformation of agricultural systems to ensure food security and environmental sustainability in a changing climate . Climate-resilient agricultural techniques will enhance agricultural productivity under future climate changes conditions (both crop production and livestock). Ecosystem-based adaptation interventions (such as riverbank restoration, agroforestry, reforestation and fire management) will enhance the supply of ecosystem goods and services, further increasing agricultural productivity (for example through reduced soil erosion and increased fodder for livestock) while also preventing land degradation and thereby reducing the impacts of floods and droughts. The enhanced supply of ecosystem goods and services will, in turn, provide communities with options for alternative climate-resilient livelihoods, reducing their need to engage in unsustainable practices such as expanding slash and burn agriculture to generate income. The realization of locally relevant climate resilient infrastructure investments will reduce direct losses of damages and indirect costs of disruption, while contributing to build and sustain climate-resilient communities and environments. Investments in key gaps in forecast and observing equipment in Northern Ghana will transform the skill of weather forecasts. Climate advisories for drought and flood hazards will support farmers to optimise planting methods and schedules for higher yields.

These climate change adaptation interventions — targeted to circumvent and/or disrupt northern Ghana-specific climate change impacts — would be introduced to smallholder farmer communities in a coordinated manner throughout the three northern regions of Ghana. The implementation of these climate change adaptation interventions will scale up and strengthen EbA, combined with associated community training, extension services and awareness-raising, would increase opportunities for knowledge and technology exchange between communities, and thereby promote autonomous upscaling of these interventions. Technical assistance (TA), capacity building and performance incentives to District Assemblies will further promote the upscaling and sustainability of these interventions, adaptation will be integrated into regional and district development plans, and the incremental costs for adaptation absorbed through earmarked financial top ups to regular districts budgets. With improved capacity, sub-national administrations, in particular District Assemblies, will be better placed to access additional national and international climate finance as well. This should in principle lead to integrated climate change responses at the local level and provide the continuity and maintenance capacity that local infrastructure and governance interventions often lack. Overall, these approaches would transition the smallholder farming communities of northern Ghana away from unsustainable agricultural and land-use practices, while promoting more climate-resilient, productive and sustainable livelihoods.

Project objective

The proposed project objective is to enhance the climate resilience of vulnerable smallholder farming communities in northern Ghana by improving food security and enhancing the agro-based rural economy.

Project outcomes

The project's outcomes will strengthen institutional and regulatory systems for climate-responsive planning and development, enabling government agencies and local institutions to effectively integrate climate adaptation into policies and strategies. The project supports the adoption of climate-resilient agricultural practices, alternative livelihoods, and access to financial resources, ensuring farmers can sustainably adapt to climate impacts. Additionally, the project improves knowledge and learning processes, fostering informed decision-making and empowering communities with the tools and information needed to scale up climate adaptation efforts across the region.

Barriers to the adaptation solution and project objective

Several barriers to the preferred adaptation solution have been identified for northern Ghana but can be categorised into three main barriers. These are listed below and explained in detail in Section 7 of Annex 2: Feasibility Study.

- **Barrier 1:** Limited institutional, technical and financial capacity for planning, budgeting, implementing and monitoring climate change adaptation interventions.
- **Barrier 2:** Limited knowledge and awareness of climate resilient methods and ecosystem linkages in disaster impact;
- **Barrier 3:** Inadequate access for communities to climate information and best practices for applying early warning data in agricultural practices.
- **Barrier 3:** Limited access for local governments and communities to financial resources to finance adaption interventions.

Project approach to overcoming the barriers to adaptation

The proposed project will overcome the abovementioned barriers to increase the climate resilience of smallholder farming communities in northern Ghana. This will be achieved through five interconnected outputs, described in detail in Section 1 of Annex 2: feasibility Study. The overall theory of change for this project is presented in Figure 1 below.

Barrier 1 will be addressed by strengthening the technical and institutional capacity of regional and district government structures in northern Ghana to plan, budget, implement and monitor climate change adaptation interventions. The policy work will take place in year 4-5 of the project, once results can be harvested and used for planning purposes. The EPA will execute this output, based on experience gathered in the execution of the country's National Adaptation Planning process, due to be completed in 2025.

Training on climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods will be provided to extension officers (Activity 2.1) and local communities (Activity 2.2), thereby addressing Barrier 2. Extension officers and local communities will implement these climate change adaptation interventions, allowing them to gain practical knowledge of these practices (Activity 2.4). Thirdly, training materials and knowledge products capturing best practices and lessons learned, including those related to the use of fire by communities from other initiatives such as the SLWMP, AF and IFAD projects will be disseminated (Activity 5.1), allowing knowledge of climate change adaptation to be shared with institutions and individuals who were not directly involved in project implementation. Finally, the project will conduct community-level knowledge-sharing and awareness-raising events (Activity 5.2) for both beneficiary and non-beneficiary communities in northern Ghana. By increasing the knowledge of climate change adaptation in non-beneficiary communities, these events will promote the autonomous upscaling of climate change interventions promoted by the project.

Barrier 2 will be addressed by convening of knowledge-sharing and awareness-raising events (Activity 5.2) with participants from: i) district-level government staff¹³; ii) regional-level government staff¹⁴; iii) national-level government staff¹⁵; iv) climate change adaptation project and programme managers; v) representatives of academic and research institutions; vi) NGO staff; vii) representatives of multilateral development agencies; and viii) project beneficiaries. These events will promote information sharing and encourage coordination between current and future climate change adaptation projects and programmes. A robust monitoring and evaluation strategy will be put into place (Activity 5.1) to future facilitate the generation and dissemination of knowledge products (Activity 5.2).

Barrier 3 will be addressed by the new data information and management system established under Activity 1.2. The existing drought early warning systems in Ghana have limited coverage and inadequate channels for information transfer and dissemination. As a result of these limitations, smallholder farmers in northern Ghana are unable to apply climate information to their agricultural practices and take proactive measures against drought impacts. By providing these farmers with quality climate data and early warning against drought, the project will enable a shift away from reactive methods for drought management — which currently contribute to crop losses, thereby increasing climate vulnerability.

The project will contribute to overcoming Barrier 4 by increasing the capacity of the most vulnerable smallholder farmers to form farmer-based organisations (FBOs), strengthening financial management capacity, links to value chains being developed in parallel thereby facilitating linkages to agricultural credit and insurance (Activities 4.1–4.4). The implementation of climate-resilient agricultural practices (Activity 2.4) will generate consistently higher incomes for smallholder farmers, allowing them to finance adaptation interventions once the project has finished. Approximately

¹³ E.g. extension officers, representatives from District Assemblies and DEMCs.

¹⁴ E.g. staff from regional EPA offices, Regional Department of Agriculture office and REMCs.

¹⁵ E.g. staff from the MESTI, MoFA, EPA, MLGCRA, MoF and the NDA.

US\$270,000 and US\$1.7 million in income is expected to be generated by investments in climate resilient agriculture and agroforestry (elaborated in Annex 3a).

Together, the interventions mentioned above overcome the barriers to climate change adaptation in northern Ghana, thereby contributing to a paradigm shift within the agricultural sector towards climate-resilient sustainable development.

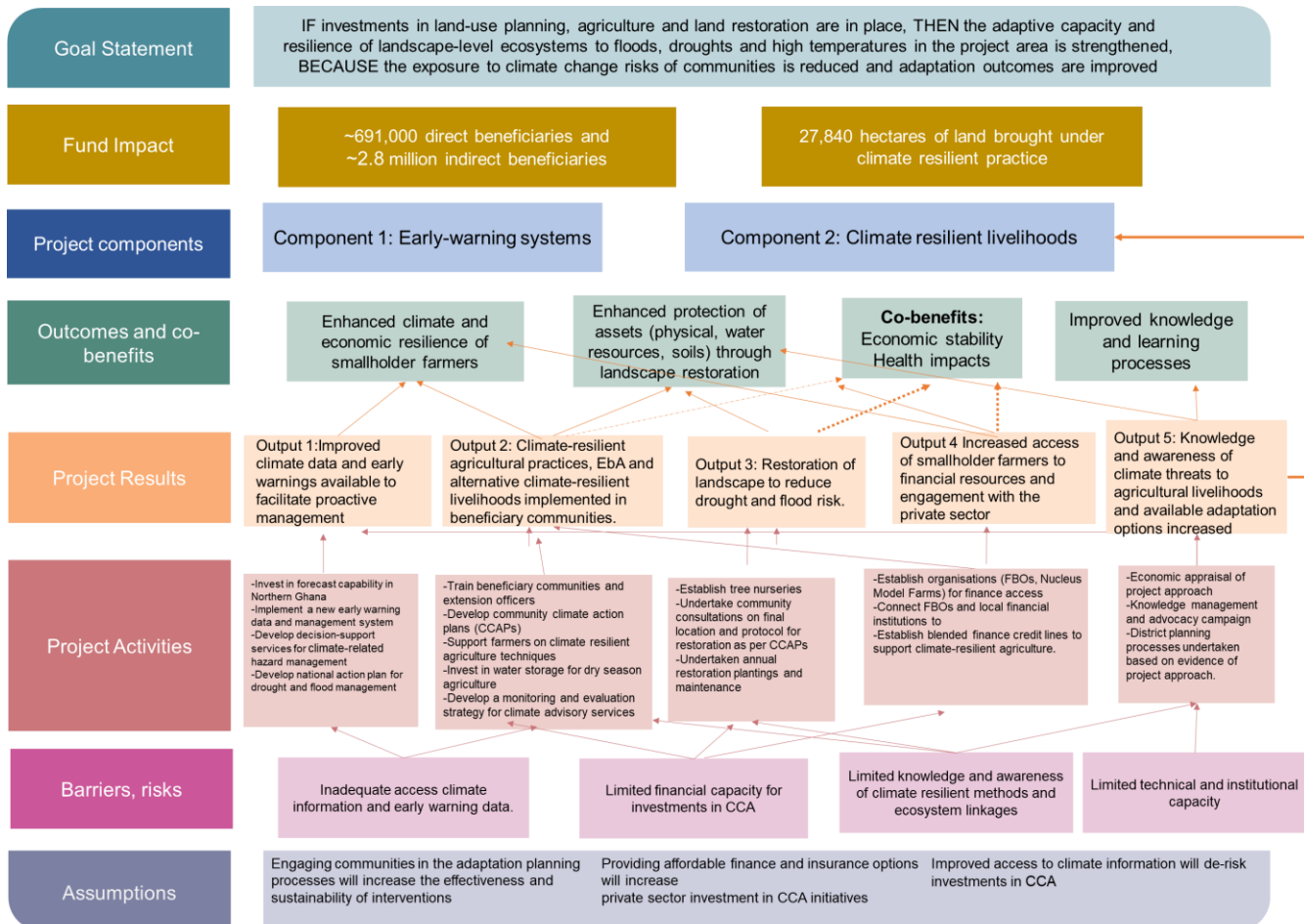


Figure 1: Theory of change diagram

The project will work at three levels: community level on planning and implementation systems as well as at the Regional and National level regarding investments in weather forecast capability that will serve the Northern Sector, complementing investments in forecasting capability in the South of the country as well as establishing a national action plan for drought and flood hazard management. The community level integrated ecosystem-based adaptation approach will inform the District level planning and budgeting systems. The project will build capacity and awareness of District planning officers to scale up the project approach. Combined with associated community training, extension services and awareness-raising, this will increase opportunities for knowledge and technology exchange between communities, and thereby promote autonomous upscaling of these interventions.

Specific efforts will be made to support women farmers and female-led households who are particularly vulnerable to climate impacts due to limited land tenure and financial access. Farmer-based organisations will be trained in financial literacy and savings management with a focus on empowering women and creating equitable access to credit and insurance schemes. Through this financial strengthening, FBOs, particularly those led by women, will enable the farmer groups to connect to the local banking sector thereby promoting higher returns that can be reinvested.

The relevant outputs will be supported by the two executing entities (EEs), namely the: i) Environment Protection Agency (EPA) of the Ministry of Environment, Science, Technology and Innovation (MESTI); and ii) Ghana Meteorological Agency (GMet) of the Ministry of Communication and Digitization. The EPA will be supported by the Ministry of Food and Agriculture as implementing partner (IP) for output 1. The GMet will be supported by the Water

Resources Commission as implementing partner and by UNEP-Danish Hydraulic institute (DHI) to provide technical advisory services for output 1.

B.2 (b). Outcome mapping to GCF results areas and co-benefit categorization

Output number	GCF Mitigation Results Area (MRA 1-4)				GCF Adaptation Results Area (ARA 1-4)			
	MRA 1 Energy generation and access	MRA 2 Low-emission transport	MRA 3 Building, cities, industries, appliances	MRA 4 Forestry and land use	ARA 1 Most vulnerable people and communities	ARA 2 Health, well-being, food and water security	ARA 3 Infrastructure and built environment	ARA 4 Ecosystems and ecosystem services
Output 1	☐	☐	☐	☐	☒	☒	☐	☒
Output 2	☐	☐	☐	☐	☒	☒	☒	☒
Output 3	☐	☐	☐	☐	☐	☒	☐	☒
Output 4	☐	☐	☐	☐	☒	☒	☐	☐
Output 5	☐	☐	☐	☐	☒	☒	☐	☐

Co-benefit number	Co-benefit					
	Environmental	Social	Economic	Gender	Adaptation	Mitigation
Co-benefit 1	☒	☐	☒	☐	☐	☐

B.3. Project/programme description (max. 2500 words, approximately 5 pages)

Geographies of intervention and target beneficiaries:

The proposed project will be implemented in the North East, Upper East and Upper West Regions of northern Ghana. As described above, the five northern regions are especially exposed to negative climate change impacts. Additionally, the northern populace is extremely sensitive to climate variability and change and has limited capacity to adapt to any changes in climate¹⁶. It is for these reasons of excessive exposure, elevated sensitivity and limited adaptive capacity that the northern regions of Ghana are considered the most vulnerable regions of the country to climate change¹⁷. See Section 3 of Annex 2 — Feasibility Study,¹⁸ for a detailed analysis that identifies the three northern regions as the most vulnerable to climate change impacts.

Within the northern regions of Ghana, implementation of the proposed project will be focused on eight of the 42 districts that comprise northern Ghana), namely:

- Jirapa, Lambussie, Lawra and Wa West in the Upper West Region;
- Binduri and Garu in the Upper East Region; and
- East Mamprusi and Yunyoo-Nasuan in the North East Region.

These districts were selected for project implementation using a comprehensive, quantitative assessment of district-specific vulnerability to climate change — incorporating exposure, sensitivity and adaptive capacity of communities — and a rigorous stakeholder engagement process (see Section 3 of Annex 2: Feasibility Study, and Annex 7h: Stakeholders Engagement Plan).

The districts are partly in the Sudan Savanna Zone, located in the north-eastern corner of Ghana, with the majority of the zone located in Burkina Faso and Mali. It is characterised by fire-swept short grasses interspersed with low-density

¹⁶ See 'Climate rationale and context' above for more information on sensitivity and adaptive capacity.

¹⁷ Please see Annex 2 'Feasibility Study': Section 3 for a full description of how climate change vulnerability was calculated.

¹⁸ Section 3: Climate change vulnerability of the districts and regions of Ghana.

woodland. Grass cover is low, and some areas of land are bare and severely eroded. The Sudan Savanna Zone receives the least rainfall of Ghana's agro-ecological zones (~940 mm per annum) and only one rainfall season. Other Districts are in the Guinea Savanna Zone (or Interior Savanna Zone), located north of the Transitional Zone. This is the largest agro-ecological zone in Ghana, covering the northern half of the country. The zone is characterised by wooded grassland, consisting of a ground cover of grasses of variable height, interspersed with fire-resistant, deciduous, broad-leaved trees. In general, tree cover and height decrease from south to north along a gradient of decreasing rainfall. The Guinea Savanna Zone receives ~1,100 mm of rainfall per year, mostly during one rainfall season.

From those districts not currently receiving support, and taking climate change vulnerability scores into account, the final project districts were selected by national, regional and district level stakeholders from: i) the National Designated Authority (NDA) of Ghana within the Ministry of Finance (MoF); ii) members of the inter-ministerial Technical Advisory Committee (TAC) to the NDA; iii) staff of the Directorate of Crop Services from the Ministry of Food and Agriculture (MoFA); iv) representatives of the Environmental Protection Agency (EPA) from the Ministry of Environment, Science, Technology and Innovation (MESTI); v) Ministry of Local Government, Chieftaincy and Religious Affairs (MLGCRA); and vi) members of civil society organisations.

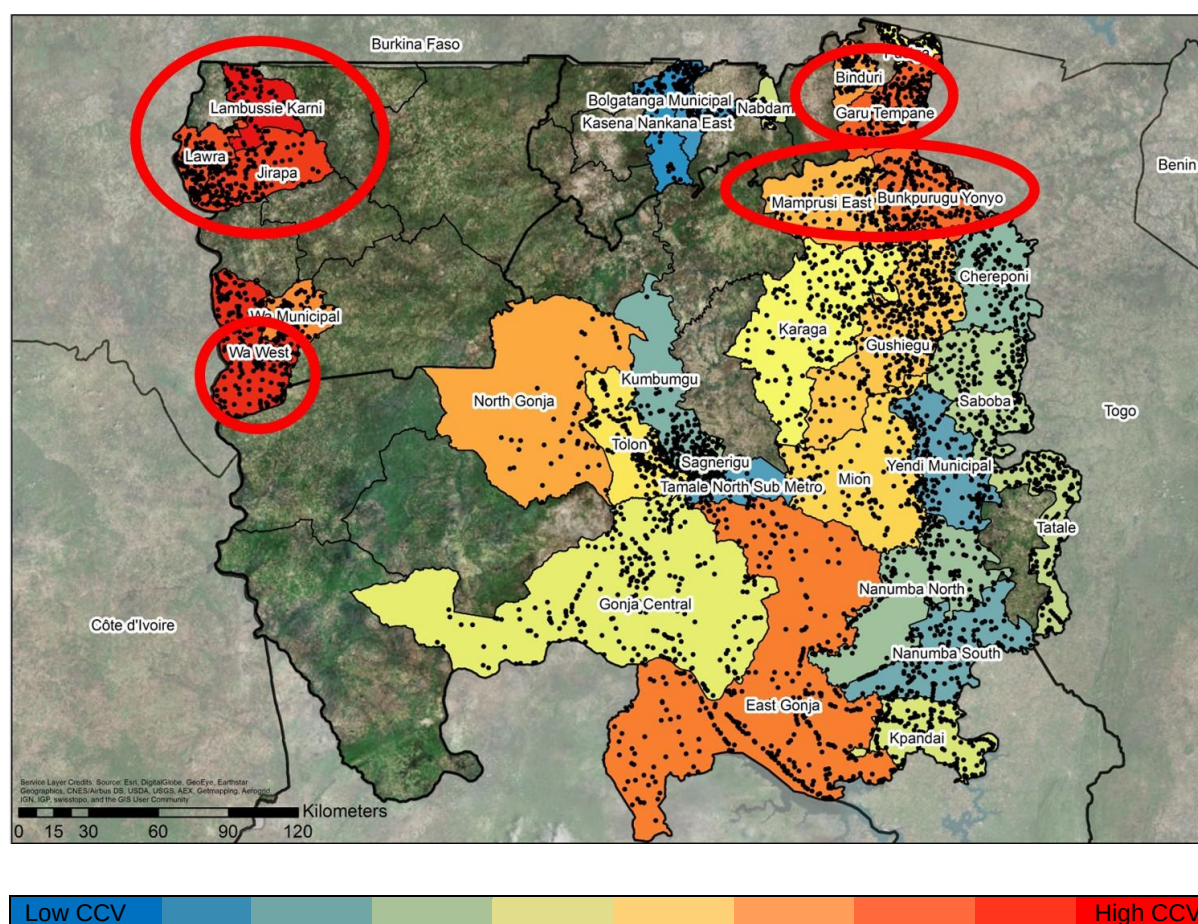


Figure 2. Target districts of the project within the three northern regions. Districts that are not depicted in this map are receiving support from the SLWMP and Adaptation Fund projects and thus were not considered eligible for selection.

Within each district, 15 communities will receive direct support from the project. Each community will receive support for three consecutive years (**Table 1**). Across the eight districts, the project will, therefore, support 120 communities over seven years. The direct beneficiary communities will be selected based on a rigorous set of selection criteria and comprehensive consultations at the national, regional, district and community levels (see Annex 7h: Stakeholder Engagement Plan). The selection criteria include *inter alia*: i) high vulnerability to climate change¹⁹; ii) proximity²⁰ to at least five other vulnerable communities to promote knowledge and technology transfer to these vulnerable

¹⁹ i.e. high exposure and sensitivity and low adaptive capacity.

²⁰ <5km

communities; iii) a willingness to participate; and iv) favourable land availability and access. Beneficiary communities will also be selected to ensure a representative geographic coverage across each district. The selection process has and will continue to involve consultations with: i) representatives from MESTI, MoFA and the NDA at the national level; ii) EPA and Department of Agriculture staff from the North East, Upper East and Upper West regional offices; iii) zonal EPA officers within target districts; iv) District Assemblies (DA)²¹ from the eight target districts; v) leaders from potential beneficiary communities (including chiefs and chief mothers); and vi) beneficiary community members.

Table 1. Timeframe for community support for a single district. A total of 15 communities will be supported in each district.

Community	Year						
	1	2	3	4	5	6	7
Community 1 – 3							
Community 4 – 6							
Communities 7 – 9							
Communities 10 – 12							
Communities 13 – 15							

At the time of writing, four communities within each district have been selected based on the criteria described above. These communities were selected following detailed community consultations (see Annex 7h: Stakeholder Engagement Plan and Section E.5.3) that took place during the development of this Funding Proposal. Three of the four communities already identified will be selected to receive support during the first year of project implementation. The remaining community, as well as the additional communities that will also be identified through the selection process described above, will receive support from the second year of project implementation onwards.

To ensure adaptive learning from implementation when moving from one community to the next (**Error! Reference source not found.**2) lessons learned will be systematically captured from each community after the first three years of direct support and applying those insights to subsequent communities. The project implementation team will conduct evaluations with community leaders, local authorities and participants to assess the effectiveness of interventions, identify challenges and document successful practices, particularly those related to climate vulnerability reduction, knowledge transfer, and land access. These findings will inform adjustments to the selection criteria, support strategies and community engagement methods, ensuring that the approach evolves and improves as the project progresses across the 120 communities. This iterative learning process will foster a dynamic, responsive project implementation that becomes more tailored to local contexts over time.

Project objective:

The proposed project objective is to enhance the climate resilience of vulnerable smallholder farming communities in northern Ghana by improving food security and enhancing the agro-based rural economy²².

The project will be implemented in eight districts in northern Ghana that have been specifically chosen because: i) of their high vulnerability to climate change impacts (see Annex 2: Feasibility Study, Section 3); and ii) they are currently not receiving support from other ongoing initiatives.

The project's objective will be achieved by the successful implementation of project activities to achieve five outputs comes as outlined in the Theory of Change and detailed in Section 13 of Annex 2: Feasibility Study and summarised below:

Component 1: Early warning systems

Output 1: Improved climate data and early warnings made available to facilitate proactive drought and flood management.

²¹ Consultations with District Assemblies focused on district extension officers, planning officers, disaster risk officers and fire services.

²² Project activities will enhance the agricultural productivity of smallholder farmers, allowing farmers to sell excess produce to increase their household income.

Ghana's early warning system will be enhanced by a range of interventions that will collectively facilitate access to improved information on floods and droughts (Activity 1.1), expand hydrometeorological and groundwater monitoring observation networks and expand national radar coverage, particularly in underserved Northern region which will significantly improve the accuracy of GMet's numerical weather prediction model and capacity to generate high skill forecast products (Activity 1.2), build capacities to equip personnel with advanced skills in drought and flood monitoring, assessment, and response, incorporating the latest technologies and best practices (activity 1.3) . as well as establishing a robust national framework for generating skill advisories and disseminating, *inter alia*, climate-related hazard management information to local communities (Activity 1.4). These efforts will be supported by a national action plan for drought and flood hazard management (Activity 1.5).

Activity 1.1: Implement a new early warning data information and management system to provide access to improved data sources and new datasets on floods and droughts

Activity 1.2: Enhancing hydrometeorological and groundwater monitoring observation networks

Activity 1.3: Capacitate key technical staff at national, regional and district levels, including GMet, HYDRO and WRC, for drought and flood services delivery

Activity 1.4: Establishing a robust national communication framework for disseminating DSS and climate-related hazard management to communities.

Activity 1.5: Implement national action plan for coordinating drought and flood hazard management in the agricultural sector

Component 2: Climate-resilient livelihoods

Output 2: Climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods implemented in beneficiary communities.

The proposed project will strengthen capacity for and implement climate-resilient agriculture, landscape restoration for drought and flood management and alternative climate-resilient livelihoods in beneficiary communities. The proposed project will focus on training of 200 extension officers (of which 30% of women) on climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods, and support the implementation of appropriate climate change adaptation interventions drawn from a menu of possible interventions identified from climate risk assessments and by beneficiary communities with an aim to reduce their vulnerability to climate change. Each beneficiary community will be trained on climate change interventions and will develop/finalise their own Climate Change Adaptation Plan (CCAP). The specific adaptation interventions identified within the CCAP – drawn from the menu of climate change adaptation interventions (See section 13, Annex 2: Feasibility Study)– will then be implemented in each community with the assistance of district extension officers.

Activity 2.1: Train extension officers on climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods.

Activity 2.2: Train beneficiary communities in northern Ghana on climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods.

Activity 2.3: Develop community climate action plans (CCAPs) in collaboration with beneficiary communities.

Activity 2.4: Implement climate change adaptation interventions, including climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods, identified in the CCAPs in beneficiary communities.

Activity 2.5: Develop a monitoring and evaluation strategy for climate advisory services in northern Ghana to improve the accuracy and appropriateness of advisories for smallholder farmers.

Output 3: Restoration of landscape to reduce drought and flood risk.

At the farm level, a variety of nature-based solutions or nature-based infrastructure can be implemented, including reforestation efforts through tree planting to reduce soil erosion and increase infiltration capacity and establishing riparian buffer zones along the riverside to reduce flood impacts. Encouraging the adaptation of flood-based farming measures to store flood water to be used during dry spells and drain the land after rainfall or floods will preserve agricultural production. During the project lifespan, agroforestry and restoration interventions will be implemented in 120 communities. Estimating ~1000 people per community, the direct beneficiaries of these interventions have been calculated by assuming that all 1000 individuals in each of the 120 target communities will benefit. In addition, 2000 hectares of riparian land in the 8 Districts will be restored to protect against floods²³.

²³ The nature-based infrastructure mentioned in the text (reforestation efforts through tree planting to reduce soil erosion and increase infiltration capacity and establishing riparian buffer zones along the riverside to reduce flood impacts) involves potential impact on Indigenous Peoples', particularly in terms of access to and use of their territories and the natural

The proposed project will:

Activity 3.1 Implement land restoration on communal land in 120 communities as per CCAPs

Output 4: Increased access of smallholder farmers to financial resources and engagement with the private sector.

Output 4 will bridge this gap by increasing the capacity of the most vulnerable smallholder farmers, such as women farmers, to form farmer-based organisations (FBOs) and manage finances and thereafter access agricultural credit and insurance. The project will connect the local level demand for credit to international blended finance initiatives which UNEP is brokering. In doing so, this output will contribute to GCF Outcome A7.0 — Strengthened adaptive capacity and reduced exposure to climate risks. GCF proceeds will not be used to finance credit lines and guarantees through financial intermediaries.

Technical assistance will be provided by GIRSA a Government non-banking financial institution established to de-risk agricultural financing and stimulate increased lending to the agricultural sector, with the project reimbursement of experts' time and logistical costs (rental of training venue and catering). GIRSA confirmed its interest in partnering for the project and agreed to lead the technical assistance to local FIs on the benefits of EBA/CRA for the creditworthiness of farmers.

In order to create a credit score database on FBOs and their members that are targeted by the project and have never borrowed before, an entity such as Financial Access, an NGO with specific expertise in the creation of smallholder farmer credit databases would be partnered with the EPA and the project. The entity would also train LFIs on its scoring methodology and use of the database. Financial Access has already confirmed its strong interest in creating a credit score database for both farmers and FBOs targeted by the project, to be made available as an online dashboard to LFIs. Financial Access could also provide technical assistance to LFIs on the Financial Access scoring methodology and use of the dashboard.

Activity 4.1: Establish farmer-based organisations (FBOs) and Village Savings and Loans Associations (VSLAs) that can access credit and insurance for farming and non-farming livelihood activities.

Activity 4.2: Connect FBOs and local financial institutions to improve access of beneficiary communities to credit and insurance products.

Activity 4.3: Establish a blended finance model to provide credit lines to support climate-resilient agriculture.

Output 5: Knowledge and awareness of climate threats to agricultural livelihoods and available adaptation options increased to inform the upscaling of climate change adaptation across northern Ghana.

To promote the upscaling and replication of project interventions, Output 5 will focus on activities that contribute to knowledge sharing and awareness-raising at national and local levels. By promoting the upscaling and replication of project interventions outside of beneficiary communities, the project will facilitate widespread and sustained behavioural transformation in smallholder farming communities in northern Ghana. The project will specifically integrate findings and experiences of two other approved GCF project: the UNDP-Forestry Commission project which does not have a knowledge management output and the AfDB project which has a small activity on knowledge management. The proposed project will increase the prominence of this output in its proposal by adding an economic valuation component to measure ecosystem services arising from all three GCF projects, drawing on the methodologies and experiences from our Environmental Economics work. Furthermore, the drought and flood management plan at the national level (Output 1) will feature as part of awareness and training workshops with District Assemblies in order to mainstream in the 8 District Medium Term Development Plans and will be used as part of the instruments to guide the preparation of the CCAPs at the community level as an integral part of the CCAPs.

Activity 5.1: Generate and disseminate knowledge products capturing best practice and lessons learned to inform the upscaling of climate change adaptation across northern Ghana.

Activity 5.2: Conduct knowledge-sharing and awareness-raising events.

Activity 5.3 Conduct district awareness and training workshops with District Assemblies to integrate CCAPs and EbA in District development plans and District environment plans and budgets

resources, rivers and water points available in the area. Therefore, this component requires measures to ensure access to rivers is secured for Indigenous Peoples. During the inception phase of the project's implementation, UNEP will consult Indigenous Peoples in the area to find out their customary use of the area and resources. In case the land and resources located in the area in which the nature-based infrastructures would be placed, securing FPIC will be required.

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

The proposed project will be implemented over a seven-year period. The **United Nations Environment Programme (UNEP)** will serve as the **Accredited Entity (AE)** and the following entities will serve as the **Executing Entities (EEs)**: i) Government of Ghana represented by the Ministry of Environment, Science and Technology (MESTI) delegating to the **Environment Protection Agency (EPA)**, which will provide overall lead coordinating and executing role for the project, and for Activities 2.1 2.2, 2.3, 2.4, 2.5 , in close collaboration and coordination with the Ministry of Food and Agriculture (MOFA) as implementing partner (IP), and lead on Activities 3.1, 4.1, 4.2, 4.3, 5.1, 5.2 and 5.3; ii). Government of Ghana represented by the Ministry of Communication and Digitalization's **Ghana Meteorological Agency (GMET)** for Activities 1.1, 1.2, 1.3, 1.4, 1.5 . with the UNEP-Danish Hydraulic Institute (DHI) providing technical advisory services and Ghana's Water Resources Commission serving as implementing partner (IP) for national and district level engagement and coordination.

Accredited Entity

As the **Accredited Entity**, UNEP will carry out fiduciary and safeguards oversight and provide the necessary scientific expertise and technical support to the project formulation, start up, implementation, evaluations and closure. In addition, a UNEP representative will fulfil the role of **Task Manager (TM)**, taking responsibility for project oversight and supervision, including undertaking at least two missions per year to the project. As AE, UNEP will ensure that the project activities are aligned with both GCF and UNEP's policies. Operating policies and procedures will follow UNEP's programme manual, which includes provisions for financial management and procurement.

UNEP is implementing several large integrated programmes funded by the GEF (the largest being USD98M for the net zero, nature positive accelerator integrated programme). We have also implemented several projects in Ghana as part of regional and global projects. UNEP has extensive experience in the design and implementation of initiatives that promote EbA — which delivers adaptation as well as mitigation benefits — and has championed tools and methodologies that the proposed project will rely on at a national and global level. These include Vulnerability Impact Assessments, Natural Capital Accounting, among others. An example of UNEP's operating experience in the region is the Stimulating Community Initiatives in Sustainable Land Management project (SCI-SLM, 2009–2014)²⁴. The overall objective of the SCI-SLM was to refine ways of stimulating the improvement and upscaling of community-based sustainable land management initiatives and to institutionally embed SCI-SLM approaches at local and regional levels in four pilot countries in Africa, including Ghana. UNEP was also active in Ghana under the Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety (BMUB) UNEP/UNDP/WRI GCF Readiness Programme. Under this programme, UNEP supported Ghana in capacity development with regards to National Designated Authority (NDA) capacity and decision-making about project prioritisation. It also supported national entity accreditation capacity building, as well as the accreditation application submission and was selected as an AE for this project when it was determined that national accreditation might delay the submission of this proposal. Specific projects implemented by the UNEP in Ghana include *inter alia*: i) addressing transboundary concerns in the Volta River basin and its downstream coastal area (2007–2012)²⁵; and ii) CC-DARE — Climate Change and Development — Adapting by Reducing Vulnerability (2008–2012)²⁶. In addition to the aforementioned projects, UNEP is the Delivery Partner for Ghana's NAP Readiness project GHA-RS-003 "Enhancing multi-sector planning and capacity for effective adaptation in Ghana". UNEP's central role in Ghana's NAP process allows for close alignment of the proposed project with the country's adaptation processes, including the central- and district-level systems for adaptation planning being established through the NAP Readiness project. Where possible, existing structures established under the NAP Readiness project will be used by the proposed project to maximise efficiency and collaboration. The project will also benefit from UNEP's Regional office for Africa's participation in the UN Country Team (UNCT) monthly coordination meetings, hosted by the UN Resident Coordinator Office, with the aim of ensuring effective coordination of UN programmes and projects executed by Government ministries and agencies with support from UN agencies (e.g. UNDP, FAO, etc.), and benefit from lesson learned and avoid duplication of efforts.

Executing Entities

²⁴ UNEP – 2016 – Stimulating Community Initiatives in Sustainable Land Management (SCI-SLM). Terminal Evaluation, Final Report. Available at: <https://wedocs.unep.org/rest/bitstreams/10610/retrieve>

²⁵ UNEP – 2011 – Addressing transboundary concerns in the Volta River basin and its downstream coastal area. Mid-term evaluation. Available at: <https://wedocs.unep.org/rest/bitstreams/806/retrieve>

²⁶ UNEP-UNDP – 2013 – CC-DARE – Climate Change and Development – Adapting by Reducing Vulnerability. Terminal Evaluation, Final Report. Available at: <https://wedocs.unep.org/rest/bitstreams/818/retrieve>

The **Environmental Protection Agency (EPA)** of the Ministry of Environment, Science, Technology and Innovation (MESTI) will serve as the main **Executing Entity (EE)** of the project. The EPA is the lead institution for UNFCCC activities in the country and is the main Country Implementation Institution for the technical coordination of activities on climate change and other environmental conventions ratified by Ghana. The EPA has a proven record as an executing entity in Ghana, having been instrumental in the implementation of large-scale environmental programmes, including *inter alia* the GEMP and SLWMP.

The Ghana Meteorological Agency (GMet) will implement Output 1. GMet have technical expertise with respect to climate variability, climate change, seasonal forecasting and drought assessment.

UNEP will sign separate Project Cooperation Agreements (PCAs) with the EPA and GMet to establish clear roles and responsibilities for the execution of project activities²⁷; and to ensure that the project is executed in line with GCF and UNEP rules, policies and requirements. The PCAs will establish clear roles and responsibilities for the respective parties for the delivery of the proposed activities, the schedule and conditions for instalments, the determination of the prevailing fiduciary standards and the terms and conditions for arbitrations and termination of contract. The PCAs will include specific obligations for the EEs on project execution, financial management, personnel administration and reporting, as well as on arbitration and liability terms. The EEs will be required to comply with UNEP rules, policies and procedures on procurement. The EPA and GMet will be accountable to the AE for the effective use of resources during project implementation. The EEs will supervise, oversee and manage project implementation, as well as report on project progress; and to ensure that project activities are well coordinated and aligned with national priorities. UNEP, as Accredited Entity for the project, has undertaken a capacity assessment of the EPA particularly by reviewing the elements of governance and public accountability. UNEP is confident that the EPA has the relevant fiduciary and financial management capacity to act as the EE. All operating procedures will align with the UNEP's programme manual, which includes provisions for financial management and procurement. GMET and EPA will sign an Implementation Agreement that sets out the reporting relationship (GMET to EPA) as well responsibilities on delivering project targets.

UNEP will also sign a PCA with the UNEP-DHI Centre on Water and Environment (UNEP-DHI Centre) which is a United Nations Environment Programme (UNEP) centre of expertise. UNEP-DHI Centre is dedicated to supporting UNEP's work in helping national governments improve the management, development and use of their freshwater resources in the face of climate challenges. The UNEP-DHI Centre is hosted by DHI, an independent, not-for-profit, international advisory and research organization. DHI has more than 60 years of experience in water resources management and is represented by over 1,000 staff in more than 25 countries. For the upcoming project, UNEP-DHI Centre will draw upon DHI's cutting edge hydrological and climatic modelling tools, which are globally recognized and used by governments and leading international organizations in many regions of the world. UNEP-DHI was the implementing partner in the pilot project that looked at the flood and drought early warning system for the Volta Basin, covering Benin, Burkina Faso, Cote D'Ivoire, Ghana, Mali and Togo, which ended in 2017. UNEP-DHI is now providing technical assistance to a project focused on reversing the deterioration of ecosystems and water resources in the Volta River Basin. The goal of the project is to support integrated ecosystem-based development in the Volta River Basin by enhancing transboundary governance and restoring and conserving ecosystems to ensure sustainable livelihoods. Funded by the Global Environment Facility (GEF) through UN Environment Programme and IUCN, the five-year project is being implemented by the Volta Basin Authority (VBA) and the Global Water Partnership-West Africa (GWP-WA) between 2023-2028.

Structure for implementation and governance

At the national level, the EPA will receive guidance from a **Project Steering Committee (PSC)** and technical advice from the **National Climate Change Committee (NCCC)**. The PSC will be responsible for providing project oversight and advisory support for project implementation. This includes reviewing annual work plans and project reports, and approving changes to project targets, activities or timelines. Bi-annual PSC meetings — along with *ad hoc* meetings when deemed necessary — will be held to take management decisions, discuss project performance against indicators and provide technical/strategic guidance to the Project Management Unit (PMU). The PSC will be co-chaired by MESTI and UNEP and comprise representatives from the: i) MoFA; ii) Ministry of Finance (MoF); iii) Forestry Commission; iv)

²⁷ The PCA will determine requirements for the disbursement and management of funds to the EE, as well as establish agreed upon supervisory roles.

Water Resources Commission; v) Ministry of Local Government, Chieftaincy and Religious Affairs (MLGCRA); vi) Ministry of Land and Natural Resources; vii) Savannah Accelerated Development Authority (SADA); viii) GMet; ix) NADMO; x) Ministry of Gender, Children and Social Protection (MoGCSP); and xi) Civil Society Organizations (CSOs) in Agriculture²⁸. Gender balance will be ensured in the PSC, and women's associations, ethnic minority groups and Indigenous Peoples will be represented through inclusion of CSOs advocacy groups that will be selected by them²⁹, or directly by traditional authorities nominated during the inception period. National implementing entities from Ghana accredited, or currently seeking accreditation, with the GCF, namely the Social Investment Fund and EcoBank, will also be invited to observe PSC meetings. The EPA will be the Secretariat to the PSC.

The **Project Management Unit (PMU)** will be in the Regional EPA office located in Tamale. A Project Manager (PM) will be competitively recruited to the PMU, along with a Financial Officer (FO), a Procurement and Administrative Officer (PAO), A Monitoring and Reporting Specialist (MRO) and Social and Environment Safeguards and Gender Specialist (ESS/GS). The Deputy Executive Director (Technical) of EPA will be assigned the role of Project Director and will be the primary focal point in the EPA for the project. A UNEP Chief Technical Advisor, based in the PMU will be responsible for providing technical advisory support to the PM and PMU in the overall planning, implementation and monitoring of the project, and ensure programmatic and technical direction of the project to achieve intended objectives, outputs and deliverables. PM responsibility will include coordinating with executing entities and overall management of the project, regional and district staff to ensure efficient execution at the local level. The responsibilities of the FO and PAO will include: i) providing administrative, logistical and financial support to the PMU at the national level; ii) assisting the PMU and the UNEP Finance and Procurement Officer (FPO) in writing and disseminating project reports; and iii) ensuring that project activities and tasks are carried out following government and UNEP best practices and procedures. In particular, the FPO — in cooperation with the UNEP Finance Management Officer (FMO) — will be responsible for managing all financial transactions, assuring alignment with project work plans and national priorities.

National, regional and local stakeholders

At the regional (sub-national) level, the project will coordinate with three regional EPA offices and Department of Agriculture offices located in Nalerigu (North East Region), Bolgatanga (Upper East Region) and Wa (Upper West Region). In each region, the EPA and Department of Agriculture will receive guidance from Local Steering Committees (LSCs) comprising the: i) Regional Coordinating Council; ii) Regional Planning Office; iii) Regional Director of Agriculture; iv) Regional Director of the EPA; v) District Coordinating Directors of project districts; vi) District Directors of Agriculture of project districts; vii) Regional Fire Officers; viii) District Forest Officers; and ix) District Planning Officers of the project districts. Additional members may be included in the LSCs if and when required.

The EPA will establish an Implementation Agreement (IA) with the Ministry of Food and Agriculture (MoFA) with regards to the management and execution of agricultural interventions. A Project Implement Manual will be established that defines roles and mandates of the respective parties to execute the project in accordance with the financing proposal. The MoFA Agricultural Extension Agents will be the primary link between district government structures and beneficiary communities. This will include capacity building activities from the respective institutions for farming communities, as well as feedback and inputs from farming communities to project implementing parties in shaping the information services and their delivery modes to target the needs of farmers. Utilising the extension services communication channels will be particularly important given their importance in communicating technological advances and available resources from the national level institutions to the local community level (further details are provided in Annex 2: Feasibility Study). The need to equip extension officers with better climate information that can then be transferred to local communities has been echoed in community level consultations with farmers. In beneficiary communities, Climate Change Action Groups (CCAGs) and Farmer Based Organisations (FBOs) will provide a link between community members and district-level government structures and will serve to coordinate community-level activities.

The GMet will establish an Implementation Agreement (IA) with the Water Resources Commission (WRC) with regards to integrating hydrological information, in particular groundwater monitoring, into early warnings made available to facilitate proactive flood and drought hazard management. A Project Implement Manual will be established that defines roles and mandates of the respective parties to execute the project in accordance with the financing proposal.

²⁸ In 2017, the Government of Ghana began restructuring the SADA into the Northern Accelerated Development Authority (NADA) with the intention of accelerating economic and social development in the Northern Development Zone.

²⁹ Organisations that may represent Indigenous Peoples on the PSC could include the Centre for Indigenous Knowledge and Organizational Development (CIKOD) and the Pastoral Rights Protection Network Ghana as these are the foremost organisations in Ghana advancing the rights of pastoralists. These organisations would only be included based on a clear and formal nomination from traditional authorities representing potentially affected Indigenous Peoples. If IP groups prefer representation by other traditional authorities on the PSC, these will be prioritised by the project.

In establishment of the national action plan and coordination mechanism, as well as training activities under Output 1, the project will also seek to include and link to other relevant stakeholders along the agricultural-value chain — such as technology providers and insurance actors — where found to be beneficial to the farming communities, to ensure that the project maximises its benefits to local communities and agricultural sector as a whole. Further details are provided in the Stakeholder Engagement Plan (Annex 7h) and the Gender Action Plan (Annex 8).

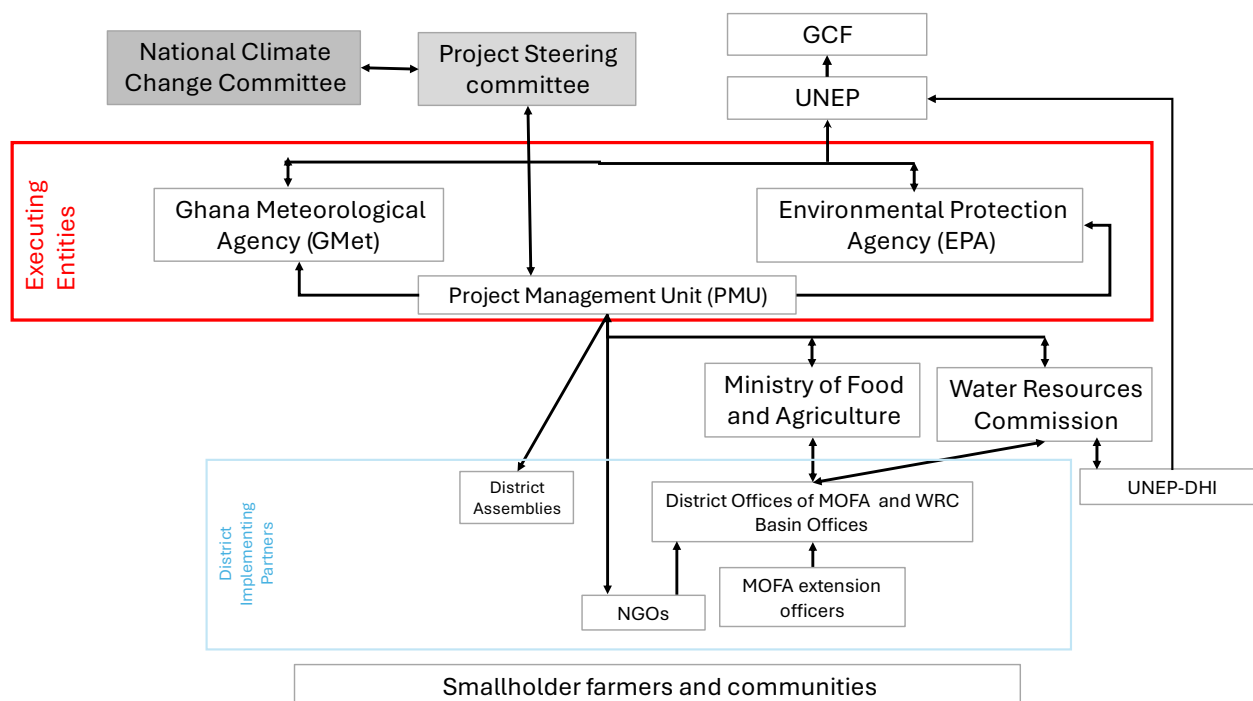


Figure 3. Implementation arrangements for the project. Black arrows depict reporting lines.

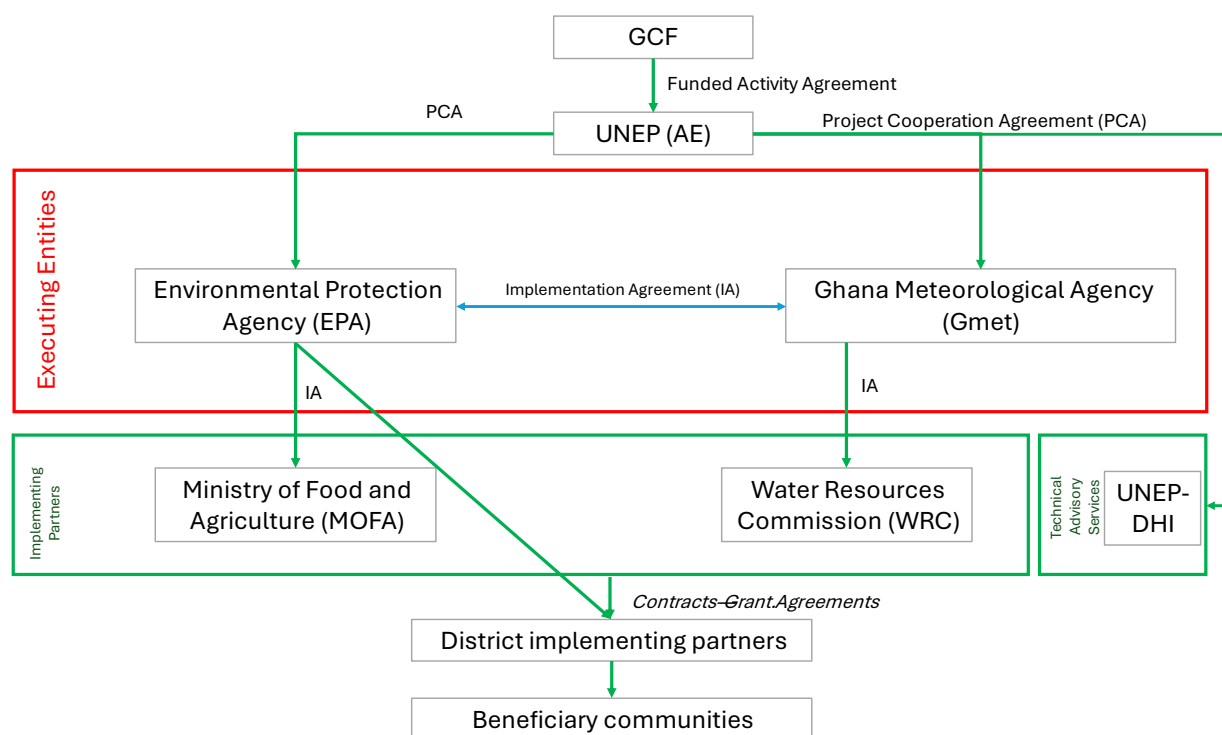


Figure 4. Flow of funds and contractual arrangement for the project, green arrows indicate the flow of funds and contractual arrangement, while blue arrows indicate contractual arrangement only between the relevant parties.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

Building the climate-resilience of the Ghanaian agricultural sector is vital to protect the livelihoods of smallholder farmers in northern Ghana. Shifting rainfall and temperature regimes are expected to reduce already suboptimal crop yields. Consequently, food security and income generation of the country's most vulnerable people will be negatively impacted. These negative impacts will be compounded by — and further enhance — severe environmental degradation resulting from rapid population growth, dependence on natural resources and the use of unsustainable agricultural techniques. Without effective adaptation action, smallholder farmers will likely become increasingly vulnerable to the current and future effects of climate change, with serious repercussions for both the northern and Ghanaian economy. However, climate change adaptation in Ghana is hampered by a lack of financial resources, institutional and technical capacity needed to implement effective adaptation interventions.

Without investment into adaptation interventions, the livelihoods of ~10 million men and ~12 million women (~75% of the population) who are reliant on agriculture will remain at risk to the threats of climate change. However, the lack of financial resources hampers climate change adaptation in Ghana. The agricultural sector in Ghana is reliant on foreign funding, with ~65% of total public expenditure in support of food and agriculture funded by external sources. These include, among others, the World Bank, USAID and the European Union³⁰. The need for external funding is because of the current state of the Ghanaian economy. Despite a period of rapid economic growth between 2000 and 2013, graduating Ghana to lower-middle-income status, the country's economy has seen a decline in growth over the past seven years³¹. Increasing debt, high inflation and currency depreciation have contributed to the deterioration of the economy, leading to a current account deficit of -9.3%³² and a debt-to-GDP ratio of 58%³³. It is estimated that without interventions, the cumulative damages of weather events in four of the eight implementation districts will reach US\$680.2 million by 2050 (elaborated in Annex 3: Economic and Financial Analysis). These damages will further constrain the GoG's fiscal space for financing adaptation and resilience interventions. Additional challenges to Ghana's economic outlook include high domestic financing costs, technical problems in the oil and gas sectors, energy problems

³⁰ FAO. 2014. Analysis of Public Expenditure in Support of the Food and Agriculture Sector in Ghana, 2006-2012: A preliminary assessment. Technical notes series, MAFAP, by Ghins, L., Rome.

³¹ IMF Survey – 2016 – Ghana: The Bumpy Road to Economic Recovery. Available at: www.imf.org/.

³² World Bank – 2017 – Trading markets in Ghana – Country Private Sector Diagnostic. Available at: http://www.ifc.org/wps/wcm/connect/ac42c20a-c82c-48b7-8432-221c0e066e2a/CPSD-Creating-Markets-in-Ghana-Nov-2017_v1.pdf?MOD=AJPERES&CID=IFC_LI_IFC_EN_EXT.

³³ <https://www.bloomberg.com/news/articles/2019-05-28/ghana-s-debt-at-the-highest-in-four-years-as-revenue-undershoots>

related to state-owned enterprises and continued weak commodity prices³⁴. These factors have caused an unfavourable fiscal climate for the Ghanaian government, resulting in its inability to fund or secure private-sector loan financing for large-scale investments in innovative climate change adaptation measures. Therefore, given the current economic situation in Ghana, the contribution of funds by the GCF is critical for the successful implementation of the proposed project. Without investment into climate-resilient agriculture and EbA interventions, the livelihoods of the ~75% of the population of northern Ghana that are reliant on smallholder farming are at risk to the threats of climate change.

Funds are being requested by the GoG in the form of a grant without repayment contingency, given the focus of the project on the most vulnerable communities in Ghana. The grant funds will be used to provide the inputs necessary for project implementation while the EPA and MOFA will bear a large share of the project staffing costs. Grant finance is needed to fund the strengthening of technical and institutional capacity to develop, implement and coordinate climate change adaptation at regional and district level in northern Ghana because of inadequate financial resources³⁵. Although District Assemblies generate some revenues on their own (i.e. through district taxation), they barely manage to collect sufficient funds to cope with the many development priorities they face³⁶. This is especially so in northern Ghana where many of the inhabitants depend on subsistence rain-fed agriculture for their livelihood³⁷. The limited budgets are spent on priority projects, such as health, road construction and education with inadequate resources available for additional activities such as climate change adaptation³⁸. Given the current economic situation in Ghana, the contribution of funds by the GCF is critical for the successful implementation of the proposed project.

Given the focus of the project on the most vulnerable communities in Ghana, there is currently limited investment from the private sector in these approaches. Consequently, northern Ghana has limited access to the financial resources necessary for the adoption of appropriate climate-resilient practices to adapt to climate change. Smallholder farmers are unable to leverage private sector finance as financial institutions are unwilling to grant loans to them for agricultural activities because of the high-risk nature of rain-fed agriculture in the region^{39,40}. Instead, the banks employ joint liability lending structures and lend to registered farmer-based organisations (FBOs)⁴¹.

The sustainable development of northern Ghana's agricultural sector requires substantial investment into adaptation practices such as climate-resilient agriculture and EbA to build the climate-resilience of vulnerable farming communities. The initial returns of such investments are largely at the individual level, with returns being re-invested by farmers into agricultural production and used to cover their basic needs, leaving limited scope for cost recovery for early investors. A GCF grant is, therefore, being sought by the GoG to invest in climate change adaptation and bring about a paradigm shift in the agro-based rural economy of northern Ghana. Through this project, GCF support will accelerate northern Ghana's transition towards climate-resilient sustainable development, facilitating economic growth and securing the livelihoods of vulnerable smallholder farming communities.

The Government of Ghana has secured co-financing to the value of US\$6,987,681 through in-kind contributions and inclusive of a cash co-financing of US\$250,000. In-kind contributions consistent of staff time, offices and equipment that will be provided by the EPA and Directorate of Crop Services to implement this project.

B.6. Exit strategy (max. 500 words, approximately 1 page)

³⁴ World Bank – 2017 – Ghana Country Overview. Available at: <http://www.worldbank.org/en/country/ghana/overview>.

³⁵ Sova C, Chaudhury AS, Nelson W, Nutsukpo DK, Zougmore RB. 2014. Climate change adaptation policy in Ghana: Priorities for the agriculture sector.

³⁶ Mosello B, Adamtey R, Obuobie E (eds). 2017. *Making water infrastructure investment decisions in a changing climate. A political economy study of river basin development in Ghana*. Overseas Development Institute. London.

³⁷ Ibid.

³⁸ Ibid.

³⁹ Fearon J. 2000. Economic analysis of soil conservation practices in northern Ghana. MPhil, University of Ghana, Accra.

⁴⁰ Antwi-Agyei P. 2012. Vulnerability and adaptation of Ghana's food production systems and rural livelihoods to climate variability. DPhil, University of Leeds, Leeds.

⁴¹ Gallenstein RA, Mishra K, Sam AG, Miranda MJ. 2019. Willingness to pay for insured loans in northern Ghana. *Journal of Agricultural Economics* 70: 640-662.

The project will work at three levels: community level on planning and implementation systems as well as at the Regional and National level regarding investments in weather forecast capability that will serve the Northern Sector, complementing investments in forecasting capability in the South of the country as well as establishing a national action plan for drought and flood hazard management. The community level integrated ecosystem-based adaptation approach will inform the District level planning and budgeting systems. The project will build capacity and awareness of District planning officers to scale up the project approach. Combined with associated community training, extension services and awareness-raising, this will increase opportunities for knowledge and technology exchange between communities, and thereby promote autonomous upscaling of these interventions.

Fundamental aspects of the project's sustainability are listed below.

- The project will create capacity in District Authorities to replicate and maintain climate adaptation activities after the initial investment period, through the integration of climate change adaptation into regional and district medium-term development plans as well as through the consolidation of a country-based financing mechanism to channel climate finance at the local level.
- Beneficiary communities will receive extensive training on climate-resilient agricultural practices and the implementation of on-the-ground EbA interventions. In addition to building the adaptive capacity of these rural communities and allowing them to become self-sufficient under climate change conditions, an enabling environment will be created for the long-term maintenance and upscaling of proposed project activities. This will be further supported by the introduction of alternative climate-resilient livelihoods that add value to agricultural products, thereby further promoting sustainability. This will include improved storage, processing equipment and access to finance.
- A grassroots approach to building climate-resilience will ensure that activities are needs-based, locally appropriate and supported by local communities. An extensive community engagement process — including the development of Community Climate Action Plans — will be carried out. Throughout this process women and youth will be targeted, particularly through the introduction of alternative climate-resilient livelihoods, to ensure their participation and support for project interventions. This will promote the long-term commitment of community members after the project's implementation period has elapsed.
- The project will increase the access of smallholder farmers to financial resources by establishing FBOs that can access credit and insurance for farming and non-farming livelihood activities. In addition, the project will provide smallholder farmers in the FBOs, training in business and financial management. This enterprise approach will ensure that communities are equipped with the financial resources, business knowledge and financial capacity required to sustain and upscale project interventions after the support phase. The project will develop bankable deals for de-risking finance for access by smallholder farmers through the Agri3 Fund currently being established as a partnership between UNEP, Rabobank and the Dutch Government.

An extensive knowledge-sharing programme will support the project's M&E activities, which will allow non-beneficiary communities to learn from the experiences of nearby target communities. In addition, annual knowledge-sharing conferences will function as a platform for discussions on the lessons learned and sharing of new ideas and research on EbA to ensure the sustainability of interventions under future climate change conditions. Subsequently, this knowledge base will be used as a resource to leverage additional funds from public and private sector investors to maintain and replicate project interventions after the project's funding period, while further strengthening cross-sectoral collaboration.

C. FINANCING INFORMATION						
C.1. Total financing						
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)	Total amount			Currency		
	63.211141			million USD (\$)		
GCF financial instrument	Amount	Tenor	Grace period	Pricing		
(i) Senior loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(ii) Subordinated loans	<u>Enter amount</u>	<u>Enter years</u>	<u>Enter years</u>	<u>Enter %</u>		
(iii) Equity	<u>Enter amount</u>			<u>Enter %</u> equity return		
(iv) Guarantees	<u>Enter amount</u>	<u>Enter years</u>				
(v) Reimbursable grants	<u>Enter amount</u>					
(vi) Grants	63.211141					
(vii) Results-based payments	<u>Enter amount</u>					
(b) Co-financing information	Total amount			Currency		
	6.987681			million USD (\$)		
Name of institution	Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
Government of Ghana – Environment Protection Agency	<u>Grant</u>	<u>0.25</u>	<u>million USD (\$)</u>	<u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Government of Ghana – Environment Protection Agency	<u>In kind</u>	<u>1.988967</u>	<u>million USD (\$)</u>	<u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Government of Ghana – Ministry of Food and Agriculture	<u>In kind</u>	<u>2.848714</u>	<u>million USD (\$)</u>	<u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Government of Ghana – Water Resources Commission, Ministry of Sanitation and Water Resources	<u>In kind</u>	<u>0.5</u>	<u>million USD (\$)</u>	<u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
Government of Ghana – Ghana Meteorological Agency	<u>In kind</u>	<u>1.4</u>	<u>million USD (\$)</u>	<u>Enter years</u>	<u>Enter%</u>	<u>Options</u>
(c) Total financing (c) = (a)+(b)	Amount			Currency		
	<u>70.198822</u>			<u>million USD (\$)</u>		

(d) Other financing arrangements and contributions (max. 250 words, approximately 0.5 page)	N/A
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C.2. Financing by component

Component	Output	Indicative cost million USD (\$)	GCF financing		Co-financing		
			Amount million USD (\$)	Financial Instrument	Amount million USD (\$)	Financial Instrument	Name of Institutions
<u>Component 1: Early warning systems</u>	Output 1: Improved climate data and early warnings made available to facilitate proactive flood and drought hazard management.	15.358090	13.458090	Grants	1.900000	In-kind	GMet,
<u>Component 2: Climate-resilient livelihoods</u>	Output 2: Climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods implemented in beneficiary communities.	25.917811	22.819097	Grants	3.098714	In-kind	EPA & MoFA
	<u>Output 3:</u> Restoration of landscape to reduce drought and flood risk.	<u>20.448344</u>	<u>19.823957</u>	Grants	0.624387	In-kind	<u>EPA</u>
	Output 4: Increased access of smallholder farmers to financial resources and engagement with the private sector.	1.889342	1.071100	Grants	0.428	In-kind	EPA
					0.390242	In-kind	
	Output 5: Knowledge and awareness of climate threats to agricultural livelihoods and available adaptation options increased to inform the upscaling of climate change adaptation across northern Ghana.	2.855148	2.464906	Grants	0.390242	In-kind Choose an item.	EPA

	<u>M&E Costs</u>	1.392462	1.314414	<u>Grants</u>	0.78048	In-kind	<u>EPA</u>
	<u>Project Management Costs</u>	Enter amount 2.337625	Enter amount 2.259577	<u>Grants</u>	Enter amount <u>0.78048</u>	In-kind Choose an item.	EPA
Indicative total cost (USD)		70.198822	63.211141		6.987681		

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☐ No ☒

The proposed project will build the technical and institutional capacity of regional and district government staff and local smallholder farming communities to implement and monitor climate change adaptation projects under Outputs 1 - 5 of the project. Output 1 will transition farmers to a proactive planning and management approach to climate hazards. Under Output 2, 3 and 4 of the project, capacity-building activities will focus on training extension officers on climate-resilient agriculture and EbA (Activity 3.1); beneficiary communities on: i) climate-resilient agriculture, EbA and alternative climate-resilient livelihoods (Activity 3.2); ii) basic business and financial management (Activity 3.2); and iii) processes for applying for credit and insurance from LFIs (Activity 4.2). Capacity-building activities conducted under Output 3 will allow communities to become financially self-sufficient, creating opportunities for further investments into climate-resilient livelihoods.

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's [Initial Investment Framework](#).

D.1. Impact potential (max. 500 words, approximately 1 page)

The proposed project will contribute to the GCF's fund-level adaptation objective of increased climate-resilient sustainable development by reducing the vulnerability of smallholder farmers in northern Ghana through the implementation and adoption of climate-resilient agriculture, ecosystem-based adaptation and alternative climate-resilient livelihoods. The number of direct and indirect beneficiaries, disaggregated by gender, and the project's contributions to the GCF's outcome-level objectives and specific results areas are described and quantified below⁴².

Adaptation impact potential through increased resilience

The proposed project will contribute to three GCF adaptation results areas (ARAs), namely: i) ARA1 — increased resilience of the most vulnerable people and communities; ii) ARA2 — increased health, well-being, food and water security; ARA3 - Infrastructure and built environment and iv) ARA4 — increased resilience of ecosystems and ecosystem services. Specific contributions to these result areas are described in the following sub-sections.

Direct beneficiaries of the proposed project will include the 691,125 rural crop farmers who will benefit from improved coverage by Early Warning Systems (Output 1), as well as 6920 farmers benefitting from improved climate-resilient agricultural practices and 120,000 farmers benefitting from agroforestry and restoration (Output 3)

Indirect beneficiaries comprises the population of the three target regions in northern Ghana — **2,861,674** people — who will benefit from the enhanced food and water security and improved resilience to climate hazards delivered by the proposed project through the improvement of climate data and early warnings dissemination (Output 1). This information will be used to, *inter alia*, coordinate national-level drought and flood risk planning and responses, which will improve the climate-responsiveness of national planning in the future. For more information on the project's outputs, see Section 13 of Annex 2: Feasibility Study.

ARA1 – Increased resilience of the most vulnerable people and communities

⁴² Detailed information on the assumptions and estimates of the impact potential of the proposed project are provided in the project's results framework and economic analysis (Section E of this document and Annex 3, respectively).

Ghana's early warning system will be enhanced by a range of interventions that will collectively facilitate access to improved information on floods and droughts (Activity 1.1), expand hydrometeorological and groundwater monitoring observation networks and expand national radar coverage, particularly in underserved Northern region which will significantly improve the accuracy of GMet's numerical weather prediction model and capacity to generate high skill forecast products (Activity 1.2), build capacities to equip personnel with advanced skills in drought and flood monitoring, assessment, and response, incorporating the latest technologies and best practices (activity 1.3) . as well as establishing a robust national framework for generating skill advisories and disseminating, *inter alia*, climate-related hazard management information to local communities (Activity 1.4). These efforts will be supported by a national action plan for drought and flood hazard management (Activity 1.5). Although the interventions will be primarily designed to support direct beneficiaries (e.g., rural crop farmers), indirect beneficiaries—such as local businesses, families, and surrounding communities—will also gain access to early warning data, which will enhance their ability to plan and reduce disaster risks, thus improving resilience.

ARA2 – Increased health, well-being, food and water security

Ghana's early warning system will be enhanced under Output 1 to facilitate proactive drought and flood management. Access to up-to-date climate information, facilitated by the mobile application (Activity 1.4), will benefit 691,125 rural crop farmers (52% women and 48% men), across the country who will be empowered to effectively plan for and respond to drought and flood hazards, effectively reducing their food insecurity. Proactive measures that farmers will be supported to undertake via this intervention include, *inter alia*: i) timing crop planting for optimum yields; ii) shifting to drought- and flood-resistant crops; and iii) relocation of agricultural efforts to more suitable sites that will minimise crop losses. These proactive measures will reduce the impacts of extreme weather events on food production, leading to greater food security for rural households. Additionally, improved management of water resources, coupled with climate-resilient agricultural practices, will ensure more reliable water availability for both agricultural and household use, contributing to improved health and well-being by reducing the risks of food and water shortages during periods of climatic stress. Furthermore, by reducing the risks of crop losses and supporting improved agricultural practices, the food security of broader rural populations, including those not directly targeted by the interventions, will also improve. Additionally, local ecosystems' stability from proactive management of agricultural lands will contribute to improved water security in rural areas.

Food security will also be improved through activities under Output 2 by the introduction of climate-resilient agricultural interventions across northern Ghana, including measures such as crop rotation and intercropping, slash and mulching, conservation tillage, contour ploughing, contour bunding, vegetative barriers, ridging, organic composting and zai pits, cover cropping, introducing climate-resilient seed varieties, adapted planting calendar and dry season gardening. These strategies will benefit 6,920 farms (and 13, 840 hectares), representing an average of 2% population in the 8 Districts.

Investments under Output 3 will improve the integrity of soils, supporting water infiltration to the aquifers, accumulation of nutrients and moisture in the soils and preventing soil erosion which, coupled with heavy rainfall events, leads to flooding and loss of physical and natural assets.

ARA3 - Infrastructure and built environment

Investments under Output 2 and Output 3 will improve the resilience of the land and livelihoods of households to the impacts of extreme weather events on infrastructure and the built environment within the eight districts and 120 beneficiary communities. The planned investments in water storage tanks and small-scale water infrastructure under Output 2 in the 120 beneficiary communities will be designed in order to withstand extreme weather events. The restoration activities within the landscape will also serve to mitigate against loss to infrastructure and the built environment.

ARA4 – Increased resilience of ecosystems and ecosystem services

The site-level restoration efforts implemented under Output 3 — including agroforestry, small-scale communal fodder banks, riverbank restoration, flood risk reduction and fire management — are expected to improve the resilience of 14,000 ha of vulnerable ecosystems and ecosystem services. A further 120 beneficiary communities, comprising a total 120,000 people (assuming an average size of 1,000 people per community), will benefit from land restoration and agro-forestry on communal land.

The Economic modelling carried out for the economic and financial assessment estimated that restoration of 14,000 ha of degraded land in strategic locations in four Districts would reduce flood risk by 46.7%. The budget allows for restoration of 14,000 ha across the 8 Districts (Output 3). Impacts against this indicator would be verifiable in longer

time periods than the project grant. The project will reduce the risk of floods, benefiting populations outside the immediate target area. Restoration and flood risk reduction will enhance ecosystem services such as soil fertility and water regulation that are crucial for livelihoods in adjacent communities, resulting in broader resilience to climate change impacts.

GCF Outcome-level enabling environment indicators

The proposed project will be evaluated against several relevant GCF impact assessment factors, as specified in the GCF Integrated Results Management Framework. It is anticipated that the project will achieve two Outcome-level indicators, described below⁴³.

- Core Indicator 5: Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner; and
- Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards.

Degree to which GCF investments contribute to strengthening institutional and regulatory frameworks for low emission climate-resilient development pathways in a country-driven manner.

The project will build capacity and awareness of District planning officers to scale up the project approach. Combined with associated community training, extension services and awareness-raising, this will increase opportunities for knowledge and technology exchange between communities and thereby promote autonomous upscaling of these interventions.

At the field level, strengthened institutional development will be achieved by providing REMCs, DEMCs, as well as Regional EPA and Department of Agriculture offices with equipment required for effective and efficient delivery and monitoring of climate change adaptation and agricultural extension services. Furthermore, the project will train 200 extension officers (30% of which are women) in the target districts on climate-resilient agricultural processes, EbA and alternative climate-resilient livelihoods. Knowledge will be extended to smallholder farmers in target communities — and other communities not targeted by the project. In addition, , the utilization of grants for purposes that include both capacity building and TA aimed at adaptation mainstreaming on the one hand, and a country-based, progressively institutionalized local adaptation financing mechanism, create a holistic and integrated approach to tackle the issue of climate finance deployment at the sub-national level and boosting of resilience building efforts at the community level.

A monitoring programme will be designed to monitor project progress towards its targets and evaluate the socio-economic impacts of the various climate change adaptation interventions being implemented. The monitoring programme will be designed in collaboration with relevant national research institutions, such as University for Development Studies, Kwame Nkrumah University of Science and Technology, Forest Research Institute of Ghana Soil Research Institute and Ghana Meteorological Agency. The monitoring programme will build an evidence base for the further implementation of climate change adaptation interventions in northern Ghana.

Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards.

An awareness-raising campaign will be implemented in both beneficiary and non-beneficiary communities in northern Ghana. The campaign will increase awareness about appropriate climate change adaptation interventions and will promote the upscaling of these interventions outside of the project target sites. The awareness-raising campaign will be developed by a suitably qualified national organisation and will utilise locally-appropriate communication and knowledge dissemination mechanisms, such as video documentaries and phone-in radio shows. All of the awareness-raising activities included in the campaign will be appropriate to local circumstances, taking into account literacy levels and access to different types of media. Various knowledge products detailing best practice and lessons learned through the project will be developed. These knowledge products will include *inter alia*: i) policy briefs; ii) climate change adaptation intervention implementation guidelines; iii) training manuals; and iv) reports. These knowledge products will then be disseminated to relevant stakeholders, including through the annual knowledge-sharing conferences (Activity 5.2) and the Ghana Climate Change Data Hub.

⁴³ A detailed description of the project activities that will be implemented under each output to achieve the GCF project outcomes is presented in Section E.6 of the Funding Proposal.

Climate data and early warnings against drought and floods will be provided to the crop-farming household either via new mobile applications or through local community groups, enabling a proactive response to drought and flood management.

At least three national organisations will be empowered to use new data products and the decision-support system to strengthen adaptive capacity to climate risks.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

The proposed project will catalyse a paradigm shift towards climate-resilient agriculture and natural resource-based livelihoods in northern Ghana. This will be achieved by facilitating the widespread adoption by smallholder farmers of climate-resilient agricultural practices, EbA approaches and alternative livelihoods. Details on how the project will contribute to GCF's paradigm shift objective are presented below.

Potential for scaling up and replication

Decision-support services will be introduced under **Output 1** to empower farmers and planning officials to respond effectively to climate change-induced flood and drought hazards in a coordinated and strategic manner. Investments in weather forecast capability that will serve the Northern Sector will complement investments in forecasting capability in the South of the country as well as establishing a national action plan for drought and flood hazard management. Finally, a national framework for disseminating climate information to communities will be implemented to promote national and grassroots knowledge-sharing. The plan will include a national coordination mechanism to promote collaboration between national and regional stakeholders. A national action plan for coordinating drought and flood hazard management in the agricultural sector will be delivered to ensure that EbA interventions can be replicated beyond the targeted areas.

Regarding **Outputs 2 and 3**, Within the eight project districts, there is an estimated 240,000 ha of farmland on which climate-resilient agriculture and livestock management can be practiced (representing 12–66% of the district area)⁴⁴. This represents a potential 10-fold increase in the potential impact from this project. Regarding protection from fluvial floods, there are 8,700 ha of buffer zone protection concerning riverine forest protection and 9,200 ha of river buffer protection. A monitoring and evaluation strategy will be implemented under **Output 5** to enable best practices and lessons learned during EbA implementation to be documented and disseminated, further enabling future replication. The model can be replicated in 43 districts in the Northern Savannah zone for drought and flood management.

The Economic modelling carried out for the economic and financial assessment estimated that restoration of 14,000 ha of degraded land in strategic locations in four Districts would reduce flood risk by 46.7% in the 4 Districts. The budget allows for restoration of 14,000 ha across the 8 Districts (Output 3).

The implementation of climate-resilient agricultural practices (**Output 3**) is expected to generate consistently higher incomes for smallholder farmers, allowing them to finance adaptation interventions once the project has finished. Farmer-based Organisations and their financial management capacity, and by linking to value chains and the Ghanaian financial sector for loans and insurance. A monitoring and evaluation strategy will be implemented under **Output 4** to enable best practices and lessons learned during EbA implementation to be documented and disseminated, further enabling future replication.

The knowledge generated will be used as a resource to leverage additional funds from public and private sector investors to maintain and replicate project interventions after the project's funding period, while further strengthening cross-sectoral collaboration (**Output 5**).

Potential for knowledge sharing and learning

A monitoring and economic valuation programme will be designed to monitor project progress towards its targets and evaluate the socio-economic impacts of the various climate change adaptation interventions being implemented. The monitoring programme will be designed in collaboration with relevant national research institutions, such as University for Development Studies, Kwame Nkrumah University of Science and Technology, Forest Research Institute of Ghana Soil Research Institute and Ghana Meteorological Agency. A climate change intervention monitor will be placed in each targeted community to collect relevant information stipulated in the monitoring programme. Monitoring and evaluation will focus primarily on *inter alia*: i) the achievement of project targets (see Section E); ii) socio-economic benefits of project interventions; iii) changes in crop productivity under different climate-resilient agriculture and EbA scenarios; iv) socio-economic benefits and uptake of alternative climate-resilient livelihoods; and v) environmental co-benefits of

⁴⁴ Tables 5 and 9 from Section xx of the Feasibility assessment.

project interventions. The monitoring programme will build an evidence base for the further implementation of climate change adaptation interventions in northern Ghana.

Based on the results of the monitoring programme, various knowledge products detailing best practice and lessons learned through the project will be developed. These knowledge products will include *inter alia*: i) policy briefs; ii) climate change adaptation intervention implementation guidelines; iii) training manuals; and iv) reports. These knowledge products will then be disseminated to relevant stakeholders, including through the knowledge-sharing and awareness-raising events (Activity 5.2) and the Ghana Climate Change Data Hub.

Knowledge-sharing and awareness-raising events with participants from: i) district-level government staff⁴⁵; ii) regional-level government staff; iii) national-level government staff; iv) climate change adaptation project and programme managers; v) representatives of academic and research institutions; vi) NGO staff; vii) representatives of multilateral development agencies; and viii) project beneficiaries. These events will promote information sharing and encourage coordination between current and future climate change adaptation projects and programmes.

Annual knowledge-sharing conferences will be convened to promote information sharing and encourage coordination between current and future climate change adaptation projects and programmes in Ghana. The awareness-raising campaign will be developed by a suitably qualified national organisation and will utilise locally appropriate communication and knowledge dissemination mechanisms, such as video documentaries and phone-in radio shows. All of the awareness-raising activities included in the campaign will be appropriate to local circumstances, taking into account literacy levels and access to different types of media.

Knowledge-sharing events will also be organised by district staff and extension officers at each beneficiary community. At each knowledge-sharing event, interested community members from five nearby communities (approximately 100 people), along with extension officers from neighbouring districts, will be brought to a beneficiary community to view climate change adaptation demonstration sites and learn from beneficiary community members. Event participants will receive basic theoretical and practical training on climate change adaptation. These events will promote the autonomous upscaling of climate change interventions promoted by the project in non-beneficiary communities.

Training on climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods will be provided to District officers, extension officers, local communities.

An improved climate information and early warning services will be established, focusing on the institutional framework and technological delivery system for early warning forecasts. A national coordination mechanism will be established to promote collaboration between national and regional stakeholders. Advisories developed based on early warning forecasts and climate data generated by the delivery system will complement the M&E system. The project will develop useful climate advisories in northern Ghana by facilitating the integration of local knowledge and the experience of end-users into the production and dissemination processes.

By promoting the upscaling and replication of project interventions outside of beneficiary communities, the project will facilitate autonomous and widespread and sustained behavioural transformation in smallholder farming communities in northern Ghana.

Contribution to the creation of an enabling environment including regulatory framework and policies

Climate data and early warnings against droughts and floods will be provided to the crop-farming household either via new mobile applications or through local community groups, enabling a proactive response to drought and flood management. The focus will be on enabling local institutions and farmers to shift from the current paradigm of reactive planning (disaster response) to proactive planning practices with the help of available early warning information and decision support tools for climate hazards, droughts and floods. A national framework for disseminating climate information to communities will be implemented. A national action plan for coordinating drought and flood hazard management in the agricultural sector will be delivered to ensure that EbA interventions can be replicated beyond the targeted areas. A national coordination mechanism will be established to promote collaboration between national and regional stakeholders.

A monitoring and evaluation strategy will be implemented under **Output 3** to enable best practices and lessons learned during EbA implementation to be documented and disseminated, further enabling future replication. The project will

⁴⁵ E.g. extension officers, representatives from District Assemblies and DEMCs.

convene knowledge-sharing and awareness-raising events under Outputs 4 and 5, targeting representatives from the government, civil society and beneficiary communities. Technical capacity building will be provided to extension officers and local communities under Outputs 1 and 3, respectively, to enable recipients to gain practical knowledge of climate change adaptation practices. An economic valuation component to measure ecosystem services will support the strengthening of the enabling environment.

D.3. Sustainable development (max. 500 words, approximately 1 page)

The proposed project will use GCF funding to contribute towards the achievement of 13 of the 17 UN Sustainable Development Goals (SDGs)⁴⁶ in northern Ghana, including: 1) No Poverty; 2) Zero Hunger; 3) Good Health and Well-being; 5) Gender Equality; 6) Clean Water and Sanitation; 8) Decent Work and Economic Growth; 9) Industry, Innovation and Infrastructure; 10) Reduced Inequalities; 11) Sustainable Cities and Communities; 12) Responsible Consumption and Production; 13) Climate Action; 15) Life on Land; and 17) Partnerships for the Goals. The sustainability of contributions to the SDGs will be ensured through a multifaceted exit strategy, as described in Section B.6. Furthermore, the project will advance the work of the UN Country Team (UNCT) in Ghana by contributing to three of the results areas, namely: i) Inclusive, Resilient and Sustainable Economic Growth and Transformation, ii) Accessible, Equitable, Inclusive, Sustainable Quality Basic Social Services and iii) Durable Peace and Security in Ghana and the Subregion (UNSDP, 2023-2025). The UNSDP — which is a strategic partnership agreement between the UN and the GoG — reflects Ghana's national goals and its commitments to global development initiatives and sets out the UN system's collective contributions to help the Government and other stakeholders achieve these goals.

In addition, the following co-benefits are expected from project interventions:

- Reforestation through an EbA approach will be achieved through the planting of climate-resilient indigenous tree species and natural regeneration⁴⁷ in previously degraded areas. The reforested areas will provide canopy cover, which will reduce soil erosion and capture runoff. Forest restoration through EbA will yield considerable co-benefits for local communities such as: i) providing NTFPs; ii) improving the dry-season flow of rivers; iii) increasing the recharging of aquifers; and iv) reducing community exposure to climate-related environmental hazards. These co-benefits will enhance the resilience of local communities to climate change while also diversifying local livelihoods.
- The project activities are expected to sequester approximately 1.2 million tCO₂e over the lifespan of the project.
- The proposed project will benefit northern Ghana's most vulnerable people and least developed regions. This will be particularly important in coming decades as the GoG works to attain high-income status, as well as to being classified as a newly industrialised country, by reversing the development divergence between the northern and southern regions of the country. The agricultural production of ~142,000 households in northern Ghana will be secured against the adverse effects of climate change, which will lead to increased income generation and growth of the agro-based rural economy. Even at suboptimal levels of production, the agricultural sector accounted for ~20% of Ghana's GDP in 2017⁴⁸. Increases in agricultural productivity across a large proportion of northern Ghanaian farms as a result of climate-resilient agriculture and ecosystem-based adaptation interventions have the potential to increase the country's GDP considerably and contribute to economic development. In addition, by increasing the resilience of post-harvest storage and processing of agricultural products, there will be a: i) decrease in wastage and loss of agricultural produce; ii) diversification in products that may be sold, which will include products that may be sold nationally and internationally; and iii) decrease in food shortages, as produce can be stored for use during the dry season.

At least 50% of project beneficiaries are women. EbA interventions will be focussed on the empowerment of women, which is essential to increase their climate-resilience — as well as that of their communities. Shifts in gender roles and increases in women's contributions to household incomes have been shown to lead to greater respect for and recognition of women in Ghana⁴⁹. Women will be targeted particularly for the implementation of alternative climate-resilient livelihoods — which will include value-addition activities such as shea butter production, small ruminant rearing, beekeeping and soap making. Lastly, women and the youth will be considered and included in the dissemination of early warning climate information.

D.4. Needs of recipient (max. 500 words, approximately 1 page)

⁴⁶ <http://www.un.org/sustainabledevelopment/sustainable-development-goals/>

⁴⁷ Natural regeneration will be accomplished by reducing slash-and-burn and indigenous tree harvesting activities.

⁴⁸ <https://data.worldbank.org/indicator/NV.AGR.TOTL.ZS?locations=GH>

⁴⁹ Webb J. – 2015 – Gender dynamics in a changing climate: how gender and adaptive capacity affect resilience. *Care Climate Change*.

The GoG acknowledges the urgency of adapting to climate change in northern Ghana. The Climate Change Vulnerability (CCV) index⁵⁰ calculated in the Feasibility Study⁵¹ confirms the GoGs concerns and indicates that the Upper West Region (CCV=0.61) is the most vulnerable in Ghana, followed by the North East (CCV=0.26) and Upper East (CCV=0.26) Regions. Northern Ghana is particularly at risk because of the substantial development gap between the north and the south of the country. This has isolated northern Ghana from the socioeconomic development that has been prevalent in the south. As a result, there has been financial instability and a rise in poverty in the north⁵². Smallholder farmers in northern Ghana — who rely heavily on rainfed agriculture and natural ecosystems for their livelihoods — are particularly vulnerable to the effects of climate change.

The GOG is currently unable to fund adaptation projects that can significantly increase the climate resilience of the agricultural sector in northern Ghana without external assistance. This is because of the government's limited fiscal capacity caused by, among others, a recent slow-down in economic growth, high inflation, currency depreciation and high domestic financing costs⁵³. Indeed, the GOG has indicated that it currently has the fiscal capacity to provide ~6% of the funds required to implement the country's INDCs⁵⁴. Debt-to-GDP ratio was 64% in 2019. The lack of public financing for climate change adaptation has been exacerbated by the limited availability of alternative sources of funding from local capital markets. Private financial institutions have been unwilling to provide loan finance to individual smallholder farmers in northern Ghana because of their general lack of collateral and high default risk⁵⁵. Instead, the financial institutions have exclusively adopted joint liability lending structures restricted to servicing only registered FBOs⁵⁶. Consequently, smallholder farmers in northern Ghana — many of whom are not organised into registered FBOs — have limited access to local capital markets from which they can obtain loans to finance adaptation interventions. The proposed project will contribute to the financing of climate change adaptation in northern Ghana by: i) providing funding for the implementation of climate-resilient agriculture and EbA interventions in target districts; and ii) catalysing future private sector funding by organising smallholder farmers into registered FBOs that can access loans from LFIs to finance adaptation interventions, and by increasing their access to international blended finance.

In addition to financial support, the GoG requires strengthening and capacity building of regional and district institutions to enable them to manage climate change adaptation projects and integrate climate change adaptation into development planning. Presently, most district-level institutions in Ghana have limited capacity to manage and coordinate climate change adaptation projects. There is also limited integration of climate change adaptation into medium-term development plans in most regions and districts of Ghana. The proposed project will strengthen the technical capacity of regional and district institutions in northern Ghana to implement and monitor climate change adaptation projects by: i) training DEMC members in target districts on managing and coordinating climate change adaptation projects; and ii) facilitating the integration of climate change adaptation into regional and district medium-term development plans.

D.5. Country ownership (max. 500 words, approximately 1 page)

The Government of Ghana (GoG) recognises the current and future threats of climate change and has prioritised adaptation in the country's national development strategies. In this regard, the GoG has adopted a National Climate Change Adaptation Strategy (NCCAS, 2010-2020)⁵⁷ that is designed to facilitate the protection of Ghana's current and future development from climate change impacts by strengthening its adaptive capacity and building the resilience of society and ecosystems. The NCCAS outlines Adaptation objectives in nine development sectors, including: i) livelihoods; ii) energy; iii) agriculture; iv) health; v) early warning; vi) fisheries management; vii) land use; and viii) water. The implementation of the NCCAS will form part of the decentralised planning and implementation system. Ministries, Departments and Agencies (MDAs) at the national level are responsible for policy, planning, monitoring and evaluation of development programmes, with MESTI being the supervisory agency. It aims to enhance the adaptive capacity and resilience to climate change in Ghana through: i) improving society's awareness of and preparedness for future climate

⁵⁰ On a relative scale of -1 (low vulnerability) to 1 (high vulnerability).

⁵¹ Please see Annex 2 'Feasibility Study' for details on the calculation of the Climate Change Vulnerability index.

⁵² Ashaley-Nikoi J – 2013 – Bridging the Ghanaian Development Divide – Geographical Dual Economy of Ghana. University of Ghana Legon, Faculty of Social Sciences. Available at: <http://www.egn.org.gh/categories/developmental-economics/51-bridging-the-ghanaian-development-divide>.

⁵³ Please see Section B.5 for a rationale for GCF involvement.

⁵⁴ Republic of Ghana – 2015 – Ghana's intended nationally determined contribution (INDC) and accompanying explanatory note.

⁵⁵ Antwi-Agyei P. 2012. Vulnerability and adaptation of Ghana's food production systems and rural livelihoods to climate variability. DPhil, University of Leeds, Leeds.

⁵⁶ Gallenstein RA, Mishra K, Sam AG, Miranda MJ. 2019. Willingness to pay for insured loans in northern Ghana. *Journal of Agricultural Economics* 70: 640-662.

⁵⁷ UNEP and UNDP – 2012 – National Climate Change Adaptation Strategy. CC DARE

change; ii) mainstreaming of climate change into national development to reduce climate risks; iii) increasing the robustness of infrastructure development and long-term investments; iv) increasing the flexibility and resilience of vulnerable ecological and social systems to enhance their adaptive capacity; and v) fostering competitiveness and promoting technological innovation. The proposed project contributes directly to the NCCAS's objectives of: i) improving awareness and preparedness to future climate change; ii) mainstreaming climate change adaptation into development planning; and iii) increasing the resilience of communities and ecosystems to climate change.

In addition to the NCCAS, there are several policies and plans that aim to increase climate change adaptation and resilience in Ghana, including the: i) National Climate Change Policy (NCCP, 2015–2020)⁵⁸; ii) Nationally Determined Contributions (NDC, 2020-2030)⁵⁹; iii) National Climate-resilient Agriculture and Food Security Action Plan (NCAFSAP, 2016-2020); and iv) National Adaptation Planning Readiness Proposal (NAP Readiness Proposal, 2020–2024). The NCCP (2015–2020) has five main policy sectors which include: i) Agriculture and Food Security; ii) Disaster Preparedness and Response; iii) Natural Resource Management; iv) Equitable Social Development; and iv) Energy, Industrial and Infrastructural Development. These policy areas have been subdivided into ten programme areas that address the critical issues of climate change in Ghana. The ten programme areas will include *inter alia*:

- developing climate-resilient agriculture and food security systems;
- building climate-resilient infrastructure;
- increasing the resilience of vulnerable communities to climate-related risks;
- improving management and resilience of ecosystems;
- minimising impacts of climate change on access to water; and
- addressing gender issues in climate change.

The proposed project contributes to four of the NCCP's ten programme areas: i) developing climate-resilient agriculture and food security systems; ii) increasing resilience of vulnerable communities to climate-related risks; iii) minimising impacts of climate change on access to water and sanitation; and iv) addressing gender issues in climate change; addressing climate change and migration.

The 2018 NDC agriculture chapter is centred around three main agricultural priorities: conservation agriculture, climate smart technologies to increase fisheries and livestock productivity; post-harvest management and food processing. Complementary actions on water conservation and irrigation development, agroforestry are highlighted as needing government attention. The project aligns with four of the seven priority adaptation actions in the NDC, specifically: i) agriculture resilience building in climate-vulnerable landscapes; ii) value addition-based utilisation of forest resources; iii) integrated water resources management; and iv) increasing the resilience of women and vulnerable people. Thirdly, the proposed project contributes to six of the eight action areas in the NCAFSAP, viz.: i) institutional capacity building for research and development; ii) development and promotion of climate-resilient cropping systems; iii) adaptation of livestock production systems; iv) water conservation and irrigation systems; v) post-harvest management; and vi) risk transfer and alternative livelihoods. Lastly, the project will complement the following three priorities in the NAP Readiness Proposal: i) training of government officials on climate risk analysis; ii) development of adaptation plans; and ii) mainstreaming of climate change adaptation into medium-term development plans.

The proposed project is fully country-driven as it was prioritised by the National Designated Authority (NDA) in June 2016 using a transparent prioritisation tool developed under the GCF Readiness Programme. It will help to achieve adaptation priorities in the agricultural sector as well as other wider objectives regarding disaster risk reduction, water resource management and ecosystem management in line with various policies and strategies in line with the provisions of the Ghana Strategic Investment Framework for Sustainable Land Management (GSIF 2011-2025). GSFI objectives are to: i) mainstream and upscale sustainable land-management decisions; ii) secure ecosystem services; and iii) improve rural livelihoods in Ghana. Activities under the auspices of GSIF will be designed to reverse land degradation to sustain high agricultural production and ensure food security, whilst at the same time maintaining ecosystem integrity.

⁵⁸ Ministry of Environment, Science and Technology (MEST), Republic of Ghana. 2014. Ghana National Climate Change Policy. Available at: <https://s3.amazonaws.com/ndpc-static/CACHES/NEWS/2015/07/22/Ghana+Climate+Change+Policy.pdf>.

⁵⁹ Government of Ghana – 2015 – Ghana's intended nationally determined contribution (INDC) and accompanying explanatory note. UNFCCC. Available at: http://www4.unfccc.int/ndcregistry/PublishedDocuments/Ghana%20First/GH_INDC_2392015.pdf.

In addition, extensive consultations with national, regional, district and community level stakeholders (including local government, private sector and civil society representatives, women and other vulnerable groups) were conducted in-country by the GoG during the development of the proposal (see Annex 7h). It is important to note that while these consultations were extensive, there were challenges in identifying and engaging with Indigenous Peoples present on the project landscape. This gap has been identified and will be addressed during the inception period and through the operationalization of an Indigenous Peoples Planning Framework (Annex 6C). The Environmental Protection Agency (EPA) of the Ministry of Environment, Science, Technology and Innovation (MESTI) will serve as the Executing Entity (EE) of the project in close collaboration with the Directorate of Crop Services of the Ministry of Food and Agriculture (MoFA), the Ghana Meteorological Agency (GMet) and the Ministry of Local Government, Chieftaincy and Religious Affairs (MLGCRA). The EPA has a proven track record as an EE in Ghana, having been instrumental in the implementation of large-scale environmental programmes including, *among others*, the GEMP and SLWMP (see Section B.4). UNEP — which has extensive experience in the design and implementation of initiatives that promote EbA — was nominated by the NDA in December 2016 to be the accredited entity for the project.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

The inclusion of GCF funds in the proposed project will overcome the financial barriers to building the climate resilience of vulnerable smallholder farming communities in northern Ghana. As described in Section B.5, the GoG lacks the financial capacity to initiate climate change adaptation projects without external financial support. Furthermore, the focus of the project on the most vulnerable communities in Ghana limits the potential for private investment into climate change adaptation initiatives. However, with GCF support and using the cost-effective approach of EbA, the GoG will be able to increase the climate-resilience of the agro-based rural economy in northern Ghana effectively. There is a growing body of scientific evidence demonstrating the cost-effectiveness of past initiatives that have included an EbA approach to climate change adaptation^{60,61}.

A financial analysis was carried out (Annex 3a and b) to assess potential financial returns of some of the proposed project interventions. Many of the project activities are centred on institutional and technical capacity building, making them of a public goods nature. These activities will, therefore, not yield a direct return on investment. The financial analysis, therefore, focuses on those activities that involve on-the-ground interventions to improve the climate resilience of agriculture, land management and rural livelihoods. A cost-benefit study was conducted based on cost estimations for the implementation of interventions and the project's predicted outputs (Annex 3b). The cost estimates were based on current market values. A discount rate was then applied to determine the net present value (NPV) of each intervention. The financial analysis demonstrated positive NPVs for proposed project interventions such as: i) maize-soybean crop rotation (NPV = US\$563); ii) Cassia-maize intercropping (NPV = US\$465); iii) mango-maize intercropping (NPV = US\$463); and iv) Shea nut processing (NPV = US\$4-US\$84). An additional analysis was conducted to determine the discounted Benefit to Cost Ratio (BCR) of both NBS and hybrid interventions for the period 2024–2050 (Annex 3a). This assessment revealed that both interventions are economically viable, with an estimated BCR ranging between 2.09 and 3.44 for Lambussie, 2.19 and 3.78 for Lawra, 2.21 and 3.85 for Jirapa and 2.74 and 4.32 for WA West. The Internal Rate of Return (IRR) ranges from 17–38% across all locations (Table 2, elaborated in Annex 3).

Table 2. Results of the cost-benefit analysis, BCR (discounted and undiscounted) and IRR, by location and source.

	SSP1-2.6		SSP3-7.0		SSP5-8.5	
	NBI vs Reference	Hybrid vs Reference	NBI vs Reference	Hybrid vs Reference	NBI vs Reference	Hybrid vs Reference
Lambussie						
BCR	2.09	3.31	2.07	3.17	2.25	3.44
BCR discounted	1.67	2.36	1.55	2.14	1.70	2.34
IRR	23.1 %	28.9 %	17.4 %	22.9 %	19.6 %	24.5 %
Lawra						
BCR	2.24	3.64	2.19	3.48	2.41	3.78
BCR discounted	1.78	2.59	1.63	2.35	1.83	2.58
IRR	25.8 %	32.4 %	18.7 %	25.4 %	21.8 %	27.4 %

⁶⁰ De Groot et al. 2013. Benefits of investing in ecosystem restoration. *Conservation Biology* 27: 1286-1293.

⁶¹ Rao et al. 2013. An economic analysis of ecosystem-based adaptation and engineering options for climate change adaptation in Lami Town, Republic of the Fiji Islands. A technical report by the Secretariat of the Pacific Regional Environment Programme. Apia, Samoa. Available at: http://ian.umces.edu/pdfs/ian_report_392.pdf

Jirapa						
BCR	2.31	3.70	2.21	3.53	2.49	3.85
BCR discounted	1.85	2.62	1.65	2.37	1.92	2.63
IRR	27.8 %	32.7 %	19.2 %	25.5 %	23.9 %	28.5 %
WA West						
BCR	2.76	4.14	2.74	4.02	2.98	4.32
BCR discounted	2.20	2.95	2.05	2.71	2.24	2.93
IRR	35.7 %	37.9 %	26.5 %	30.3 %	28.2 %	31.7 %

The design of proposed project interventions has drawn substantially from the best practices and lessons learned from a range of local and international development projects and studies. Additionally, stakeholder consultations with representatives from regional- and district-level government departments, as well as communities from the target districts, have provided insights on the local appropriateness of international best practices. The best practices and lessons learned include the: i) Foundational Support Initiative which promotes community engagement; ii) use of Regional, District and Community Environmental Management Committees; iii) the use of EbA as a framework to integrate sustainable land management and climate change adaptation interventions; iv) importance of integrating gender considerations into project interventions; and v) promotion of autonomous upscaling of project interventions through knowledge sharing between beneficiary and neighbouring communities. This will include knowledge sharing between pastoral and sedentary communities as possible, as many rural communities increasingly practise mixed livelihood strategies (which often includes a combination of traditional pastoral practices and local sedentary farming practices or petty trading). These knowledge sharing processes will be facilitated through an Indigenous Peoples Plan, given that pastoralists in Ghana are predominantly constituted of Indigenous groups such as the Fulani. These lessons learned and best practices are discussed in detail in Section 10 of Annex 2, Feasibility Study. The project will also draw from the substantial experience of UNEP to maximise the benefits gained from past experiences and best practices. The development of a Gender Action Plan and an Environmental and Social Management Framework — guided by UNEP's internal safeguards — will ensure that interventions are in line with GCF requirements and safeguards.

E. LOGICAL FRAMEWORK

E.1. Project/Programme Focus

- ☐ Reduced emissions (mitigation)
☒ Increased resilience (adaptation)

E.2. GCF Impact level: Paradigm shift potential (max 600 words, approximately 1-2 pages)

Assessment Dimension	Current state (baseline)		Potential target scenario (Description)	How the project/programme will contribute (Description)
	Description	Rating		
Scale	Climate change adaptation interventions in Ghana typically have a localised geographic scope and benefit a small proportion of smallholder farmers. This can be attributed to limited institutional and technical capacity and low access to funding for upscale interventions. The Government of Ghana (GoG) is exposed to substantial institutional capacity challenges, particularly at the regional and district levels, where weak coordination and	Low	In the target scenario, Ghana would have a robust and scalable climate adaptation framework extending to national, regional and local levels. Government departments and agencies would have the technical and institutional capacity to integrate climate considerations into all sectoral policies and programs, thereby reducing reliance on external aid and mobilising domestic resources for resilience initiatives. A coordinated approach among NGOs, CSOs, and CBOs would ensure a broader impact, with streamlined knowledge transfer from international programs to local levels. Enhanced collaboration and communication among stakeholders would facilitate the upscaling of successful interventions, leveraging collective knowledge and	Knowledge management and communication will also be facilitated under Output 1, which will establish a national framework for disseminating climate-related hazard management information to stakeholders at both national and community levels. Opportunities for knowledge and technology exchange between communities and between projects will be strengthened, thereby promote autonomous upscaling of EbA interventions. The drought and flood management plan under Output 1 will serve to inform integration of drought and flood management into district development planning and budgeting processes across the country.

	collaboration between government sectors hinder the effective management of adaptation projects. Additionally, government staff and district extension officers are inadequately trained and lack the technical expertise to implement and monitor on-the-ground interventions. Most adaptation projects in northern Ghana strongly rely on donor funding with minimal local investment, limiting their capacity to expand. High implementation costs and the isolated nature of these projects restrict their scalability. The lack of coordination among stakeholders exacerbates this issue, as national institutions are not equipped to facilitate collaboration and knowledge sharing, preventing the effective upscaling of successful interventions. As a result, the scalability of		experiences. Strategic planning and investment would address financial barriers, ensuring resources are available to expand initiatives beyond their initial scope. This comprehensive, country-driven strategy would enhance the scalability of climate adaptation interventions, promoting transformational change and resilience across Ghana.	Decision-support services will be introduced under Output 1 to empower farmers and planning officials to respond effectively to climate change-induced flood and drought hazards in a coordinated and strategic manner. These efforts will be supported by a national action plan for drought and flood hazard management, The plan will include a national coordination mechanism to promote collaboration between national and regional stakeholders. The Economic modelling carried out for the economic and financial assessment estimated that restoration of 14,000 ha of degraded land in strategic locations in four Districts would reduce flood risk by 46.7% in the 4 Districts. The project budget allows for restoration of 14,000 ha across the 8 Districts (Output 3). Scaling up of the restoration effort to reduce flood risk will be achieved through the knowledge management approach in (Output 5) and blended finance credit lines (Output 4). Climate-resilient livelihoods introduced under Output 2 which will be supported by the creation of blended finance credit lines under Output 4 to facilitate access to finance for adopting and upscaling livelihoods beyond the project lifespan. Moreover, Output 5 includes dedicated knowledge-sharing and awareness-raising campaigns to ensure that best practices and lessons learned throughout
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	climate change adaptation interventions in Ghana is low, inhibiting the spread of adaptation technologies and transformational change across the region.			the proposed project can be developed and disseminated with relevant stakeholders and other adaptation projects across Ghana.
Replicability	Replicating successful interventions in Ghana is hindered by limited institutional and technical capacity for implementing and monitoring climate adaptation interventions, insufficient knowledge transfer and low coordination and communication between projects, government and academic institutions. At the national level, institutions are ill-equipped to facilitate the coordination between adaptation initiatives and encourage collaboration to catalyse the upscaling of interventions. Without	<u>Low</u>	The potential target scenario for replicability in Ghana involves a robust institutional, planning and budgeting framework that facilitates the strategic replication of successful climate adaptation interventions across sectors and regions. Enhanced government capacity would improve coordination, fostering collaboration and knowledge sharing among stakeholders, including government bodies, academic institutions, and local communities. Effective communication mechanisms would ensure that initiatives benefit from collective knowledge and experiences. Institutions would facilitate coordination and upscaling of interventions, supported by strategic planning and investment to overcome financial barriers.	Under Output 1, a dedicated decision-support platform for weather and hydrological early warning services will be developed for national and local agencies. Finally, a national framework for disseminating climate information to communities will be implemented to promote national and grassroots knowledge-sharing. A national action plan for coordinating drought and flood hazard management in the agricultural sector will be delivered to ensure that EbA interventions can be replicated beyond the targeted areas. Within the eight project districts, there is an estimated 240,000 ha of farmland on which climate-resilient agriculture and livestock management can be practiced (representing 12–66% of the district area)⁶². This represents a potential for a 10-fold increase in impact from adopting the project approach across the Districts. Regarding protection from fluvial floods, there are some 9000 ha of buffer zone protection from riverine forests possible,

⁶² Tables 5 and 9 from Section xx of the Feasibility assessment.

	such communication and coordination between stakeholders, climate change adaptation initiatives in Ghana will likely continue to operate in isolation, not benefiting from the collective knowledge of ongoing or completed ones.			implying a four-fold increase in impact potential from the project approach, which will be up-scaled through the District planning process. A monitoring and evaluation strategy will be implemented under Output 5 to enable best practices and lessons learned during EbA implementation to be documented and disseminated, further enabling future replication. Restoration interventions for flood and drought management under Output 3, can be replicated in 43 districts across the Northern Savannah zone. Capacity-building activities conducted under Output 5 will create an enabling environment for the GoG to replicate and maintain project activities after the initial investment period.
Sustainability	Low institutional capacity within the Government of Ghana (GoG), particularly at the regional and district levels, hampers the continuation of successful adaptation interventions after the project. While providing a basic governance structure, the existing Regional and District Environmental	<u>Low</u>	In the envisioned target scenario for sustainability in Ghana, adaptation interventions would evolve into self-sustaining practices beyond the project's initial timelines, built upon a strong institutional, community and financial foundation. The project's alignment with national strategies ensures country ownership and ongoing support. Regional and district-level institutions will be strengthened by capacity building, enabling them to plan and implement climate change adaptation initiatives independently. A grassroots approach	Under Output 1 , a dedicated decision-support platform for weather and hydrological early warning services will be developed for national and local agencies. Finally, a national framework for disseminating climate information to communities will be implemented to promote national and grassroots knowledge-sharing. A national action plan for coordinating drought and flood hazard management in the agricultural sector will be delivered to ensure that

	Management Committees often lack the necessary resources and technical expertise to coordinate and manage adaptation interventions effectively. This challenge is compounded by insufficient training for government staff on integrating climate change adaptation into local development planning and insufficient coordination among different sectors. At the grassroots level, smallholder farming communities in northern Ghana face substantial sustainability challenges because of poverty, which limits their capacity to adopt efficient but costly farming technologies and practices, such as irrigation, mechanised tilling, and using fertilisers and resistant crop varieties. Many farmers rely on low-technology equipment because they cannot		will ensure that interventions are locally appropriate and supported by beneficiary communities, with active participation from women and youth to secure community buy-in. Collaboration with private sector entities and financial institutions will support the development of value chains and concessional financial products, enhancing economic resilience. Finally, knowledge-sharing and monitoring activities, including annual conferences, will facilitate the dissemination of lessons learned and innovative practices, ensuring the ongoing replication and sustainability of project interventions.	EbA interventions can be replicated beyond the targeted areas. Capacity-building activities conducted under Output 5 will create an enabling environment for the GoG to replicate and maintain project activities after the initial investment period. A monitoring and evaluation strategy will be implemented under Output 5 to enable best practices and lessons learned during EbA implementation to be documented and disseminated, further enabling future replication.
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	afford more expensive machinery, and the high costs of agricultural inputs further exacerbate this issue. Additionally, insufficient credit and agricultural insurance access hinders their capacity to finance modern agricultural technologies and recover from extreme weather events. This lack of financial resources and knowledge prevents farmers from increasing agricultural yields and improving food security, ultimately undermining the sustainability of climate-resilient practices.					
E.3. GCF Outcome level: Reduced emissions and increased resilience (IRMF core indicators 1-4, quantitative indicators)						
The proposed project will target the following GCF Results Areas: i) ARA1 Most vulnerable people and communities; ii) ARA2 Health, wellbeing and food and water security; iii) ARA3 Infrastructure and built environment and iv) ARA4 Ecosystems and Ecosystem Services.						
GCF Result Area	IRMF Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final ⁶³	

⁶³ The final target means the target at the end of project/programme implementation period. However, for core indicator 1 (GHG emission reduction), please also provide the target value at the end of the total lifespan period which is defined as the maximum number of years over which the impacts of the investment are expected to be effective.

ARA1 Most vulnerable people and communities	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline and project monitoring (interim and final evaluations) informed by household surveys and reports.	0 people with increased resilience to climate change impacts relating to droughts and floods.	340,000 direct beneficiaries, 1.4 million indirect beneficiaries	691,125 direct beneficiaries (52% women and 48% men), 2.9 million indirect beneficiaries (51% women and 49% men)	- Capacity building, technologies and practices introduced are continued beyond the project lifespan and are effective at delivering tangible adaptation benefits. - Baselines will be validated during project inception.
	<u>Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options</u>	Baseline report and project monitoring (interim and final evaluations) informed by household surveys to confirm uptake and income generation of climate-resilient livelihood options.	0 people benefitting from the adoption of climate-resilient livelihood opportunities.	3, 640 people (men and women) whose households have adopted climate-resilient livelihoods options.	6920 (men and women) farm owners who have adopted climate-resilient livelihoods options.	- Ecosystem-based adaptation leads to increased resilience and enhanced livelihoods of vulnerable people. - Climate-resilient agricultural strategies will improve the resilience of ecosystems and ecosystem services. - Local communities actively adopt livelihood options introduced by the project.⁶⁴ - Livelihood options are successful and generate income. - Baselines will be validated during project inception.
	<u>Supplementary 2.4: Beneficiaries (female/male) covered</u>	Random sampling; project level field surveys comprising interviews and workshops with local communities	0 people benefit from coverage of early warning systems	300,000 people benefit from coverage of early warning systems	691,125 direct beneficiaries (52% women and 48% men), 2.8 million	Following capacity building, farming communities will access, utilize and contribute to the new drought early

⁶⁴ For additional information on the livelihood options please see Section 13.4 of Annex 2: Feasibility Study.

	<u>by new or improved early warning systems</u>				indirect beneficiaries (51% women and 49% men)	warning climate information. • Baselines will be validated during project inception.
<u>ARA2 Health, well-being, food and water security</u>	<u>Core 2: Direct and indirect beneficiaries reached</u>	Baseline and project monitoring (interim and final evaluations) informed by household surveys and reports	0 people with increased resilience to climate change impacts relating to droughts and floods.	340,000 direct beneficiaries, 1.4 million indirect beneficiaries	691,125 direct beneficiaries (52% women and 48% men), 2.9 million indirect beneficiaries (51% women and 49% men)	- Capacity building, technologies and practices introduced are continued beyond the project lifespan and are effective at delivering tangible adaptation benefits. - Baselines will be validated during project inception.
<u>ARA2 Health, well-being, food and water security</u>	<u>Supplementary 2.2: Beneficiaries (female/male) with improved food security</u>	Baseline report and project monitoring (interim and final evaluations) informed by random sampling and project level field surveys comprising interviews and workshops with local communities	0	60,000 people whose households benefit from improved food security (indirect beneficiaries)	120,000 people benefit from improved food security (indirect beneficiaries) ⁶⁵	- The national and local proactive approach to drought management, including local capacity building in vulnerable communities, will increase the food security of crop farming households vulnerable to climate change. - Baselines will be validated during project inception.
<u>ARA3: Infrastructure and built environment</u>	<u>Core 3: Value of physical assets made</u>	Baseline and project monitoring (interim and final evaluations)	0	USD3,430,000 of water	USD6,860,000 of water	Livelihoods and interventions are

⁶⁵ Assuming a population of 1,000 in each of the 120 targeted communities.

	<u>more resilient to the effects of climate change and/or more able to reduce GHG emissions</u>	informed by household surveys and reports. Field surveys and GIS mapping of changes in infrastructure and built environment across project intervention sites		storage assets made more resilient to effects of climate change.	storage assets made more resilient to effects of climate change	successful and lead to the protection and enhancement of economic assets in the project areas
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Supplementary 3.1 Change in expected losses of economic assets due to impact of extreme climate-related disasters</u>	Baseline, mid-term and terminal environmental and socio-economy data gathering surveys and analysis The long duration of the dry season experienced in northern Ghana results in prolonged periods of food shortages. High temperatures in northern Ghana already contribute to reduced water-retention capacity and frequent drying of water bodies — such as wells, dams, and streams — that serve as sources of water for both livestock and household use, including for dry season gardening. Increased temperatures linked to climate change exacerbate the problem through the expansion of ranges of the weeds,	High losses	At least a 20% change in expected losses of economic impacts due to impact of extreme events.	At least a 20% change in expected losses of economic impacts due to impact of extreme events.	

		pests and fungi that cause food spoilage. Conversely, extreme rainfall events often result in severe flooding across many areas of northern Ghana, destroying crops, taking lives and damaging assets and public infrastructures, as well as reducing storage capacity of reservoirs in northern Ghana.				
<u>ARA4 Ecosystems and ecosystem services</u>	<u>Core 4: Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice</u>	Household surveys and reports Field surveys and GIS mapping of changes in land cover across project intervention sites.	0 ha restored, conserved or managed in a climate-resilient manner	4,000 ha of smallholder agroecosystems strengthened in response to climate change through the introduction of climate-resilient agricultural and EbA.⁶⁶ <u>Breakdown:</u> 3,460 ha of agricultural land made more resilient to climate change through the	27,840 ha of smallholder agroecosystems strengthened in response to climate change through the introduction of climate-resilient agricultural interventions and EbA. <u>Breakdown:</u> 13,840 ha of agricultural land made more resilient to climate change	• EbA interventions are successful and lead to the protection and enhancement of ecosystem function.

⁶⁶ Assuming ~115 ha of agricultural land per target community

				adoption of climate-resilient agricultural practices. 3,500 ha of degraded Guinea Savanna restored through EbA interventions	through the adoption of climate-resilient agricultural practices.⁶⁷ 14,000ha of degraded Guinea and Sudan Savanna restored through EbA interventions.⁶⁸	
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E.4. GCF Outcome level: Enabling environment (IRMF core indicators 5-8 as applicable)

The proposed project will target two of the four enabling environment indicators included in the GCF Integrated Results Management Framework (IRMF), namely: i) Core Indicator 5; and ii) Core Indicator 8.

Core Indicator	Baseline context (description)	Rating for current state (baseline)	Target scenario (description)	How the project will contribute	Coverage
Core indicator 6: Degree to which GCF investments contribute to technology deployment, dissemination,	There is limited technical capacity to produce reliable observation and forecasting data for drought risk assessment and early warning	Absence of hydrometeorological data generated from northern Ghana	Early warning forecast skills are improved by investments in meteorological technology, data and analysis.	The project will support GMet in increasing its network of hydrometeorological stations in northern Ghana and its capacity	<u>Three target Regions of northern Ghana</u>

⁶⁷ Estimating an average farm size of ~2 ha⁶⁷, across the 13,840 ha where CSA will be implemented.

⁶⁸ Assuming ~115 ha of natural ecosystems per target community

development or transfer and innovation;	systems in Ghana. This is compounded by the absence of a national drought early warning system.		Evidence of people taking up new technologies and practices on climate resilient agriculture and livelihoods.	to issue timely and detailed early warning advisories and forecasts to target beneficiaries.	
Core indicator 8: Degree to which GCF investments contribute to effective knowledge generation and learning processes, and use of good practices, methodologies and standards	Communities have limited awareness of climate change and appropriate adaptation responses. Smallholder farmers do not have the capacity to apply climate information to their agricultural practices and adopt proactive responses to drought impacts.	<u>low</u>	At least three national organisations are empowered to use new data products and the decision-support system to strengthen adaptive capacity to climate risks. Communities are made aware of climate threats and appropriate responses, and local farmers are using climate information in agricultural decision making. Climate data and early warnings against drought and floods will be provided to the crop-farming household either via new mobile applications or through local community groups, enabling a proactive response to drought and flood management.	The project will convene knowledge-sharing and awareness-raising events under Outputs 4 and 5, targeting representatives from the government, civil society and beneficiary communities. Technical capacity building will be provided to extension officers and local communities under Outputs 1 and 3, respectively, to enable recipients to gain practical knowledge of climate change adaptation practices.	<u>National level (one country)</u>

E.5. Project/programme specific indicators (project outcomes and outputs)						
Project/programme results (outcomes/ outputs)	Project/programme specific Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions / Note
				Mid-term	Final	
Output 1: Improved climate data and early warnings made available to facilitate proactive flood and drought hazard management.	Number of males and females reached by [or total geographic coverage of] climate-related early warning systems and other risk reduction measures established/strengthened	Random sampling; project level field surveys comprising interviews and workshops with local communities across Ghana	No national flood and drought early warning system in place	National flood and drought early warning system implemented and used by National organisations	National flood and drought early warning system implemented and used at all levels from National organisations and local farmers across Ghana	The new DSS with early warning system and other decision-making information will be used (with training support) by national organisations and local crop farmers to strengthen adaptive capacity to climate risks
Output 2: Climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods implemented in beneficiary communities.	Change in the capacity of extension officers (gender-disaggregated) to implement climate-resilient agriculture, EbA and alternative climate-resilient livelihoods.	Training scorecard Interviews with extension officers Attendance registers from training events	Extension officers in target districts have not been trained on climate change and EbA	n/a	200 extension officers (of which at least 30% are women) have increased capacity to implement climate-resilient agriculture, EbA and alternative climate-resilient livelihoods.	At least 30% of extension workers are women.

	Number of Community Climate Action Plans developed and validated by beneficiary communities	Review of CCAPs Interviews with extension officers	Target communities have not developed CCAPs	40 Community Climate Action Plans developed and validated by beneficiary communities with the participation of both men and women.	120 Community Climate Action Plans developed and validated by beneficiary communities with the participation of both men and women.	
Output 3: Restoration of landscape to reduce drought and flood risk	Economic value (USD) of avoid / prevented damage/ loss to physical assets (buildings, infrastructure, agricultural land, etc) as a result of landscape restoration.	Baseline, mid-term and terminal environmental and socio-economy data gathering surveys and analysis	Baseline value of physical assets (buildings, infrastructure, agricultural land, etc) . Delayed onset of the wet season results in a shortening of the growing season for rainfed agriculture. Crops, mostly cultivated in floodplains of the rivers, have not fully matured like before when the water levels in the river rise and the fields flood.	At least a 20% increase in economic value (USD) of assets made more resilient (compared to baseline)	At least a 50% increase in economic value (USD) of assets made more resilient (compared to baseline)	

			This results in harvest losses. Extreme rainfall events often result in severe flooding across many areas of northern Ghana, destroying crops, taking lives and damaging assets and public infrastructures, as well as reducing storage capacity of reservoirs in northern Ghana, causing them to dry-up quicker in the dry season as well as increase the risk of dam collapse, leading to downstream flooding.			
	Hectares of natural resources brought under improved low-emission and/or climate-resilient management practice	Field surveys and GIS mapping of changes in land cover across project intervention sites. Household surveys and reports	0 ha restored, conserved or managed in a climate-resilient manner	3,500 ha of degraded Guinea Savanna restored through EbA interventions	14,000 ha of degraded Guinea and Sudan Savanna restored through EbA interventions.	Assuming approximately 115 ha of natural ecosystems per target community (120 communities total).

Output 4: Increased access of smallholder farmers to financial resources and engagement with the private sector	Number of farmer-based organisations (FBOs) established in beneficiary communities	Registration documents Interviews with extension officers Interviews with FBOs	No FBOs established in target communities	80 farmer-based organisations established in beneficiary communities	240 farmer-based organisations established in beneficiary communities (2 in each beneficiary community)	
	Number of VSLAs established in beneficiary communities	Interviews with VSLAs Interviews with extension officers	0	80 VSLAs established in beneficiary communities	240 VSLAs established in beneficiary communities (2 in each beneficiary community)	Community members are willing to work together in VSLAs.
	Percentage of FBOs established in beneficiary communities that have been able to access agricultural credit.	Interviews with FBOs and LFIs Review of financial records	0	10% of FBOs established in beneficiary communities that have been able to access agricultural credit.	30% of FBOs established in beneficiary communities that have been able to access agricultural credit.	
Output 5: Knowledge and awareness of climate threats to agricultural livelihoods and available adaptation options increased to inform the upscaling of climate change adaptation across northern Ghana.	Number of community-level knowledge and awareness events	Household surveys and reports Interviews with extension officers Review of awareness-raising material	0	80 community-level knowledge and awareness events organised	240 ⁶⁹ community-level knowledge and awareness events organised	Communities outside of the target communities are willing to attend knowledge and awareness events.

⁶⁹ 2 events in each targeted community

Project/programme co-benefit indicators						
Co-benefit 1: Increased and diversified household income	Percentage of households who report diversified sources of income at target sites.	Household surveys across 120 target communities.	0%	10%	50%	Communities are willing to adopt livelihood diversification initiatives. Improved climate-resilient agricultural practices result in increased household income.
E.6. Project/programme activities and deliverables						
Activities	Description	Sub-activities			Deliverables	
Activity 1.1: Implement a new early warning data information and management system to provide access to improved data sources and new datasets on floods and droughts.	A new data and information management system, combined with an enhanced service delivery platform for hydrometeorological services (Activity 2.2), will facilitate the cross-sectoral integration of up-to-date climate information and early warnings to enable proactive as opposed to reactive management. Easier access to improve data sources and new datasets facilitated by the improve early warning system will address the barriers highlighted by stakeholders. This data information system will introduce innovative modelling approaches, based on remote sensing data, and service delivery platforms. Additionally, with improved access to quality climate data information, farmers will be more actively involved in planning processes pertaining to flood and drought management. Proactive measures that farmers will be supported to undertake via this intervention include, inter alia: i) timing crop planting for optimum yields; ii) shifting to drought-resistant crops; and iii) relocation of agricultural efforts to more suitable sites that will minimise crop losses.	1.1.1 Implementing science-based methods for merging the existing Hydromet information with satellite-based information for improved accuracy of the data, and producing new seasons forecast within Ghana as a web-based portal. 1.1.2 Development, validation and implementation of a web-based service delivery and decision-support (DSS) platform for weather and hydrological early warning services to be used by national and local agencies, including water resource allocation models, risk knowledge services, drought and flood hazard monitoring and EWS.			Web-portal portal implemented Mobile-based application implemented One workshop to produce recommendations (1.1.1a) Two working sessions to enable access to satellite-based information (1.1.1b) Four working sessions to implement science-based methods (1.1.1c) One working session to produce a seasonal forecast (1.1.1d) Two working sessions to develop and implement a web-based service (1.1.2a) Two working sessions to implement risk knowledge services (1.1.2c) Two working sessions to test hazard monitoring (1.1.2d)	

			Three working sessions to validate the DSS platform (1.1.2e)
Activity 1.2: Enhancing hydrometeorological and groundwater monitoring observation networks.	The accuracy and coverage of climate data will be improved under this activity by enhancing hydrometeorological, groundwater observation networks and groundwater observation networks in the country, particularly northern Ghana.	1.2.1 Validation of assessment of current gaps and installation of a S-band radar station, solar-powered and hydrometeorological and groundwater monitoring stations to improve network coverage. 1.2.2 Development of a sustainability plan, operation and maintaining plan and institutional framework for the enhanced hydrometeorological and groundwater monitoring network	Installation of 120 groundwater monitoring wells. Installation of 20 solar energy technology systems. Expansion of the meteorological network by installing 1 S-band radar to cover northern Ghana based in Tamale; 10 Synoptic Automatic Weather Stations; 20 agrometeorological stations; 163 rainfall stations; and 100 hydrological equipment Sustainability plan for and hydrometeorological and groundwater monitoring network.
Activity 1.3: Capacitate key technical staff at national, regional and district levels, including GMet, HYDRO and WRC, for drought and flood services delivery.	The technical capacity of officials within GMet, HYDRO and WRC will be improved by delivering specialised training programs, focussed on the agricultural sector. Training will also focus on the use of the Decision Support System (DSS) tools and services.	1.3.1 Production of documentation, technical user guides and training materials for the developed services within the established service DSS platform for future use, including for future staff and internal capacity building activities. 1.3.2 Organization of training workshops on the use of the developed platform with key government organisations at national and local level, with focus on training key technical staff on the use of the DSS services. 1.3.3 Dissemination of outreach products as videos, online guidance, brochures etc. for the implemented services to other potential regional and local users.	Documentation, technical user guides and training materials. Two training workshops. Six feedback sessions on disseminated outreach products.
Activity 1.4: Establishing a robust national framework for disseminating DSS and	National and community-based Early Warning System communication mechanisms will be established to capacitate local communities in the use of the DSS information and flood and	1.4.1 Adapt and supplement existing training materials on climate and drought-related hazard management,	New and revised training materials on climate and drought-related hazard management (1.4.1a)

climate-related hazard management to communities.	drought planning established under Activities 1.1 and 1.2.	including integrating local farmers' traditional knowledge⁷⁰ on drought management and adaptation measures, and training extension officers, and farmers 1.4.2 Training local community members and farmers on the use of the available DSS tools and services, and national-local coordination mechanisms for early warning for droughts and climate-related hazards, including implementation of a mobile-based application supporting the use of the National DSS services. 1.4.3 Collect feedback from government and local community stakeholders on the implementation of the DSS and undertake awareness raising workshops with other vulnerable communities adjacent to project sites.	Three training workshops on climate-related hazard management (1.4.1b) Three training workshops with farmers on climate-resilient farming practices (1.4.1c) Four small workshops on integrating local farmers' traditional knowledge into drought management (1.4.1d) Two training workshops on the new knowledge management platform (1.4.2a) Two training workshops to support the mobile-based crowd sourcing application (1.4.2b) Six training workshops to gather feedback (1.4.3a) Five awareness-raising workshops (1.4.3b)
Activity 1.5: Implement national action plan for coordinating drought and flood hazard management in the agricultural sector.	The existing National Drought Management Plan, delineating roles and responsibilities for key institutions, is set to be put into action. This implementation will activate the national coordination mechanism designed for managing flood and drought-related hazards. The plan provides a structured framework for coordinating efforts across various entities, streamlining responsibilities to enhance the efficiency of drought management. By executing this plan, the nation aims to foster collaboration among key institutions, ensuring a unified and organized approach to addressing the challenges posed by drought.	1.5.1 Review and provide recommendations for strengthening the existing strategic framework for drought- and flood-related hazard management, and develop a national coordination mechanism for responding to drought- and flood related hazards in the agricultural sector, 1.5.2 Produce an action plan for integration of climate information into policy and decision-making processes for drought and flood management, including actions to clearly define responsibilities for long term planning and proactive measures, and initiate implementation of the action plan.	Three stakeholder engagement workshops to produce recommendations (1.5.1a) Recommendations for strengthening the existing strategic framework for drought-related hazard management. Two workshops to develop national coordination mechanism (1.5.1b) National coordination mechanism

⁷⁰ The project seeks to leverage the traditional knowledge of local farmers, rather than Indigenous Peoples' knowledge. Should Indigenous Peoples' Knowledge be identified for use during implementation, any uptake would be managed through the IPP and would ensure both FPIC and appropriate benefit sharing mechanisms are in place.

	The implementation of the National Drought Management Plan signifies a proactive step towards a comprehensive and well-coordinated strategy, emphasizing the importance of synchronized actions in managing and mitigating the impacts of drought-related hazards.	1.5.3 Undertake gender analysis for strengthening project implementation at all levels and mainstreaming of gender considerations in training and rollout activities and provide gender-sensitive training on drought- and flood-hazard management within the agriculture sector for key regional branches.	Three workshops to inform the action plan (1.5.2a) and three workshops to obtain inputs on implementing the action plan (1.5.2b) Action plan for integrating climate information into policy and decision-making processes Five small workshops to support the gender analysis (1.5.3a) Gender analysis Six training workshops (1.5.3b and 1.5.3c)
Activity 2.1: Train extension officers on climate-resilient agricultural practices EbA and alternative climate-resilient livelihoods.	The project will train extension officers within targeted districts on: i) climate-resilient agriculture, EbA and alternative climate-resilient livelihoods ii) development of Community Climate Action Plans (CCAPs); iii) the implementation of a dynamic menu of climate change adaptation to northern Ghana. Training district extension officers to plan and implement climate change adaptation interventions will ensure that the officers are prepared to facilitate the implementation of climate change adaptation interventions in target communities. Training programmes will be designed and conducted by appropriate national departments within the GoG, particularly the EPA, Department of Agriculture and Extension Services Division. The training will build on existing extension service manuals and guidelines and will include both theoretical and practical components. One training workshop per year will be conducted in each of the eight target districts for the duration of the project to ensure that extension	2.1.1: Develop training materials and guidelines for extension officers on climate change adaptation, flood and drought hazard monitoring, CRA and EbA. 2.1.2: Develop a training scorecard to measure the effectiveness of the training delivered to extension officers. 2.1.3: Deliver two training session per year for extension officers in each of the eight target districts.	Training materials and guidelines. Sixteen training workshops (2 per district) per year.

	officers remain up to date with adaptation technologies and that newly recruited officers also benefit from the training.		
Activity 2.2: Train beneficiary communities in northern Ghana on climate-resilient agricultural practices and EbA interventions.	The project will train beneficiary communities in northern Ghana on CRA, EbA and alternative climate-resilient livelihoods. Firstly, sensitisation workshops will be held in the first year that the project provides support to a community. District, Regional and National government staff will facilitate the community sensitisation workshops. Following this, staff from relevant departments of the GoG ⁷¹ and extension officers will train beneficiary communities on climate change and the climate change impact pathways most relevant to them as smallholder farmers. Extension officers will then introduce the menu of climate change adaptation interventions to communities. Targeted communities will then be trained to implement climate change adaptation interventions identified within the menu of interventions. In addition, extension officers will organise farmers into special Common Interest Groups (CIGs) and provide targeted training on specific adaptation technologies from the menu of climate change adaptation interventions. Finally, demonstration sites will be established by extension officers with willing community members to demonstrate various climate change adaptation interventions that can be implemented at the household level.	2.2.1: Organise project sensitisation workshops in each of the targeted communities. 2.2.2: Develop training materials for smallholder farmers on climate change adaptation, CRA and EbA. 2.2.3: Develop a training scorecard to measure the effectiveness of the training delivered to smallholder farmers. 2.2.4: Deliver training to targeted communities on: i) climate change impacts; ii) use of DSS tools and services; and iii) menu of climate change adaptation interventions. 2.2.5: Establish CIGs and deliver targeted training on specific climate change adaptation interventions. 2.2.6: Establish five demonstration sites in each targeted community.	A community sensitisation workshop in each of the 120 targeted communities (i.e. 120 sensitisation workshops). A community training workshop in each of the 120 targeted communities. (i.e. 120 training workshops). A targeted training workshop in each of the 120 targeted communities. (i.e. 120 training workshops) during the second and third year of project support. Five CRA and EbA demonstration sites in each targeted community (i.e. 600 demonstration sites).
Activity 2.3: Develop community climate action plans (CCAPs) in collaboration with beneficiary communities.	Under this activity, each target community will finalise — with the assistance of extension officers — a CCAP that: i) identifies the most serious climate threats experienced or likely to be experienced by the community; ii) recognises current coping strategies utilised by	2.3.1: Organise workshops in each of the targeted communities to develop/finalise a CCAP.	5-day validation workshops for each of the 120 targeted communities. Validated CCAPs for each of the 120 targeted communities.

⁷¹ For example, staff within the EPA, specializing in education.

	the community; iii) selects and justifies potential adaptation interventions from the project's menu of climate change adaptation interventions that will help build climate resilience; and iv) details implementation and budgeting requirements to implement potential adaptation interventions. The CCAPs will be validated during 5-day workshops facilitated by five extension officers. In addition to validation by target communities, adaptation interventions detailed in each CCAP will be reviewed against evaluation criteria⁷² by DEMCs and REMCs and will then be approved by the PMU and Directorate of Crop Services. National staff from EPA and MoFA will attend 1 CCAP development workshop in each district during the first year of the project in which they will guide regional and district extension officers. The development of CCAPs began with the community consultations conducted during the development of this proposal (see Annex 2). This activity will develop and finalise CCAPs for each of the 120 targeted communities.	2.3.2: Review adaptation interventions in each CCAP against evaluation criteria⁷³ by DEMCs, REMCs, PMU and Directorate of Crop Services.	
Activity 2.4: Implement climate-resilient agricultural practices identified in the CCAPs in beneficiary communities.	The project will provide technical and input⁷⁴ support to beneficiary communities to implement the climate change adaptation interventions taken from the menu of interventions and identified as required and locally appropriate in their CCAPs. The interventions will be implemented by the local	2.4.1: Implement climate change adaptation interventions, including EbA and climate-resilient agriculture in target communities. 2.4.2: Provide technical support to communities to implement and maintain climate change adaptation interventions.	13,840⁷⁷ ha of agricultural land made more resilient to climate change.

⁷² Evaluation criteria will include: i) consistency with budget limits; ii) consistency with EbA interventions from the agreed menu of climate change adaptation interventions; iii) clearly established land ownership/tenure rights; iv) appropriate implementation duration; v) the consistency of the CCAP with environmental and social safeguards; vi) a realistic implementation plan and budget; and vii) the signature of the responsible extension officer.

⁷³ Evaluation criteria will include: i) consistency with budget limits; ii) consistency with EbA interventions from the agreed menu of climate change adaptation interventions; iii) clearly established land ownership/tenure rights; iv) appropriate implementation duration; v) the consistency of the CCAP with environmental and social safeguards; vi) a realistic implementation plan and budget; and vii) the signature of the responsible extension officer.

⁷⁴ Permissible inputs will include *inter alia*: i) land preparation; ii) simple farming tools; iii) fertiliser sprayers; iv) tree seedlings; v) fencing materials; vi) water storage tanks; vii) climate-resilient seeds; viii) fertilisers; ix) herbicides; and x) pesticides.

⁷⁷ Assuming ~115 ha of agricultural land per target community

	communities with technical support from district extension officers. The community will not receive funding directly, but instead will be provided with the equipment and/or inputs⁷⁵ – and training through Activity 2.1 – required to implement each intervention. The details of the extent, implementation needs and budgeting requirements to implement each adaptation intervention will be drawn from the relevant CCAP. Each community will be allocated the same fixed amount to implement its priority adaptation interventions⁷⁶ over a period of three years. The proposed project will cover input costs for the implementation of climate change adaptation interventions in each target community. Input costs will cover, among other things: • small-scale agricultural equipment such as spades, wheelbarrows, pick-axes and hoes for improved ploughing and composting • climate-resilient seed varieties • nursery establishment costs • seedlings for agroforestry and riverbank protection • land preparation • labour for sowing, tree planting, weeding, digging compost and Zai pits, contour bunding, ridging, firebreak cutting etc. • fruit trees for intercropping • fencing to protect interventions against livestock • organic fertilisers and insecticides • water tanks • improved post-harvest storage devices • small-scale water infrastructure • fodder banks and communal rangelands.		
Activity 2.5: Develop a monitoring and evaluation	This activity will improve the effectiveness of climate advisories in northern Ghana by	2.5.1: Develop an M&E framework for climate advisories services in northern Ghana.	M&E framework for climate advisories.

⁷⁵ Communities will not receive direct access to funds, but instead the executing entity will be responsible for the procurement and delivery of inputs to the beneficiaries.

⁷⁶ The interventions implemented will be consistent with the criteria described in Output 3.2.

strategy for climate advisory services in northern Ghana to improve the accuracy and appropriateness of advisories for smallholder farmers.	facilitating the integration of local knowledge and the experience of end-users into the production and dissemination processes. This process will be centred on the development of a community-based monitoring and evaluation (M&E) system targeting end-users of climate advisories. The M&E will serve to ground truth climate advisories and assess their uptake by end-users. Community surveys will be held on an annual basis throughout project implementation to gather primary data from smallholder farmers on their perceptions of climate advisories and to consider links between local farmers' traditional knowledge and climate science. Finally, a feedback mechanism will be established within key government institutions — including the Ghana Meteorological Agency (GMet), the Ministry of Food and Agriculture (MoFA), the Ministry of Environment, Science, Technology and Innovation (MESTI), Environmental Protection Agency (EPA), the National Climate Change Committee (NCCC), and the Council for Scientific and Industrial Research (CSIR) — to improve the integration of the M&E data in the production and dissemination of climate advisories.	2.5.2: Conduct annual community surveys with smallholder farmers in the target communities to ground truth climate advisories and identify the needs of the communities. 2.5.3: Assess linkages between indigenous knowledge and climate science to identify entry points for integrating indigenous knowledge into climate advisories. 2.5.4: Establish a feedback mechanism with producers of climate advisories to improve the generation and dissemination of advisories using data collected through the M&E system.	18 community surveys. Feedback mechanism between climate advisories and M&E system.
Activity 3.1 Implement land restoration on communal land in 120 communities as per CCAPs	The project will establish tree nurseries to support reforestation in 120 communities; Undertake community consultations, and secure the FPIC of Indigenous Peoples when involved, on final location for restoration as per CCAPs and agree on modalities for plantings, maintenance and protection of the sites and Undertaken annual restoration plantings and maintenance in agreed areas as per CCAPs.	3.1.1. Establish community tree nurseries to support reforestation in 120 communities 3.1.2 Undertake community consultations on final location for restoration as per CCAPs and agree on modalities for plantings, maintenance and protection of the sites 3.1.3 Undertaken annual restoration plantings and maintenance in agreed areas as per CCAPs	14,000⁷⁸ ha of degraded Guinea and Sudan Savanna agroecological systems restored through EbA interventions and improved land management.

⁷⁸ Assuming ~115 ha of natural ecosystems per target community

Activity 4.1: Establish farmer-based organisations (FBOs) and Village Savings and Loans Associations (VSLAs) that can access credit insurance for farming and non-farming livelihood activities.	The project will establish two FBOs in each of the 120 communities targeted by the project, i.e. a total of 240 FBOs. Eight districts will be targeted by the project, engaging an average of 15 communities per district. The FBOs will meet the minimum requirements for receiving joint liability agricultural loans and insurance identified during a survey of LFIs conducted for the AFAWA programme. The project will provide support to members of the established FBOs to register their organisations with MoFA (Sub-activity 4.1.1). Furthermore, members of the established FBOs will be trained on basic business and financial management techniques, including records and bookkeeping as well as the processes for applying for credit and insurance from LFIs. Extension officers will deliver this training during the first year that each beneficiary community receives support to ensure that the FBOs meet the eligibility criteria set by LFIs for joint liability lending. The project will train established FBOs to set up Village Savings and Loans Associations (VSLAs, Sub-activity 4.1.2). The purpose of establishing VSLAs is twofold. Initially, the VSLAs will provide simple savings and loan facilities to these communities who currently do not have access to formal financial services. Secondly, operating the VSLAs will enable them to maintain up-to-date financial records which will enable the FBOs to meet the eligibility requirements of LFIs for joint liability loans and insurance. The recommended blended finance model (see Activity 4.3) will set up a climate-resilient loan (CRL) program as an exit strategy to finance the continued implementation of EBA/CRA measures after the expiry of the GCF project,	4.1.1: Establish two FBOs in each targeted community (total of 120 communities across 8 districts) through which smallholder farmers can apply for agricultural credit and insurance. Train members of the FBOs on basic business and financial management techniques including records and bookkeeping as well as the process for applying for credit and insurance from LFIs. 4.1.2: Train FBOs on the establishment and operation of VSLA.	240 FBOs established (2 in each beneficiary community). 240 VSLAs established (2 in each targeted community).
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	both by FBOs that benefited from GCF activities as well as new, future ones.		
Activity 4.2: Connect FBOs and local financial institutions to improve access of beneficiary communities to credit and insurance products.	The project will facilitate engagement between FBOs and local financial institutions to improve access of beneficiary communities to credit and insurance products, see discussion under Output 4 on GAIP and GIRSAL, and risk mitigation instruments for agricultural lending. Several LFIs in Ghana, including Agricultural Development Bank and EcoBank ⁷⁹ , run outreach programmes that assist farmers in accessing credit. These programmes: i) help farmers to establish organisations or cooperatives; ii) provide financial management training; and iii) take farmers through the loan application process. However, these programmes tend to focus on central and southern Ghana where agricultural production is higher and there are more commercial farming operations. The project will work with relevant LFIs to extend these services to beneficiary communities in northern Ghana, see discussion under Output 3 on partnering with financial institutions.	4.2.1: Include representatives from LFIs in the training and awareness-raising events organised through the project. 4.2.2: Conduct a road show where representatives from LFIs are taken to beneficiary communities.	Knowledge products for LFIs. Two workshops with LFIs. Three roadshows with representatives from LFIs.
Activity 4.3: Establish a blended finance model to provide credit lines to support climate-resilient agriculture.	Climate-resilient loan (CRL) programs set up by one or more LFIs and funded by concessional IFI credit lines will benefit from four de-risking tools linked to the blended finance model: i) the credit score database compiled with the projects support; ii) existing GIRSAL credit guarantees for agricultural lending; iii) off-take agreements with off-takers; and iv) agricultural insurance provided by the new agricultural insurance fund. To develop the blended finance model, the project will begin by providing training to the Business Advisory Centres (BACs) to enable BACs to coach FBOs and their members on the business and financial aspects of EBA/CRA and on the identification of off-takers and	4.3.1: Provide technical assistance to the staff of the BACs using a training-of-trainers approach. 4.3.2: Provide logistical support to BACs through the provision of motorbikes that will allow BAC officers to provide TA to FBOs on-site. Fuel and maintenance of the motorbikes, as well as the allocation of BAC officer time specifically for TA to FBOs under this project will be provided by the EPA. 4.3.3: Organize annual “climate-resilient agriculture forums” convening FBOs, value chain actors, LFIs and NGOs operating in the agricultural sector in the target regions and relevant government entities. 4.3.4: Provide technical assistance to LFIs on (i) the on the business and financial aspects of EBA/CRA that	64 technical training workshops delivered to BACs (two workshops per BAC for 32 BACs) Three technical training workshops for LFIs 16 motorbikes 12 climate-resilient agriculture forums Credit score database created One insurance study produced

⁷⁹ Based on interviews with representatives from ADB and EcoBank.

	signing of off-take agreements, thereby enhancing FBO bankability for local financial institutions. BACs will be capacitated by the provision of logistical support, including motorbikes, to enable them to deliver training to FBOs. Subsequently, annual 'climate-resilient agriculture forums' will be convened to facilitate the establishment of business relationships between FBOs and value chain actors, will allow FBOs to share progress on their path to climate resilience (with focus on enhanced farm profitability) and LFIs to advertise their CRL products, loan application procedures and requirements.	make EBA/CRA-adopting FBOs more bankable than conventional ones, (ii) the design of CRL products whose terms reflect the specificities of EBA/CRA and (ii) the identification of and introduction to IFIs that could provide concessional credit lines to fund the local FIs' CRL programs. 4.3.5: Create a credit score database on FBOs and their members that are targeted by the project and have never borrowed before. 4.3.6: Produce actuarial data on the FBOs and their members that adopt EBA/CRA under the project, to be used by the new agricultural insurance fund to fine-tune its agricultural insurance policies for EBA/CRA adopting farms.	
Activity 5.1: Generate and disseminate knowledge products capturing best practice and lessons learned to inform the upscaling of climate change adaptation across northern Ghana.	A monitoring programme will be designed to monitor project progress towards its targets and evaluate the socio-economic impacts of the various climate change adaptation interventions being implemented. A climate change intervention monitor will be placed in each targeted community to collect relevant information stipulated in the monitoring programme. Monitoring and evaluation will focus primarily on inter alia: i) the achievement of project targets; ii) socio-economic benefits of project interventions; iii) changes in crop productivity under different climate-resilient agriculture and EbA scenarios; iv) socio-economic benefits and uptake of alternative climate-resilient livelihoods; and v) environmental co-benefits of project interventions. The monitoring programme will build an evidence base for the further implementation of climate change adaptation interventions in northern Ghana. Based on the results of the monitoring programme, various knowledge products detailing best practice and lessons learned through the project will be developed. These knowledge products will	5.1.1: Design monitoring programme to monitor project progress and determine the socio-economic impacts of the various adaptation interventions. 5.1.2: Implement the monitoring programme in association with local research institutions. 5.1.3: Produce annual monitoring reports. 5.1.4: Generate knowledge products on best practice and lessons learned through the monitoring programme. 5.1.5: Disseminate knowledge products, including through the Ghana Climate Change Data Hub	Monitoring programme Knowledge products

	include inter alia: i) policy briefs; ii) climate change adaptation intervention implementation guidelines; iii) training manuals; and iv) reports. These knowledge products will then be disseminated to relevant stakeholders, including through the annual knowledge-sharing conferences (Activity 5.2) and the Ghana Climate Change Data Hub.		
Activity 5.2: Conduct community-level knowledge-sharing and awareness-raising events.	An awareness-raising campaign will be implemented in both beneficiary and non-beneficiary communities in northern Ghana. The campaign will increase awareness about appropriate climate change adaptation interventions and will promote the upscaling of these interventions outside of the project target sites. The awareness-raising campaign will be developed by a suitably qualified national organisation and will utilise locally appropriate communication and knowledge dissemination mechanisms, such as video documentaries and phone-in radio shows. All of the awareness-raising activities included in the campaign will be appropriate to local circumstances, taking into account literacy levels and access to different types of media. Knowledge-sharing events will also be organized by district staff and extension officers at each beneficiary community. At each knowledge-sharing event, interested community members from 5 nearby communities (approximately 100 people), along with extension officers from neighbouring districts, will be brought to a beneficiary community to view climate change adaptation demonstration sites and learn from beneficiary community members. Event participants will receive basic theoretical and practical training on climate change adaptation. These events will promote the autonomous upscaling of climate change	5.2.1: Develop an awareness-raising campaign and communication plan to share knowledge on climate change adaptation in northern Ghana. 5.2.2: Implement an awareness-raising programme utilising locally appropriate communication and knowledge dissemination mechanisms. 5.2.3: Install signboards at project interventions sites detailing the project approach and adaptation interventions. 5.2.4: Host community knowledge-sharing events where people from non-beneficiary communities and extension officers from non-target districts are exposed to the adaptation interventions and approaches implemented by this project. 5.2.5: Produce and distribute video documentaries detailing the project approach, adaptation interventions and community success stories. 5.2.6: Share best practices and lessons learned with relevant stakeholders and other adaptation projects in Ghana. 5.2.7: Organise a knowledge-sharing conference annually to share information and strengthen coordination between climate change adaptation projects.	Awareness-raising campaign. 240 community-level knowledge-sharing events. 120 project signboards. Annual knowledge-sharing conference.

	interventions promoted by the project in non-beneficiary communities. Signboards will be installed at each project intervention site detailing the climate change adaptation interventions being implemented there. This will allow surrounding communities to continue to learn about climate change adaptation interventions being implemented outside of the formal awareness-raising and knowledge-sharing events.		
Activity 5.3 Conduct district awareness and training workshops with District Assemblies to integrate CCAPs and EbA in District Development Plans and District Environment Plans and Budgets	Regional offices, REMCs and DEMCs in target districts will be institutionally and technically strengthened to enable them to deliver and monitor climate change adaptation projects and facilitate the integration of EbA and CCAPs in District development plans and district environment plans and budgets.	5.3.1 Organise annual workshops with eight District Assemblies and DEMC to raise awareness and experiences on CCAP and EbA and promote integration in District development plans and District environment plans and budgets	Awareness raising workshops to promote integration of EbA in district development plans

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

A full time Monitoring Officer will be employed to conduct and coordinate monitoring and evaluation (M&E) activities for the project. The Monitoring Officer will design and operationalise a performance monitoring framework to track the project's progress towards achieving its targets. This will include: i) measuring performance against the project indicators (Sections E.3–5) to evaluate the progress of the project; ii) reporting the project's performance to the PSC and PMU; and iii) providing technical support to the PC. Additionally, the Monitoring Officer will be responsible for overseeing and monitoring the application of gender-disaggregated indicators and ensuring that all gender targets are met. At particular milestones throughout the project — including annual performance reports, interim evaluations and at project conclusion — the project team will undertake evidence-gathering exercises to verify the progress. These internal results verification reports will be further validated by evaluations at the mid-term and end of the project implementation, as described below. The costs for the mid-term and terminal evaluation will be covered by UNEP. During the Terminal Evaluation at the end of the project, the evaluation team will validate a sample of the data collected through the monitoring framework.

UNEP will be responsible for managing a Mid-Term Review (MTR) at the mid-point of the project's duration. The Task Manager will oversee the process of hiring an external consultant to carry out the MTR, which will provide an assessment of project performance at the project's mid-point. This will be a formative exercise and will include analysing whether the project is on track, what problems and challenges the project is encountering, and which corrective actions are required so that the project can achieve its intended outcomes by project completion in the most efficient and sustainable way. The Project Steering Committee will participate in the MTR process and develop a management response to the review's recommendations along with an implementation plan. It is the responsibility of the UNEP Task Manager to monitor whether the agreed recommendations are being implemented during the remainder of the project's operational life. The Independent Evaluation Office of UNEP will provide an assessment of the quality of the Mid-Term Review report.

An independent Terminal Evaluation (TE) will take place once the project has reached operational completion. The Evaluation Office of UNEP will be responsible for the TE, which is a summative evaluation, and will liaise with the UNEP Task Manager and relevant stakeholders throughout the process. An independent

assessment of project performance against standard evaluation criteria (e.g. strategic relevance, effectiveness, efficiency, likelihood of impact and sustainability) will be made based on documentary evidence, stakeholder interviews and, in most cases, a field mission. Each evaluation criterion will be rated using a three-point rating scheme and a weighted average will be determined to provide an overall performance rating for the project as a whole. Where there are any differences in ratings between the evaluation team and the Evaluation Office a final determination will be made by the Evaluation Office when the evaluation report is finalised.

The draft TE report will be sent to project stakeholders during a commenting process managed by the Evaluation Office. Formal comments on the report will be shared by the Evaluation Office in an open and transparent manner. This evaluation report will be publicly disclosed and will be followed by a recommendation compliance process.

The costs for results monitoring and performance evaluation are included in the 'Detailed Budget' sheet of Annex 4 under budget lines 284–285.

The EE will submit semi-annual progress reports and quarterly financial statements to the PMU and the PMU will consolidate the reports and submit them to UNEP as the AE. In turn, UNEP will submit annual performance reports and semi-annual financial reports to GCF. The detailed reporting timelines are as follows:

Under the PCAs, the EE will be required to report to UNEP as follows:

- a. Progress reports: by 30 July for January to June;
- b. Quarterly financial reports by 15 January, 15 April, 15 July, and 15 October;
- c. Annual audited statements by 30 April;
- d. Final report: within 12 months of Programme completion.

UNEP (AE) reports to the GCF:

- a. Annual Performance Reports by 1 March;
- b. Semi-annual Financial Information by 1 March and 30 Sept;
- c. Mid-Term Assessment report: halfway through Programme;
- d. Final APR: within 6 months of Programme completion;
- e. Terminal Evaluation report: within 12 months of Programme completion.

Monitoring will also be undertaken by the AE through at least two Task Manager supervision visits and field missions per year to track implementation progress and challenges and strategically plan the way forward.

F. RISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

AML/CFT and Prohibited Practices compliance were assessed in the UNEP capacity assessment of EPA. UNEP noted that there are no UN sanctions or embargoes listed against EPA or its Management team, and there appears to be no evidence of engagement in prohibited practices by EPA.

Selected Risk Factor 1

Category	Probability	Impact
Governance	Medium	Medium

Description

Secure land tenure is critical for investment in climate change adaptations. If tenure is insecure, farmers face lower expected returns due to the risk of eviction, which undermines long-term planning and sustainability. Ambiguity in land-use rights can lead to disputes between landowners, tenants and customary users, including pastoralists, resulting in tenure insecurity and reduced uptake of adaptation measures.

To address this, the project will engage both landowners and land users, including pastoralist communities, and ensure that traditional and customary land use rights are respected in line with the GCF Indigenous Peoples Policy. These rights, including access to water points and rivers, seasonal access to land grazing corridors, will be identified and safeguarded through meaningful consultations and securement of the Free, Prior and Informed Consent from Indigenous Peoples using appropriate participatory processes guided by the IPPF. Conflict-sensitive planning and local dispute resolution mechanisms will also be applied to reduce risks and ensure inclusive and sustainable implementation.

Mitigation Measure(s)

This risk will be managed through a participatory approach to planning and implementation of proposed project activities. Close engagement with local communities, GoG entities and traditional authorities in project execution will be a priority, while local officials' knowledge and experience in dealing with Ghana's land tenure situation will be invaluable. Engagement with landowners will be a priority so that landowners are made aware of the benefits of the proposed CRA and EbA interventions and the resulting climate-resilience. Furthermore, the menu of climate change adaptation interventions will allow the landowner to decide which activities are implemented on their land. This will further promote the participation and ownership of landowners in the proposed project. Consequently, the autonomous uptake of project interventions by surrounding farmers and communities will be enhanced and sustainability will be strengthened. This mitigation measure is expected to lower the probability of the risk occurring to "Low".

Selected Risk Factor 2

Category	Probability	Impact
Other	Low	High

Description

The occurrence of low community demand to implement or sustain project technologies. If the benefits of interventions are not locally relevant or the long-term benefits are not adequately explained to beneficiaries, then communities may revert to previously used and unsustainable agricultural methods. As a result, environmental degradation will continue, which will lead to further losses of livelihoods and decreased food security.

Mitigation Measure(s)

Attention will be given to locally relevant benefits in the selection of activities to encourage community buy-in. The CCAPs will ensure community participation and allow the selection of the most locally appropriate interventions from the menu of climate change adaptation interventions. Such a participatory approach to planning and implementation will build local awareness on EbA and generate interest in project interventions. The autonomous upscaling of proposed project interventions will be promoted through carefully planned site selection and the facilitation of inter-

community meetings. M&E of project interventions will be implemented, which will allow for the adaptive management of interventions. This will ensure the long-term sustainability of proposed project interventions and promote their viability.

Selected Risk Factor 3

Category	Probability	Impact
Other	Medium	High
Description		
Local communities may revert to unsustainable harvesting and farming practices after the project. In particular, trees planted during forest restoration may be cut down for fuelwood or to make way for new fields.		
Mitigation Measure(s)		
This risk will be mitigated by conducting awareness-raising campaigns in local communities about the specific economic benefits that restored climate-resilient forest ecosystems provide. Additionally, forests will be restored with tree species that offer higher financial returns to local communities than the sale of fuelwood. Fast-growing, climate-resilient species that provide valuable NTFPs will be included in plantings to demonstrate the benefits of reforestation to communities before the end of the project. These mitigation measures will adjust the probability of this risk occurring to “Low”.		

Selected Risk Factor 4

Category	Probability	Impact
Other	Low	High
Description		
The use of pesticides in interventions results in ecosystem damage through the secondary poisoning of non-target bird and mammal species.		
Mitigation Measure(s)		
The proposed project will ensure that the pesticides used by smallholder farmers have minimal impacts on agroecosystems. Furthermore, extension officers will train smallholder farmers on the safe storage, use and disposal of pesticides, as well as informing them about alternative non-toxic pest management techniques. Part Two of Ghana’s Environmental Protection Agency Act (1994) makes provisions for the effective control and management of pesticides. This is aligned with the FAO’s International Code of Conduct ⁸⁰ in terms of handling, storing, applying and disposing of pesticides. The control of pesticide use in Ghana is enforced by EPA, which is supported by the Pesticides Technical Committee. Furthermore, the proposed project will use a proactive integrated pest management system (including <i>inter alia</i> the use of pesticides and post-harvest management) to mitigate the impacts of pests on agricultural production and overall project success.		

Selected Risk Factor 5

Category	Probability	Impact
Technical and operational	Medium	High
Description		
Lack of interest or demand, amongst authorities and potential agricultural and ecosystem goods buyers, to support activities beyond the project period. This includes the risk that the increased supply of farm products and NTFPs will exceed local demand.		
Mitigation Measure(s)		
Assessments of market access for all agricultural products and ecosystem goods that will be produced through project interventions will form part of the screening process for CCAPs under Sub-Activity 2.2.2 will be conducted. These assessments will draw from the initial Market Assessment conducted under the Feasibility Study (Annex 2) to determine the long-term viability of products in terms of supply and demand. The demand for more ecosystem goods and services is likely to be stimulated over time as the supply increases and as communities become more reliant on		

⁸⁰ Available at:

http://www.fao.org/fileadmin/templates/agphome/documents/Pests_Pesticides/Code/CODE_2014Sep_ENG.pdf

them. Additionally, the post-harvest processing and storage of these products will be offered in the menu of climate resilient technologies to smallholder farmers. This will allow for the storage of products over a more extended period – especially when there is a greater supply than demand – and diversification of end-products. These end products can then be sold to various markets – local, national and international – when the demand is high, and farmers can charge higher prices. Furthermore, adequate storage will strengthen local and national food security during droughts and the long dry season. There is high-level support for post-harvest processing and storage in Ghana through the Presidential Special Initiatives (PSI) on Agriculture and Development which will de-risk the agriculture and agribusiness sector through sustainable agriculture financing aimed at rapidly ramping up agro-processing and developing new and stable markets for agricultural products. There is also potential for the small scale storage facilities that will be introduced through the proposed project to link larger regional warehousing proposed under the PSI to further enhance links with local, regional and international markets. These mitigation measures will lower the probability of the risk occurring to “Low”.

Selected Risk Factor 6

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

Limited capacity at decentralised levels to initiate and manage environmental activities; this includes District Assemblies, District Environmental Management Committees (DEMC), Community Climate Action Groups (CCAGs) and farmer-based organisations (FBO). In addition, staff at the local level may not fully understand their responsibilities under the proposed project leading to delays, and the ineffective and inefficient implementation of interventions.

Mitigation Measure(s)

The project will encourage a realistic rolling out of activities with districts, based on the capacity of the EPA and MoFA regional staff and collaborating partners. The project will also build the capacity of regional and district staff, as well as of extension officers. MoFA and GMet will be closely involved in local project activities, ensuring that the project outputs are implemented in existing local communication structures with farmers and farmer associations, through the extension officers and regional/district branches. National level entities will provide the necessary technical backstopping. This mitigation measure will reduce the probability of the risk occurring to “Low”.

Selected Risk Factor 7

Category	Probability	Impact
Technical and operational	Medium	Medium

Description

Ineffective use of public funds. Systemic weaknesses in financial management (FM) capacity and inefficient use of public funds as a result of inadequate oversight and supervisory arrangements on: i) transparency; ii) accountability; and iii) possible noncompliance with guidelines.

Mitigation Measure(s)

The EPA has a functioning Internal Audit Unit to help minimise such a risk. The GoG financial regulations and manuals are adequate for operational control. These manuals document the approval and authorisation hierarchies applicable for processing financial transactions. An external audit will also be conducted, and combined UNEP supervision will be provided throughout the life of the project. In addition, the following FM measures will be implemented: i) improving the understanding of project staff on UNEP FM processes and procedures; ii) the project shall prepare an accounting manual that clearly defines the policies and procedures to ensure consistency in financial operations and reporting; and iii) periodic Value for Money Audits on the beneficiary agencies will be conducted by the Internal Audit of the EPA to ensure that project objectives are achieved. These mitigation measures will decrease the probability of the risk occurring to “Low”.

Selected Risk Factor 8

Category	Probability	Impact
Technical and operational	Low	Medium

Description

Weakness in the procurement management capacity at the EPA, including: i) lack of proficiency for the management of procurement processes; ii) delays in the processing and submitting of procurement documents; iii) limited experience and familiarity with UNEP procurement guidelines and procedures; and iv) lack of adequate procurement staffing at the EPA.

Mitigation Measure(s)

The EPA will draw on their experience gained from implementing similar projects, including GEMP and SLWMP, which will aid in mitigating this risk. Furthermore, the following measures will be put in place to reduce procurement risk: i) the EPA will appoint an experienced project procurement officer to oversee activities and develop in-house capacity throughout project implementation; ii) capacity will be built within the EPA for procurement management in line with national procurement law; iii) regular updates of the procurement plan until project closure, with copies sent to the GCF and UNEP for review; and iv) the EPA will ensure the prompt submission of procurement documents

Selected Risk Factor 9

Category	Probability	Impact
Governance	Low	Medium

Description

Potential conflicts between stakeholders may arise, such as diverging views and understanding of project strategic direction, dissatisfaction with implementation arrangements or dissatisfaction with the approach to project implementation approach in local communities may arise. The conflict probability is generally considered to be low locally, as the proposed interventions include upgrade to centralised data and management infrastructure - i.e., no activity including direct changes to farmers land and surrounding ecosystems.

Mitigation Measure(s)

The close engagement of key stakeholders as implementing partners of the project, and participation in the national project steering committee NPSC forming the strategic direction of the project will also help avert any potential conflicts between stakeholders that may arise from lack of common understanding of the respective needs and challenges of both local stakeholders and national institutional players and project strategic direction. For local level stakeholders specifically, including farmers and farmer associations, targeted training and consultation activities will take place to inform and engage on project activities, and grievance redress mechanism will be established to ensure that any potential conflicts or issues are tackled immediately and addressed in a way that is agreeable to all parties involved.

G. GCF POLICIES AND STANDARDS

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

The project has been designed to take potential environmental and social impacts of project activities into consideration, incorporating mitigating measures to reduce the likelihood and severity of unforeseen adverse effects. All project activities have been evaluated against GCF Environmental and Social Safeguard (ESS) standards using UNEP's Safeguards Risk Identification Form (SRIF)⁸¹ as a screening tool to identify potential negative impacts. The SRIF screens the significance of environmental, social and economic risks against safeguard standards outlined in the UNEP's Environmental and Social Sustainability Framework (ESSF)⁸². The SRIF Screening Decision (Annex 6A) classified the project as moderate risk, with moderate risk factors associated with UNEP's Safeguards Standard 1: Biodiversity, Ecosystems and Sustainable Natural Resource Management and Safeguards Standard 7: Indigenous Peoples.

As the project is classified as moderate risk, an Environmental and Social Management Framework (ESMF) (see Annex 6B) has been developed. This ESMF provides explicit and systematic guidelines to identify environmental and social risks linked to project interventions and outlines the institutional and administrative pathways to address these risks. The ESMF identifies environmental and social risks and impacts that may be caused by the proposed project, including: i) introduction of harmful species; ii) increased use of agricultural chemicals; iii) increased demand for irrigation; iv) habitat conversion; v) excessive fire suppression; and v) the adverse effects of on-farm earthworks. The framework provides mitigatory measures for avoiding, minimising and mitigating the harmful risks and impacts such as: i) screening for invasive species; ii) adopting integrated pest and nutrient management approaches; iii) avoiding large-scale or diesel pump irrigation; iv) preventing the conversion of natural habitats to cropland or plantation; and v) supporting controlled early burning, rather than outright fire suppression.

Additionally, during the project development period, social risks related to ethnic minorities, inclusive engagement processes and potential exacerbation of community conflict were identified. These social risks emerge through potential for conflict between ethnic minorities (including those categorised as Indigenous Peoples per the GCF IP Policy) practicing different and sometimes competitive livelihood strategies, namely sedentary farming and semi-nomadic/transhumant pastoralism. The potential for these risks to emerge are exacerbated by the project's focus on sedentary livelihood strategies and general trend of underrepresentation of pastoral groups within government institutions and institutional processes. To address these risks, an Indigenous Peoples Planning Framework (IPPF) has been developed in accordance with the GCF Indigenous Peoples Policy. The IPPF will be implemented in the first year of project implementation and will guide the identification of Indigenous Peoples, ensure FPIC is secured prior to activities affecting them, and enable the development of site-specific Indigenous Peoples Plans (IPPs) in consultation with the relevant Indigenous Peoples. These plans will ensure that potential adverse impacts are addressed, that Indigenous Peoples are equitably included in project processes, and that their collective rights, land use practices, and their traditional knowledge are respected and safeguarded.

As the Executing Entity, the EPA will oversee the implementation/monitoring of the ESMF and IPP and provide technical guidance and specialist advice on environmental and social issues to all stakeholders⁸³. In addition, all potential delivery organisations — including private contractors — will be vetted by the EPA in terms of their environmental and social performance to ensure compliance with the ESMF. The EPA — which is responsible for ensuring compliance with environmental impact assessment (EIA) procedures under the EPA Act 1994, Act 490 — has the experience and capacity to implement the ESMF.

To guarantee effective project implementation and to address grievances that may arise as a result of project interventions, Grievance Redress Mechanisms will be established in each district, detailing the complaints procedure and the redress mechanisms for eligible⁸⁴ grievances (see Annex 6b). All community members and stakeholders will be informed of the grievance redress mechanism, including how to register a complaint. At the community level, complaints will be presented to Climate Change Action Groups linked to the proposed project. These groups will receive and record the charges, as well as issue an acknowledgement. Mediation bodies appropriate to the nature of the

⁸¹ The SRIF is a screening tool developed under UNEP's Environmental and Social Sustainability Framework (ESSF) to provide a preliminary assessment of the potential safeguard risks presented by a project.

⁸² UNEP 2015. UNEP Environment's Environmental, Social and Economic Sustainability Framework (ESSF). Available at: <https://wedocs.unep.org/rest/bitstreams/14946/retrieve>

⁸³ Funds will be allocated to the EPA for the implementation and monitoring of ESMF.

⁸⁴ Eligible grievances include: i) unequal access to project resources; ii) corruption; iii) issues of gender inequality; iv) lack of delivery of project interventions; v) exclusion of minority groups and iv) unspecified environmental or social grievances related to the project, project staff or associated activities.

grievance — including community elders, Tindanas or chiefs — will be established that will endeavour to reach mutually acceptable solutions within 20 working days. All efforts will be made to address grievances at the community level. Should any complaints arise that cannot be resolved at the first level of the Grievance Redress Mechanism, these will be referred to project staff, through Grievance Redress Committees at the district level. The Grievance Redress Committees will hold meetings with complainants to identify a solution acceptable to all parties involved. The proposed redress actions will be communicated to complainants and if satisfactory, implemented promptly. A formal complaints register will be established by the EPA in which all concerns raised through the grievance mechanism will be registered within 24 hours. The GRM is based on an existing GRM utilised by the EPA for the LoCAL project as is, therefore, well established and operating effectively. In addition to the standard project level GRM, the project will: i) include a parallel SEAH-specific survivor-centred process, which is detailed in the ESAMF; ii) ensure access to the UNEP SRM for any complainants; iii) ensure information is disseminated on how to access the GCF IRM (and notably the Indigenous Peoples Specialist within the Green Climate Fund); and finally iv) develop an Indigenous Peoples-specific GRM through the operationalization of the IPPF to ensure the provision of a culturally appropriate process that respects the traditional authority structures of any potentially affected Indigenous Peoples.

Extensive on-the-ground public consultations and stakeholder engagements were conducted during the design of the proposed project to identify the needs of the beneficiary communities and the potential social risks that the project may encounter⁸⁵. As highlighted in section D.5, it was not possible to undertake comprehensive consultations with Indigenous Peoples present on the landscape who may be impacted by the project. These groups will be mapped and consulted (with FPIC secured where required) through the implementation of the IPPF. These consultations will continue throughout the project lifespan and in accordance with an IPP in any landscape in which Indigenous Peoples are present. In addition, the inclusion of community members in the development and validation of Community Climate Action Plans (CCAPs) will facilitate the identification of community needs. The combination of community consultations, stakeholder engagements and the inclusive development of CCAPs will ensure acceptance of project activities from the beneficiary communities.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

During project preparation, a gender assessment was conducted to understand the differential vulnerability to, and impact of, climate change on women and men in Ghana. From the findings of this assessment, and an analysis of the proposed project solution and interventions, a gender action plan (GAP) was developed. The process to produce these components included: i) a desktop review of available literature; ii) gender equality studies conducted by research institutions and donor agencies; iii) details made available by the Government of Ghana; and iv) information obtained during extensive stakeholder and community consultations in 2017 and 2024.

The assessment notes that gender inequalities in Ghana disproportionately affect the adaptive capacity of women to current and future climate change, as opposed to men. The effects of climate change on households and communities will ultimately be exacerbated if the adaptive capacities of both men and women are not strengthened in parallel. Overlooking either gender will affect production and income within a household, having knock-on effects on families, children and the community as a whole. Although traditional knowledge has established coping mechanisms within communities to the harsh climatic conditions experienced in northern Ghana, it is unlikely to adapt quickly enough to overcome the effects of future climate change.

Women — including divorcees and widows — along with children and the elderly, are among the most vulnerable groups to the effects of climate change. This is largely because women and female-headed households have fewer resources to cope with reductions in food and water security, together with the effects of extreme weather events such as droughts and floods. This is particularly pertinent in the target communities, where people are often reliant on climate-sensitive natural resources for their livelihoods. Several factors contribute to the vulnerability of women to climate change, including, *inter alia*, the effects of migration, water availability, limited access to climate information or early warnings, decreased agricultural yields and reliance on climate-sensitive livelihoods.

Mainstreaming gender into development frameworks and project design — including conceptualisation, implementation, monitoring and evaluation — is therefore important to build the adaptive capacity of households and communities and reduce the social impacts of climate change on communities. To mitigate against biasing against women as project beneficiaries during implementation and subsequently, the gender action plan outlines several

⁸⁵ The potential environmental and social risks that may arise during project implementation, as well as mitigation measure to reduce these risks, are presented in Annex 6: Environmental and Social Assessment and Management Framework. In addition, summaries of all stakeholder consultations conducted during project development, as well as the stakeholder engagement plan, are presented in Annex 7h: Stakeholder Engagement Plan.

interventions to ensure the project does not unintentionally contribute to perpetuating or worsening gender equality and gender-based climate vulnerability in Ghana. The GAP seeks to promote gender equality by including activities which will increase women's representation and awareness, as well as ensuring that training programmes are gender-responsive and -inclusive. It includes clear targets, gender-responsive design features and quantifiable performance indicators to promote women's participation and benefits from the project.

The project interventions will address gender equality through several pathways, including:

- promoting women's inclusion in all project aspects, including training and awareness-raising opportunities;
- incorporating gender considerations into the development of community climate action plans, based on the specific gaps identified within each community;
- integrating women into the institutional structures and involving them in the decision-making process;
- pursuing representative participation in all consultations, workshops and validation events;
- providing women with business and financial training;
- raising awareness on the importance of equal opportunity for all people and the role women play in climate change adaptation across a community;
- sharing lessons learned through project interventions to help create a paradigm shift in the approach to gender equality in climate change adaptation projects; and
- designing and implementing gender-responsive training that considers the different learning methods and training accessibility of men and women.
- working closely with local community leaders, civil society organisations and pastoralist associations to identify and invite pastoralist representatives to participate in stakeholder consultations, workshops, and validation events.
- ensuring diversity within these groups, including equitable participation of women pastoralists, to capture a wide range of experiences and perspectives.
- ensuring that logistical and linguistic barriers to participation are addressed, enabling inclusive and effective engagement with pastoralist communities throughout the project cycle.

Gender-specific indicators defined in the GAP will be used to quantitatively assess the gender-specific benefits of project interventions on an ongoing basis. These indicators will ensure that the monitoring and evaluation process is gender-sensitive and that project interventions are effectively promoting gender equality in climate change adaptation. The results of these assessments will be incorporated into the annual Project Implementation Reports, Mid-term Report and Terminal Evaluation. The full Gender Assessment and project-level Gender Action Plan are included as Annex 8.

G.3. Financial management and procurement (max. 500 words, approximately 1 page)

The financial management and procurement within the project will be guided by UN financial regulations, rules and practices, as well as UNEP's programme manual. The financial rules of UNEP, which follow International Public Sector Accounting Standards (IPSAS), are promulgated under the Financial Regulations and Rules of the UN. Within this context, funding-allocation mechanisms are managed as per UN rules and procedures, including eligibility criteria, proposal evaluation processes, quality assurance and control, project monitoring and supervision. UNEP is audited annually by the UN Board of Auditors and has established dedicated trust funds for Green Climate Fund (GCF) resources.

The funding of the project will be conducted through a distinct grant within the Trust Fund, with the project itself being set up in the UNEP Enterprise Resource Planning (ERP) as a Work Breakdown Structure (WBS), which is itself further broken down into Work Breakdown Structure Elements (WBSEs), organised by output and outcome. The grant of the project will be linked to the WBSEs to fund the activities contributing to the delivery of specific outputs — as per the disbursement schedule detailed in the term sheet. The fee income will be managed independently and separately of the GCF project grant funds, through a dedicated grant. In line with UNEP procedures, the project will appoint a Financial Officer within the PMU who will be responsible for monitoring, reporting on and approving requests for funds on a quarterly basis. Reports to summarise the disbursement and projected demands for project funding will be prepared and submitted to a UNEP Task Manager who will undertake project supervision, in line with reporting standards and methodologies applied in past projects, such as those implemented using GEF modalities. The UN financial regulations and rules require the segregation of duties, and safeguards to ensure compliance with UN financial rules and regulations. All procurement will be undertaken in line with UN procurement regulations, rules and policies. UNEP's modalities for project implementation results in funds being transferred in tranches to the EE once the EPA has satisfied the conditions that are defined under the legal/cooperation instrument (Project Cooperation Agreement; PCA and Internal Cooperation Agreement; ICA) to be signed between UNEP Climate Change Adaptation Unit, Ecosystems Division and the EPA. The PCA will include specific obligations for EPA on financial management and reporting and will require periodic reporting from EPA to follow international financial and auditing standards. The PCA

specifically requires the audit to be undertaken by a recognised firm of certified public accountants or, for governments, by a government auditor. This auditor should state whether the GCF proceeds were covered by the scope of the audit.

As a GoG authority, the EPA follows the government's financial and procurement rules and standards. UNEP has undertaken an assessment of the EPA capacity which will guide the procurement monitoring plan which will be agreed upon between UNEP and the EPA. The procurement monitoring plan will be an annex to the PCA and will be shared with GCF upon signature, as required by the FAA. The procurement monitoring plan will also be reviewed periodically. The project's investments in equipment will be undertaken following UN procurement procedures. Finally, in line with the UNEP Programme Manual, EPA will be requested to provide an annual compliance audit covering all aspects of the project execution, including review of all expenditures incurred during the financial year.

The procurement plan for the proposed project is provided in Annex 10.

There are no UN sanctions or embargoes listed against EPA or its Management team, and there appears to be no evidence of engagement in prohibited practices by EPA. While there is no dedicated act of parliament on corruption, Ghana's penal code penalises corruption by public officials¹ and Ghana has a National Anti-Corruption Action Plan (NACAP)¹. The Ghana govt also has in place an Anti-Money Laundering Act, 2008 (Act 749)¹ which prohibits money laundering and established a Financial Intelligence Centre. The Anti-Money Laundering Act penalises money laundering and lists it as an extraditable offence.

Furthermore, the Project Cooperation Agreement with EPA will include a clause requiring EPA to contractually oblige all its sub-personnel, agents, employees, or contractors to comply with Anti-Fraud and Corruption Framework of the UN and GCF Prohibited Practices.

G.4. Disclosure of funding proposal

☒ No confidential information: The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☒ Annex 1 NDA no-objection letter(s) [\(template provided\)](#)
- ☒ Annex 2 Feasibility study - and a market study, if applicable
- ☒ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☒ Annex 4 Detailed budget plan [\(template provided\)](#)
- ☒ Annex 5 Implementation timetable including key project/programme milestones [\(template provided\)](#)
- ☒ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
[\(ESS disclosure form provided\)](#)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☒ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☒ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☒ Annex 7 Summary of consultations and stakeholder engagement plan
- ☒ Annex 8 Gender assessment and project/programme-level action plan [\(template provided\)](#)
- ☒ Annex 9 Legal due diligence (regulation, taxation and insurance)
- ☒ Annex 10 Procurement plan [\(template provided\)](#)
- ☒ Annex 11 Monitoring and evaluation plan [\(template provided\)](#)
- ☒ Annex 12 AE fee request [\(template provided\)](#)
- ☒ Annex 13 Co-financing commitment letter, if applicable [\(template provided\)](#)
- ☒ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☒ Annex 15 Evidence of internal approval [\(template provided\)](#)
- ☒ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information [\(template provided\)](#)
- ☒ Annex 18 Appraisal, due diligence or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex 22 Assessment of GHG emission reductions and their monitoring and reporting (for mitigation and cross cutting-projects)⁸⁶
- ☒ Annex 23 Beneficiary Estimation Methodology
- ☐ Annex 24 Menu of interventions
- ☐ Annex 25 Ex-Ante Carbon-balance Tool (EX-ACT)

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

⁸⁶ Annex 22 is mandatory for mitigation and cross-cutting projects.



REPUBLIC OF GHANA

MINISTRY
OF
FINANCE

P.O.Box MB 40, Ministries, Accra
Digital Address : GA-144-2024

Kindly quote this number and date on all
correspondence

My Ref. No. MOF/ERM/CFU/GCF/04/25

Your Ref. No.

Date.: 17th APRIL 2025

**RE: NO-OBJECTION LETTER IN RESPECT OF THE FUNDING PROPOSAL TITLED
“CLIMATE-RESILIENT LANDSCAPES FOR SUSTAINABLE LIVELIHOODS IN
NORTHERN GHANA” SUBMITTED BY THE UNITED NATIONS ENVIRONMENT
PROGRAMME (UNEP)**

We refer to the funding proposal titled “**Climate-Resilient Landscapes for Sustainable Livelihoods in Northern Ghana**”, submitted by the United Nations Environment Programme (UNEP) to The National Designated Authority on 8th April 2025.

2. The undersigned is the duly authorized representative of the Ministry of Finance, the National Designated Authority/ Focal Point of the Republic of Ghana.

3. Pursuant to GCF Decisions B.08/10, B.37/22, and B.41/02, the content of which we acknowledge to have reviewed, in my capacity as NDA/Focal Point, we hereby communicate our **no-objection** to the Climate-Resilient Landscapes for Sustainable Livelihoods in Northern Ghana’ as included in the funding proposal.

4. By communicating our no-objection, it is implied that:

- (a) The government of Ghana has no-objection to the Proposal; and
- (b) The Proposal is in conformity with the national priorities, strategies and plans of the Republic of Ghana.

5. We also confirm that our National process for ascertaining no-objection to the Proposal has been duly followed.

6. Notwithstanding the foregoing, we expect the United Nations Environment Programme (UNEP) to take the necessary measures to ensure that the project as described in the Proposal is implemented in a manner consistent with applicable national laws.

7. We acknowledge that this letter will be made publicly available on the GCF website.

8. Kind regards.

**PHYLLIS ADWOA FRAIKUE
NDA - FOCAL POINT
FOR: MINISTER**

**THE EXECUTIVE DIRECTOR
GREEN CLIMATE FUND
G-TOWER, 24-4 SONGDO-DONG
YEONSU-GU INCHEON CITY,
REPUBLIC OF KOREA.**

Cc: The Hon. Minister, MoF
The Hon. Deputy. Minister, MoF
The Chief Director, MoF
The Coordinating Director, MoF
The Director, ERM, MoF

Tel: +233 (0) 302 747 197
Email : info@mofep.gov.gh
Website : www.mofep.gov.gh

Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Climate-resilient landscapes for sustainable livelihoods in northern Ghana
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	United Nations Environment Programme (UNEP)
Environmental and social safeguards (ESS) category	Category B
Location – specific location(s) of project or target country or location(s) of programme	Republic of Ghana: Jirapa, Lambussie, Lawra and Wa West Districts in the Upper West Region; Binduri and Garu Districts in the Upper East Region; East Mamprusi and Yunyoo-Nasuan Districts in the North East Region.
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	Friday, May 16, 2025
Language(s) of disclosure	English for the full disclosure documents, with the executive summary available in Kusaal, Mampruli, Dagaare and Fulani.
Explanation on language	English is the official language of Ghana. Kusaal is the main spoken language in Binduri and Garu Districts in the Upper East Region. Mapruli is the main spoken language in North East Region. Dagaare is the main language spoken in the Wa West, Lawra, Jirapa and Lambussie Districts in the Upper West Region, and Fulani is spoken across all three Regions.
Link to disclosure	English: UNEPFP Ghana Disclosure Doc ESAMF ENG 14 April Clean.pdf Kusaal (Executive Summary): UNEPFP Ghana Disclosure Doc ESAMF ES Kusaal Version.pdf Mampruli (Executive Summary): UNEPFP Ghana disclosure Doc ESAMF ES Mampruli version.pdf Dagaare (Executive Summary): UNEPFP Ghana disclosure Doc ESAMF ES Dagaare Version.pdf Fulani (Executive Summary): UNEPFP Ghana Disclosure Doc ESAMF ES Fulani version.pdf
Other link(s)	UNEP Open Data: https://open.unep.org/project/GCF-GCF-GHA-XXX

	<u>Ministry of Food and Agriculture (MOFA), Ghana</u> https://mofa.gov.gh/site/index.php/projects/resilient-landscapes-for-sustainable-livelihoods-in-northern-ghana <u>Environment Protection Agency (EPA), Ministry of Environment, Science, Technology and Innovation (MESTI)</u> https://www.epa.gov.gh/new/environmental-assessment-and-audit/ <u>Ghana Meteorological Agency (GMet), Ministry of Communications and Digitalisation</u> https://www.meteo.gov.gh/news/climate-resilient-landscapes-for-sustainable-livelihoods-in-northern-ghana/
Remarks	An ESIA consistent with the requirements for a Category B project is contained in the Annex 6B “Environmental and Social Assessment and Management Framework (ESAMF)” of the UNEP Financing Proposal “Climate-resilient landscapes for sustainable livelihoods in northern Ghana”.
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity’s website	Friday, May 16, 2025
Language(s) of disclosure	English for the full disclosure document, with the executive summary available in Kusaal, Mampruli, Dagaare and Fulani.
Explanation on language	English is the official language of Ghana. Kusaal is the main spoken language in Binduri and Garu Districts in the Upper East Region. Mapruli is the main spoken language in North East Region. Dagaare is the main language spoken in the Wa West, Lawra, Jirapa and Lambussie Districts in the Upper West Region, and Fulani is spoken across all three Regions.
Link to disclosure	English: UNEPFP Ghana Disclosure Doc ESAMF ENG 14 April Clean.pdf Kusaal (Executive Summary): UNEPFP Ghana Disclosure Doc ESAMF ES Kusaal Version.pdf Mampruli (Executive Summary): UNEPFP Ghana disclosure Doc ESAMF ES Mampruli version.pdf Dagaare (Executive Summary): UNEPFP Ghana disclosure Doc ESAMF ES Dagaare Version.pdf Fulani (Executive Summary): UNEPFP Ghana Disclosure Doc ESAMF ES Fulani version.pdf
Other link(s)	<u>UNEP Open Data:</u> https://open.unep.org/project/GCF-GCF-GHA-XXX <u>Ministry of Food and Agriculture (MOFA), Ghana</u> https://mofa.gov.gh/site/index.php/projects/resilient-landscapes-for-sustainable-livelihoods-in-northern-ghana

	<u>Environment Protection Agency (EPA), Ministry of Environment, Science, Technology and Innovation (MESTI)</u> https://www.epa.gov.gh/new/environmental-assessment-and-audit/ <u>Ghana Meteorological Agency (GMet), Ministry of Communications and Digitalisation</u> https://www.meteo.gov.gh/news/climate-resilient-landscapes-for-sustainable-livelihoods-in-northern-ghana/
Remarks	An ESMP consistent with the requirements for a Category B project is contained in the Annex 6B “Environmental and Social Assessment and Management Framework (ESAMF)” of the UNEP Financing Proposal “Climate-resilient landscapes for sustainable livelihoods in northern Ghana”.
Environmental and Social Management System (ESMS) (if applicable)	
Date of disclosure on accredited entity’s website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), Indigenous Peoples Planning Framework (IPPF) (if applicable)	
Description of report	Indigenous Peoples Planning Framework (IPPF)
Date of disclosure on accredited entity’s website	Friday, May 16, 2025
Language(s) of disclosure	English for the full disclosure document, with the executive summary available in Kusaal, Mampruli, Dagaare and Fulani.
Explanation on language	English is the official language of Ghana. Kusaal is the main spoken language in Binduri and Garu Districts in the Upper East Region. Mapruli is the main spoken language in North East Region. Dagaare is the main language spoken in the Wa West, Lawra, Jirapa and Lambussie Districts in the Upper West Region, and Fulani is spoken across all three Regions.
Link to disclosure	English: UNEPFP Ghana Disclosure Doc IPPF ENG 14 April Clean.pdf Kusaal (Executive Summary): UNEPFP Ghana Disclosure Doc IPPF ES Kusaal Version.pdf Mampruli (Executive Summary): UNEPFP Ghana disclosure Doc IPPF ES Mampruli Version.pdf Dagaare (Executive Summary): UNEPFP Ghana disclosure Doc IPPF ES Dagaare Version.pdf Fulani (Executive Summary): https://open.unep.org/docs/gcf/UNEPFP%20Ghana%20disclosure%20Doc%20IPPF%20ES%20Fulani%20version.pdf

Other link(s)	UNEP Open Data: https://open.unep.org/project/GCF-GCF-GHA-XXX Ministry of Food and Agriculture (MOFA), Ghana https://mofa.gov.gh/site/index.php/projects/resilient-landscapes-for-sustainable-livelihoods-in-northern-ghana Environment Protection Agency (EPA), Ministry of Environment, Science, Technology and Innovation (MESTI) https://www.epa.gov.gh/new/environmental-assessment-and-audit/ Ghana Meteorological Agency (GMet), Ministry of Communications and Digitalisation https://www.meteo.gov.gh/news/climate-resilient-landscapes-for-sustainable-livelihoods-in-northern-ghana/
Remarks	A IPPF consistent with the requirements for a Category B project and GCF and UNEP policies on Indigenous Peoples is contained in the Annex 6C “Indigenous Peoples Planning Framework (IPPF) of the UNEP Financing Proposal “Climate-resilient landscapes for sustainable livelihoods in northern Ghana”
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Tuesday, May 13, 2025
Place	Physical copies of the ESAMF and IPPF (English for the full documents and ES in local languages) have been displayed at the EPA Regional Offices in Wa (Upper West), Bolgatanga (Upper East) and Walewale (North East), Offices of the Regional Departments of Agriculture in Wa (Upper West), Bolgatanga (Upper East) and Nalerigu (North East), 8 District Assembly Offices of the project’s Districts and at their respective offices of the Department of Agriculture in Binduri, Garu, East Mamprusi, Yunyoo-Nasuan, Wa West, Lawra, Jirapa and Lambussie Districts.
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity’s Board meeting	N/A
Date of GCF’s Board meeting	Monday, June 30, 2025

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP265

Proposal name:	Climate-resilient landscapes for sustainable livelihoods in northern Ghana
Accredited entity:	United Nations Environment Programme (UNEP)
Country/(ies):	Ghana
Project/programme size:	Medium

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The project targets eight districts in northern Ghana – identified as the most climate-vulnerable through data-driven analysis and extensive stakeholder consultations. By supporting 120 high-risk communities with adaptive, locally driven interventions, the project ensures targeted impact, strengthens resilience, and lays the foundation for scalable, climate-resilient rural development.	A potential risk is the lack of in-country presence of the accredited entity, which may lead to delays in coordination and implementation unless mitigated through strong local execution and oversight mechanisms.
All executing entities and partners are national institutions, ensuring local ownership, institutional alignment, and long-term commitment. This nationally led model builds and strengthens existing systems, supports government coordination, and enhances institutional capacity in Ghana to scale and sustain adaptation efforts independently.	

2. The Board may wish to consider approving this funding proposal in accordance with the term sheet agreed between the Secretariat and the accredited entity (AE) and, if considered appropriate, subject to the conditions set out in annex II to document GCF/B.42/02.

II. Summary of the Secretariat's assessment

2.1 Project background

3. In northern Ghana, where approximately 80 per cent of the workforce depends on rain-fed, subsistence agriculture, smallholder farmers face major challenges due to climate

variability. Prolonged dry seasons cause food shortages and water scarcity, while rising temperatures reduce water retention and increase pests and crop spoilage. At the same time, extreme rainfall leads to flooding, crop loss, infrastructure damage, and reduced reservoir capacity, heightening the risk of dam failure and downstream flooding.

4. In response, the project aims to build climate resilience among smallholder farmers in northern Ghana by improving food security and strengthening the rural economy. It focuses on climate-resilient agriculture and water storage, landscape restoration (e.g. riverbank rehabilitation, agroforestry), and climate-resilient rural infrastructure. Activities will be implemented at the community level (planning and capacity-building), regional level (weather forecasting), and national level (drought and flood action plan). Implemented over seven years in eight climate-vulnerable districts, the project will support 120 communities and provide early warning advisories to the wider population. Led by the Ghana Environmental Protection Agency (EPA), in partnership with the Ministry of Food and Agriculture, Ghana Meteorological Agency (GMet) and Water Resources Commission, the project will promote sustainable, climate-resilient livelihoods through an integrated, ecosystem-based approach.

5. The proposed funding for the project is USD 70,198,822, utilizing grant instruments. The GCF contribution to the project is USD 63,211,141 in grant. Co-financing will be provided by the Government of Ghana.

6. The project targets around 691,125 direct beneficiaries and around 2.8 million indirect beneficiaries. Together, these beneficiaries make up 9.3 per cent of the whole population of Ghana.

7. GCF categorizes the project as B (medium) for its environmental and social safeguards (ESS).

2.2 Component-by-component analysis

8. The project is structured into two components:

(a) **Component 1:** Early warning systems; and

(b) **Component 2:** Climate-resilient livelihoods.

Component 1: Early warning systems (total cost: USD 15,358,090; GCF cost: USD 13,458,090 in grants)

9. This component will strengthen Ghana's early warning system through a series of coordinated interventions. It will improve access to flood and drought information, expand hydrometeorological and groundwater monitoring networks and radar coverage – particularly in the underserved northern region – to enhance the forecasting accuracy of GMet, and build institutional and technical capacities for drought and flood monitoring, assessment, and response. It will also establish a national framework for generating and disseminating climate hazard advisories to local communities, supported by the development of a national action plan for drought and flood hazard management.

10. An output under this component is improved climate data and early warnings made available to facilitate proactive drought and flood management.

Component 2: Climate-resilient livelihoods (total cost: USD 51,110,644; GCF cost: USD 46,179,060 in grants)

11. This component will strengthen capacity for and implement climate-resilient agriculture, landscape restoration for drought and flood management, and alternative climate-resilient livelihoods in beneficiary communities. It will train 200 Extension Officers (at least 30

per cent women) on climate-resilient practices, ecosystem-based adaptation (EbA), and livelihood diversification. Communities will receive training and support to develop or finalize their own climate change adaptation plans, based on locally identified priorities and climate risk assessments. Adaptation interventions outlined in each climate change adaptation plan – selected from a menu of context-specific options – will be implemented with the support of District Extension Officers to reduce community vulnerability to climate change.

12. Outputs under this component are:

- (a) Climate-resilient agricultural practices, EbA and alternative climate-resilient livelihoods implemented in beneficiary communities;
- (b) Restoration of landscape to reduce drought and flood risk;
- (c) Increased access of smallholder farmers to financial resources and engagement with the private sector; and
- (d) Knowledge and awareness of climate threats to agricultural livelihoods and available adaptation options increased to inform the upscaling of climate-change adaptation across northern Ghana.

Monitoring and evaluation (M&E) (total cost: USD 1,392,462; GCF cost: USD 1,314,414)

Project management (total cost: USD 2,337,625; GCF cost: USD 2,259,577)

13. The project management budget will finance project management expenses, such as a Project Manager, Finance Officer and office equipment.

III. Assessment of performance against investment criteria

14. The project consistently scores medium to high against GCF investment criteria. A detailed assessment of the project's fit with GCF investment criteria is provided below.

3.1 Impact potential

Scale: Medium to high

15. The proposed project has medium to high impact potential, directly benefiting 691,125 rural farmers through enhanced early warning systems, adoption of climate-resilient agricultural practices, and large-scale agroforestry and restoration activities.

16. Indirectly, it will reach the broader population of 2.86 million in northern Ghana by improving food and water security and increasing resilience to climate hazards. Enhanced climate data and early warning dissemination will also strengthen national drought and flood response systems, contributing to more climate-responsive development planning.

3.2 Paradigm shift

Scale: Medium

17. The project will catalyse a shift from vulnerable, subsistence farming to climate-resilient, ecosystem-based livelihoods in northern Ghana. Through the adoption of climate-smart agriculture, landscape restoration, and diversified income sources, combined with strengthened early warning systems and planning capacity, communities will move from reactive to proactive climate responses.

18. Its model is designed for scale-up across 43 districts, supported by a national action plan, and integration into local development plans. A strong M&E and knowledge-sharing system will document and disseminate lessons, while training and peer-to-peer learning will promote replication and sustained behavioural change beyond the project areas.

3.3 Sustainable development potential

Scale: High

19. The programme will contribute to several Sustainable Development Goals (SDGs): SDG 1 (No poverty), SDG 2 (Zero hunger), SDG 8 (Decent work and economic growth), SDG 13 (Climate action); SDG 15 (Life on land); and SDG 17 (Partnerships for the Goals). The programme expects to realize environmental, social, economic, and gender co-benefits.

3.4 Needs of the recipient

Scale: High

20. Northern Ghana is highly vulnerable to climate change due to its reliance on rainfed agriculture, limited infrastructure, and a widening development gap. The Government of Ghana recognizes the urgency but lacks the fiscal space to fund large-scale adaptation. Smallholder farmers face limited access to credit, and local institutions lack the capacity to implement and coordinate adaptation efforts.

3.5 Country ownership

Scale: High

21. No-objection letters have been issued by the Government of Ghana. Additionally, all executing entities (EEs) and implementing partners are national entities, with technical support and oversight provided by the AE.

22. The proposed project is fully country-driven as it was prioritized by the national designated authority in June 2016 using a transparent prioritization tool developed under the GCF Readiness and Preparatory Support Programme. In addition, extensive consultations with national, regional, district and community-level stakeholders (including representatives of local government, the private sector and civil society, and women and other vulnerable groups) were conducted in-country by the Government of Ghana during the development of the proposal.

3.6 Efficiency and effectiveness

Scale: Medium to high

23. The project provides grants to strengthen capacities at the national, regional, and field levels through training sessions, land restoration, and institutional development for early warning systems. This enables smallholders to adopt and sustain climate-resilient practices beyond the programme implementation period.

24. The GCF cost of USD 91 per direct beneficiary (and USD 23 per total beneficiary) reflects moderate cost-efficiency within the GCF adaptation portfolio. The co-financing ratio of 1:0.11 is low; while this is understandable given the government's financial constraints and in-kind contribution commitments, caution is warranted until these contributions have fully materialized.

25. Scenario analysis shows that nature-based infrastructure and hybrid approaches, such as reforestation, climate-smart agriculture, and agroforestry, can substantially reduce climate-related damages and enhance community resilience. These measures lead to avoided damages and generate income benefits, with net benefits of up to USD 55.3 million (Wa West) and net present values of up to USD 68.8 million under hybrid scenarios.

26. The proposed investments are economically sound, with benefit-cost ratios ranging from 1.55 to 2.95 and internal rates of return from 17.4 to 37.9 per cent across districts. Even under conservative assumptions, the financial returns remain positive. Excluded co-benefits such as lives saved, improved nutrition, and carbon sequestration suggest even greater value, underscoring the project's strong economic and societal case.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

27. **Project overview.** The project aims to enhance the climate resilience of vulnerable smallholder farming communities in northern Ghana by improving food security and enhancing the agriculture-based economy. Specifically, the project will improve climate data and early warning systems in the Northern region of Ghana under component 1. Component 2 will focus on adoption of climate-resilient agriculture and water storage, and investments in EbA, nature-based flood-risk reduction, and climate-resilient agricultural practices, benefiting 15 communities from each of the 8 districts from 3 regions in northern Ghana. The direct beneficiaries will be identified based on the selection criteria and stakeholder consultations. Beneficiary communities will be able to select the most locally appropriate intervention from a menu of adaptation of interventions which has been made available in the project formulation phase. The potential environmental and social (E&S) co-benefits include restoration of degraded lands leading to increased canopy cover, reduced soil erosion, reduced surface run-off, improved ecosystem services and resilience, improved agricultural productivity, and reduced exposure to climate-related environmental hazards, and strengthened institutional capacity for climate adaptation, among others.

28. **E&S risk category and safeguard instrument.** The proposed project activities have been screened against the safeguards risk identification form of the United Nations Environment Programme (UNEP) and have been classified as having moderate risks associated with the UNEP safeguards standards on biodiversity, ecosystems and sustainable natural resource management and on Indigenous Peoples. The moderate risk classification is equivalent to the GCF category B classification and aligned with the accreditation level of the AE. An environmental and social assessment management framework (ESAMF) has been developed to provide systematic guidelines in further identifying E&S risks and impacts, mitigating measures, and institutional arrangements during project implementation. The following potential risks have been identified in the ESAMF: i) introduction of harmful species; ii) increased use of agricultural chemicals; iii) increased demand for irrigation; iv) habitat conversion; v) excessive fire suppression; and v) the adverse effects of on-farm earthworks. The ESAMF identifies mitigating measures following the hierarchy of mitigation.

29. **Compliance with GCF ESS standards.** The following paragraphs describe how the project complies with these standards.

30. **ESS 1: Assessment and management of E&S risks and impacts.** Since the exact locations and design of specific activities will be identified during implementation, the AE has prepared an ESAMF at funding proposal development stage. The project's ESAMF provides guidelines for the screening and assessment of potential E&S risks and impacts; identifies potential E&S risks and impacts and corresponding indicative mitigating measures; describes the implementation arrangements and capacity needs; provides details for the establishment of a grievance redress mechanism (GRM); identifies information disclosure requirements; and describes the stakeholder engagement process during the project development and implementation phases. The ESAMF also sets out activity exclusion criteria and prohibited activities, outright excluding activities that may have significant E&S risks and impacts equivalent to a category A project. During implementation, any subproject categorized as B would require a fit-for-purpose environmental and social impact assessment and environmental and social management plan which would need to be disclosed following the GCF Information Disclosure Policy. Compliance with national regulatory requirements will also be met, as required. Monitoring indicators have been developed for the ESAMF implementation and results will be reflected in the annual performance reports.

31. **ESS 2: Labour and working conditions.** The engagement of project staff and contractors will be undertaken in compliance with national labour laws and international commitments, such as International Labour Organization conventions, as well as UNEP and GCF procurement policies and principles. This will ensure that staff are not subjected to adverse working conditions, occupational health and safety risks, forced labour and child labour. The Ghana Labour Act of 2003 provides general labour protections, however, gaps with international good practice have been identified such as those regarding grievance redress mechanisms for workers and compliance with contractor obligations. As such, the UNEP safeguard standards will be applied to fill the gaps. A mandatory code of conduct will be signed by contractors and project staff. The project anticipates engagement of local labour only and no work camp establishment is envisaged. While no significant occupational health and safety risks are anticipated, measures will be in place to address any potential risks through EPA regulations and good practice. For interventions involving physical work, appropriate health and safety training, equipment and safety measures will be in place.
32. **ESS 3: Resource efficiency and pollution prevention.** Early warning infrastructure could cause construction-related impacts such as increased noise and dust levels, localized soil and vegetation disturbance, and contamination of nearby water sources. For all activities, the AE has identified this standard as triggered with a low level of risk. On output 1, as the installation of small-scale early warning equipment involves construction works, detailed mitigation measures relating to construction works will have to be detailed in the site-specific environmental and social management plans once the target sites have been identified. On output 2, the promotion of climate-resilient agricultural techniques could increase water demand and encourage the use of fertilizers and pesticides. Waste can also be generated from bags and containers from agricultural activities. To mitigate such risks, the project will provide farmers with the capacity to undertake integrated pest management, including training on the safe use of agrochemicals and waste management, storage, application, minimization and disposal. Mitigation measures should be adopted to address potential risks and impacts related to climate-resilient water infrastructure (boreholes and check dams). The activity exclusion criteria and prohibited activities list in the Environmental and Social Assessment and Management Framework identify activities likely to generate significant greenhouse gas emissions, or pollution beyond permissible limits, including uncontrolled burning or industrial emissions. No substantial consumption of energy, water or other material inputs associated with the project are anticipated. To mitigate potential impacts from small-scale climate-resilient livelihood promotion, the project will disseminate good practice guidance on pollution prevention, resource efficiency, and environmentally sound waste disposal tailored to livelihood activities. Low-input, low-pollution processing techniques through technical support and extension services will be promoted. Chemicals and materials subject to international bans and phase-outs will not be used during the proposed project interventions.
33. **ESS 4: Community health, safety and security.** Minimal health, safety and security risks are associated with the installation of early warning observation equipment and the potential for intergroup conflict between sedentary and pastoralist groups if benefits from project interventions are not managed well. The risks associated with local conflict will be reassessed and addressed at the local level through customary land-sharing practices and participatory planning processes during implementation to ensure that appropriate mitigating measures are adopted. Some sites for observation equipment may need to be fenced to limit public access for safety purposes. The project activities are not expected to result in an influx of workers requiring worker camps, use of security services, nor will they involve medium to large-scale construction. Health and safety risks related to livestock-rearing are minimal, as the project does not involve the provision or scaling-up of animal-husbandry activities, and no increase in animal populations is anticipated. Likewise, the project does not include large-scale water infrastructure or open water sources. Borehole installations will be small in scale and will

be screened to ensure they do not generate additional risks related to water or vector-borne diseases.

34. **ESS 5: Land acquisition and involuntary resettlement.** The risks are assessed to be of low significance on land acquisition and involuntary resettlement. No physical or economic displacement is anticipated from the project activities. Early warning observation equipment will be installed on government land as much as possible. In case it is found not to be feasible to install on government-owned land, formal lease on private land will be arranged through a participatory process. If the land identified is within pastoralist grazing land or traditionally owned and used land, land access will be agreed through a fully consultative process (i.e. free, prior and informed consent in the case of Indigenous Peoples' involvement). To ensure that pastoralists are not negatively affected, the project will not support any activity that results in changes in land-tenure arrangements or change existing land-use restrictions. The project will not support any activity that would lead to forced eviction.

35. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** While E&S risks from a single intervention may be considered as low risk, collectively the project is assessed as having moderate risks triggering the UNEP Safeguards Standard on Biodiversity, Ecosystems and Sustainable Natural Resource Management. The implementation of climate change adaptation interventions, including climate-resilient agricultural practices may affect local ecosystems if risks are not managed properly. In selecting sites for any activity, siting within or near protected habitats, areas supporting endangered species, and areas of high conservation value will be stringently avoided. The project will only support agricultural production in existing agricultural lands used by communities. In planning for plantation reforestation, the project will ensure alignment with any existing management plans and land-use plans, and will exclude support for any monoculture practice. No invasive species or genetically modified organism will be supported. As part of capacity-building initiatives, biodiversity conservation and risk management training practices will be incorporated into the training materials to enhance the long-term sustainability of climate-resilient agriculture.

36. **GCF Indigenous Peoples Policy and ESS 7 on Indigenous Peoples:** The AE foresees potential impact on Indigenous Peoples in regard to water, land and natural resources access. The AE has started initial engagement with representatives of transhumant pastoralists and provided an Indigenous Peoples Planning Framework, which requests systematic screening for Indigenous Peoples, consultations and ensuring free, prior and informed consent when needed. The project has excluded any change to customary land use, or restriction of use to traditional lands used by pastoralists. The AE foresees Indigenous Peoples representation at the Project Steering Committee. The AE is recommended to engage with the GCF Indigenous Peoples Advisory Group on relevant Indigenous Peoples' issues that may be encountered.

37. **ESS 8: Cultural heritage.** The proposed activities will not be located in any known cultural heritage sites and as such, no adverse impacts on cultural heritage are anticipated. However, since the project involves earthworks, chance find procedures will be adopted as a precautionary measure.

38. **Implementation arrangements.** Under the oversight role of UNEP as accredited entity, the EPA of the Ministry of Environment, Science, Technology and Innovation as the main EE will be responsible for the overall implementation of the ESAMF. EPA will be guided by a Project Steering Committee composed of representatives of various government agencies, United Nations Capital Development Fund, and civil society organizations. GMet will serve as the executing entity for output 1 on improving climate data and early warning systems. E&S focal points will be designated from each executing entity to manage safeguard implementation at operational level. To ensure effective safeguards and GRM implementation, the E&S focal points will be provided with training and technical support from EPA and the Gender and Safeguards

Specialist at the Project Management Unit (PMU) level. E&S training will also be provided for all staff involved in project implementation at the national, regional and district levels.

39. **Stakeholder engagement.** Multilevel stakeholder engagements involving representatives of the public and private sectors, women and farmer-focused non-governmental organizations, and the targeted local communities, including farmers, have been conducted by the AE together with the EE during project design and formulation. The stakeholder engagements aimed at identifying the needs, issues, risks and resolutions associated with the project activities. Key concerns raised pertain to GRMs, gender dynamics and inclusion of ethnic minorities. The key concerns raised will be addressed through the implementation of the ESAMF, Indigenous Peoples Planning Framework, Gender Assessment and Action Plan and the subsequent plans to be developed during implementation. The PMU will continue with the stakeholder consultations during project implementation as detailed in the stakeholder engagement plan.

40. **Grievance redress mechanism.** A project-supported GRM that builds on existing structures will be made available to all project stakeholders, including contractors or contracted labourers in addition to an GRM process specific to Indigenous Peoples, which will be designed, in a consultative manner with potentially affected Indigenous Peoples groups. In addition, the institutional GRM of the AE (Stakeholder Response Mechanism) and the GCF Independent Redress Mechanism will also be made available to stakeholders as additional formal channels. The GRM will not impede access to judicial or administrative remedies that may be relevant or applicable. The GRM will be disseminated to all stakeholders through meetings, printed posters and flyers in local languages posted in community centres, assembly halls, and extension offices. Full training will be provided to project implementers on the GRM implementation.

41. **Sexual exploitation, abuse and harassment safeguarding:** In accordance with the Revised Environmental and Social Policy (decision B.BM-2021/18), the AE has acknowledged sexual exploitation, abuse, and harassment (SEAH) as a relevant safeguard issue and has included risk mitigation measures in both its gender action plan and ESAMF. SEAH is recognized as a contextual risk due to project team–community interaction, gender and power asymmetries, and the participation of economically and socially vulnerable groups. Mitigation measures include mandatory SEAH prevention training for project staff, community climate action groups, farmer-based organizations, and targeted community members. The AE has also committed to conducting a dedicated SEAH risk assessment at district level during the project's inception phase. This assessment will inform context-specific mitigation planning and allow for early validation of reporting mechanisms, stakeholder sensitivities, and risk exposure across the target areas. The approach to SEAH risk mitigation is consistent with the provisions of the GCF SEAH Policy. Institutional responsibilities are clearly defined, and awareness-raising measures are embedded across relevant stakeholder levels. Training measures include both in-person and online delivery. The integration of SEAH considerations within gender, safeguard, and financial access components is noted. The planned development of the SEAH-specific GRM during the inception phase presents an opportunity to ensure that such systems are accessible, confidential and culturally appropriate.

4.2 Gender policy

42. The AE provided a gender assessment and action plan with the funding proposal and therefore, complies with the requirements of the updated GCF Gender Policy. A gender assessment has been conducted, drawing on a desk review and stakeholder consultations, and contextualized within the national gender policy environment. It identifies key gender-related barriers such as unequal access to land, limited financial inclusion, disparities in educational attainment, and underrepresentation in decision-making structures. These barriers reduce the

adaptive capacity of women and other vulnerable groups by limiting their ability to participate in, benefit from, and influence climate-resilient livelihood strategies. The assessment further notes the influence of entrenched gender norms and customary land tenure systems, which shape control over resources and reinforce gendered divisions of labour across sectors such as agriculture, shea processing and petty trade. It also considers intersectional factors, highlighting how overlapping vulnerabilities, such as disability, age and informal employment, compound exclusion from economic opportunities and climate adaptation efforts.

43. A gender action plan is provided. It presents a structured approach to addressing gender-related risks and opportunities. It outlines gender-responsive activities under each of the project's outputs and includes associated indicators, sex-disaggregated targets, timelines, roles and responsibilities, and estimated budgets. A Safeguards and Gender Specialist is expected to be embedded within the PMU to oversee the implementation, monitoring and coordination of gender-related actions, supported by institutional capacity from EPA, Ministry of Food and Agriculture, and other project partners. Additional capacity-building measures and training activities on gender mainstreaming are planned at the community and institutional levels.

44. Gender-responsive design elements are incorporated into various project components. These include support for women-led cooperatives, targeted financial literacy and mentorship programmes, participation in early warning systems, and leadership roles in knowledge-sharing initiatives. The proposal also outlines efforts to promote inclusive engagement in district-level planning processes and to ensure participation in decision-making. The use of community-level baselines, targeted leadership roles, and monitoring mechanisms suggests a commitment to tracking both participation and transformative outcomes. The Secretariat encourages the AE to continue prioritizing disaggregated impact tracking and documentation of lessons learned during implementation.

4.3 Risks

4.3.1. AE/EE capability to execute the current project (medium risk)

45. The UNEP will serve as the AE for this project. Currently, there are 10 approved FPs with UNEP, with a total committed amount of USD249 million in the GCF portfolio. The AE will collaborate with two EEs, which are government agencies namely, EPA and Gmet. Since UNEP does not have an in-country presence in Ghana and will be managing from their regional office, the capacity of the EEs will be crucial, as they will be responsible for directly managing the day-to-day operations of the project.

4.3.2. Programme specific execution risk (medium risk)

46. Monitoring of the impact for engaging financial institutions (FIs): Output 4 aims to improve smallholder farmers' access to finance by building capacity, forming farmer-based organizations (FBOs) and Village Savings and Loans Associations (VSLAs), and linking them to blended finance initiatives by international and local FIs. While there are detailed plans and analyses of existing facilities and stakeholders, the project scope is limited to the establishment of FBOs/VSLAs, and a blended finance model, rather than resulting credit lines or loan agreements - these are framed as potential exit strategies without clear indicators, making monitoring difficult. This limitation in scope highlights the need for robust monitoring mechanisms to ensure that the intended outcomes are achieved and to assess the effectiveness of capacity-building efforts as stipulated in the M&E plan.

47. Use of grant: As mentioned above, Output 4 may involve on-lending activities. Given that GCF provides a full grant, there should be a clear distinction in terms of use of grant and loans.

To mitigate this, the AE has confirmed that GCF proceeds will not be used to finance credit lines or guarantees through financial intermediaries. This assurance is crucial in maintaining the integrity of the project's financial management and ensuring that funds are utilized as intended to support smallholder farmers directly.

4.3.3. GCF's portfolio concentration risk

48. In case of approval, the impact of this proposal on the GCF portfolio risk remains non-material and within the risk appetite in terms of concentration levels, results area, single proposal or AE concentration.

4.3.4. Compliance risk (medium risk)

49. Certain project activities, such as the establishment of village savings and loan associations and a blended finance model to provide credit lines – though GCF funds will not be used to finance credit lines and guarantees through financial intermediaries – present elevated ML/TF risks.

50. The AE has assessed ML/TF risk as low given its internal controls, including its risk-based approach to engaging implementing partners on the basis of assessments of their capacity and governance capabilities, as well as its established monitoring and evaluation procedures and financial reporting mechanism, which will be applied to this project. Moreover, the AE will provide anti-money laundering/countering the financing of terrorism (AML/CFT) training to downstream partners at the start of the project. In addition, the AE's capacity assessment of EPA found that it has the requisite fiduciary and financial management capacity, including in relation to AML/CFT compliance.

51. Based on the AE's internal controls and governance mechanisms that will be applied during project implementation, and the relatively elevated ML/TF risks associated with certain activities, the overall compliance risk is determined to be medium.

Summary risk assessment	
Overall assessment	Medium
Accredited entity/executing entity capability	Medium
Project-specific execution	Medium
Compliance	Medium
GCF concentration	Within the monitoring thresholds

4.4 Fiduciary

52. As the AE, UNEP will provide fiduciary oversight and safeguard supervision, ensuring compliance with both UNEP and GCF policies. UNEP will oversee the full project cycle – including formulation, start-up, implementation, and closure – using its internal programme manual, which includes robust provisions for financial management, procurement, and project oversight. A UNEP Task Manager will lead project supervision and undertake at least two oversight missions annually. The experience of UNEP in Ghana, including as delivery partner for the National Adaptation Plan Readiness project and in implementing EbA initiatives and environmental programmes, underscores its strong technical and fiduciary capabilities and institutional presence in the country.

53. Executing functions will be carried out by Ghana's EPA and GMet, with UNEP signing formal project cooperation agreements (PCAs) with each entity. These PCAs will establish clear roles, fiduciary obligations, disbursement conditions, and reporting requirements, aligned with UNEP operational standards. The EPA, with a proven track record as an executing agency under

Global Environment Facility-funded programmes (e.g. GEMP, SLWMP), has undergone UNEP fiduciary and capacity assessments, confirming its adequacy in financial and operational management. GMet, responsible for climate and hydrological data, will implement output 2, and both entities will be held accountable for results and the effective use of project resources.

54. The project governance structure includes a national-level Project Steering Committee, co-chaired by UNEP and the Ministry of Environment, Science, Technology and Innovation, and a PMU housed within the regional office of the EPA in Tamale. The PMU will be staffed with a dedicated Financial Officer, Procurement/Administrative Officer, and an M&E specialist, and will work closely with the finance and procurement teams of UNEP. Regional implementation will be supported by local steering committees, and coordination with national ministries and district authorities will ensure vertical alignment and oversight. The EPA will serve as the Project Steering Committee secretariat and lead the day-to-day fiduciary oversight of EEs and implementing partners.

55. At the community level, implementation will be supported by agricultural extension agents and local institutions such as climate change action groups and farmer-based organizations, ensuring resource flows and information are responsive to local needs. UNEP-DHI will also be engaged through a PCA to deliver technical support using internationally recognized modelling tools. Memorandums of understanding among entities such as EPA–Ministry of Food and Agriculture and GMet–Water Resources Commission will define inter-agency roles in agricultural and hydrological interventions. This layered and contractually governed approach ensures a clear chain of accountability, appropriate use of GCF funds, and compliance with fiduciary standards across all levels of implementation.

4.5 Results monitoring and reporting

56. The proposed project aims to contribute towards three adaptation results areas, namely, the increased resilience of most vulnerable people and communities; health, well-being, food security, and water security; and ecosystems and ecosystem services. To report on the corresponding achievements, the proposal included in its logical framework five adaptation indicators:

- (a) Core indicator 2: Direct and indirect beneficiaries reached;
- (b) Supplementary 2.1: Beneficiaries (female/male) adopting improved and/or new climate-resilient livelihood options;
- (c) Supplementary 2.2: Beneficiaries (female/male) with improved food security;
- (d) Supplementary 2.4: Beneficiaries (female/male) covered by new or improved early warning systems; and
- (e) Core indicator 4: Hectares of natural resource areas brought under improved low-emission and/or climate-resilient management practices.

57. These results will be achieved by reducing the vulnerability to climate change of smallholder farmers in northern Ghana through the implementation and adoption of climate-resilient agriculture, EbA approaches, alternative climate-resilient livelihoods and locally led investments in climate resilient rural infrastructure. The project theory of change articulates the results pathways that will lead to the improved adaptive capacity of landscape-level ecosystems to flood, drought, and high temperature in the project site.

58. The project's M&E plan presents details of data sources and data collection and analytical methods for each indicator required by the Integrated Results Management Framework, including periodic surveys undertaken at the start, middle, and end of the project implementation, Geographical Information System mapping, independent verification of

reported results, focus group discussions, key informant interviews, and analysis of secondary information based on government data. The plan also allocated budget for a dedicated Monitoring and Reporting Officer. The overall M&E budget is around USD 2,337,625, which the Secretariat deems sufficient to meet the requirements of the monitoring and accountability framework, Integrated Results Management Framework, and GCF Evaluation Policy. The main risk to reporting credible results is the potential impact of any localized conflict that could affect the feasibility and/or cost of data collection on the ground.

4.6 Legal assessment

59. The Accreditation Master Agreement, which was signed on 15 December 2016 and became effective on 20 February 2017, was subsequently amended and restated by First Amendment and Restatement Agreement which was signed on 13 July 2023 and became effective on 2 August 2023 (as so amended and restated, the "AMA"),

60. The Accredited Entity has provided certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the project.

61. The proposed project will be implemented in the Republic of Ghana, country in which GCF is not provided with privileges and immunities. This means that, amongst other things, GCF is not protected against litigation or expropriation in this country, which risks need to be further assessed. Moreover, the ability of GCF to undertake redress activities and/or investigations in this country may be hindered due to the absence of privileges and immunities for relevant GCF personnel.

62. Therefore, it is recommended that the Board considers whether disbursements of GCF proceeds should only be made after GCF has obtained satisfactory protection against litigation and expropriation in the country, or has been provided with appropriate privileges and immunities for GCF and its personnel.

63. To address the matters raised in this section, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval; and
- (b) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP265

Proposal name:	Climate-resilient landscapes for sustainable livelihoods in northern Ghana
Accredited entity:	United Nations Environment Programme (UNEP)
Country/(ies):	Ghana
Project/programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Overview

1. In Northern Ghana agriculture employs roughly 80% of the region's workforce and is predominantly rainfed and relies on traditional farming methods and assessed to have "limited technological inputs." This dependence on natural climate conditions leaves farmers highly exposed to environmental shifts. The region's agroecological systems are under strain due to a combination of factors, including erratic climate linked weather patterns, environmental degradation from unsustainable land use, rapid population growth, and disruptions caused by dam spills from the Tamne and Bagre reservoirs.

2. **Exacerbating impact of climate risk on the development pathway.** Climate change is intensifying these pressures. Key trends include shorter and more unpredictable rainy seasons, increased frequency of extreme rainfall events, prolonged dry spells, rising temperatures, and heightened evapotranspiration rates. These increasing climate hazards have cascading effects such as degraded soils, reduced crop yields, loss of arable land, and more frequent bushfires. Farmers already operating at subsistence levels struggle to adapt, leading to worsening food insecurity, income instability, and deepening poverty. Without targeted interventions such as improved climate information, resilient farming techniques, and sustainable resource management the farmers' capacity to cope will further weaken.

3. **Need for strategic interventions.** The situation demands urgent action because if left unaddressed, climate-induced agricultural decline will worsen marginalization of rural communities, exacerbate food shortages, and destabilize economic activities in northern Ghana. Thus, effective solutions need integration of climate adaptation, environmental restoration, and livelihood diversification to build long-term resilience for smallholder farmers.

4. **Proposed project.** The Climate-Resilient Landscapes for Sustainable Livelihoods project in Northern Ghana, aims to enhance climate adaptation, improve agricultural resilience, and integrate climate-smart practices across vulnerable smallholder farming communities.

5. **Project objective and outcomes.** As stated in the funding proposal, the "proposed project objective is to enhance the climate resilience of vulnerable smallholder farming communities in northern Ghana by improving food security and enhancing the agro-based rural economy. The project's outcomes will strengthen institutional and regulatory systems for climate-responsive planning and development, enabling government agencies and local institutions to effectively integrate climate adaptation into policies and strategies. The project supports the adoption of climate-resilient agricultural practices, alternative livelihoods, and access to financial resources, ensuring farmers can sustainably adapt to climate impacts. Additionally, the project improves knowledge and learning processes, fostering informed

decision-making and empowering communities with the tools and information needed to scale up climate adaptation efforts across the region.”¹

6. **Project components and outputs.** The project is divided into two components, namely early-warning systems (Component 1) and climate resilient livelihoods (Component 2). The outputs (strategic interventions) are as follows: (i) improved climate data and early warnings made available for facilitating proactive management, (ii) climate-resilient agricultural practices, EbA and alternative livelihoods implemented in beneficiary communities, (iii) restoration of landscape to reduce droughts and flood risks, (iv) increased access of smallholder farmers to financial resources, and (v) increased knowledge and awareness of climate threats to agricultural livelihoods and available adaptation options.²

7. **Project summary.** This climate adaptation initiative represents a strategic investment in sustainable agricultural development, with a total project budget of USD 70,198,882 (including GCF grant funding of USD 63,211,141 and co-financing). This is a medium-scale project that has a 7-year implementation period designed to establish foundational climate resilience systems, and a 28-year lifespan to ensure lasting impacts for vulnerable farming communities. Grant financing (USD 63,211,141) is the funding modality to ensure accessibility for resource-constrained smallholders.³

1.2 Impact potential

Scale: High

8. The project is meant to improve resilience through investments in nature-based solutions such as riparian buffer restoration, agroforestry, and adaptive agricultural techniques. It also seeks to facilitate financial access for smallholder farmers through climate-resilient financing models, ensuring long-term viability beyond the funding period. Early warning systems are planned to be developed to enhance protection of communities against climate-induced disasters, further strengthening adaptive capacity.

9. **Ecosystem resilience through nature-based solutions.** The project is expected to enhance ecosystem resilience in northern Ghana by deploying large-scale Ecosystem-based Adaptation (EbA) strategies. The planned restoration of 14,000 hectares of degraded land and 2,000 hectares of riparian zones across eight districts is projected to reduce flood risks by nearly 47%, protect watersheds, and enhance soil and water retention. Other reforestation efforts using climate-resilient indigenous tree species and natural regeneration aims to reverse degradation. Agroforestry interventions in 120 communities are anticipated to benefit over 120,000 people. Fire management practices are designed to reduce the frequency and severity of bushfires exacerbated by increasing temperatures or dry seasons, further securing critical agroecosystems. These combinations of actions are intended to restore ecological integrity, support biodiversity, and buffer communities from increasing variability, frequency and intensity of climate extreme events.

10. **Resilient livelihoods and climate-smart agriculture (CSA).** The project introduces CSA practices to address climate-induced agricultural risks (e.g., erratic rainfall, extended droughts, and crop failure). The adoption of drought-resistant crops, precision irrigation, and agroforestry models is expected to promote higher yield stability and sustainable resource use. These measures are designed to safeguard food security and stabilize the incomes of smallholder farmers. The restoration of natural floodplains and creation of buffer zones aim to protect farmland and infrastructure, ensuring sustained agricultural productivity despite intensifying climate change and variability.

¹ See funding proposal section B.2

² See funding proposal section B.3

³ See funding proposal sections A.7 to A.11.

11. **Financial inclusion as resilience measure.** The project seeks to enhance farmers' creditworthiness, collective bargaining power, and other structural barriers to financial inclusion by organizing smallholder farmers into Farmer-Based Organizations (FBOs) and Village Savings and Loans Associations (VSLAs). Through capacity building and financial literacy training, farmers are expected to gain skills in accessing credit, insurance, and investments in climate-resilient practices. Strategic partnerships with local financial institutions, supported by a blended finance model, are projected to expand access to tailored financial products while de-risking lending activities. Digital financial services are planned to reach remote communities, with special provisions intended to empower women through targeted tools and training. This financial empowerment is anticipated to underpin long-term economic resilience and inclusive rural development.
12. **Upgrade of infrastructure protection and early warning systems.** The project aims to strengthen the resilience of rural infrastructure and communities through the development of localized, climate-informed early warning systems (CIEWS). CIEWS are designed to improve disaster preparedness and risk management in areas vulnerable to floods and droughts. By integrating meteorological and hydrological data with community-based dissemination channels, the project seeks to enhance protection of lives, assets, and productivity from climate shocks. Infrastructure such as roads, water facilities, and schools frequently damaged by climate extreme events is expected to benefit from improved adaptation planning and risk reduction measures.
13. **Emissions reduction co-benefits.** While primarily adaptation-focused, the project is expected to generate substantial mitigation co-benefits through sustainable land management and agro-ecological farming practices. Reforestation and agroforestry activities are projected to sequester approximately 1.2 million tCO₂e over the project's lifetime. These interventions align with Ghana's broader climate goals and offer long-term environmental dividends without compromising the project's core adaptation objectives.
14. **Broad and inclusive impact.** Overall, the project is expected to directly benefit approximately 691,125 individuals, with an additional 2.8 million people indirectly impacted through improved resilience measures. By focusing on inclusive, nature-based, and financially viable solutions, the project aims to build the adaptive capacity of communities, ecosystems, and institutions in a region facing acute climate risks. The project is designed to lay the groundwork for transformational resilience across northern Ghana.
15. The funding proposal's impact potential is assessed as high.

1.3 Paradigm shift potential

Scale: Medium to high

16. **Enabling Ghana Meteorological Agency (GMET) transformation into a modern meteorological service through digital innovation.** A successful integration or adoption of latest technologies (e.g., satellite remote sensing and AI-powered data fusion algorithms) in Output 1 (i.e., activities 1.1.1 and 1.1.2)⁴ presents a transformative opportunity to modernize the GMET and enhance its service delivery. GMET can potentially improve forecast accuracy through the intelligent merging of ground-based Hydromet data with satellite observations. This scientific advancement is designed to feed into a user-friendly web portal, making seasonal forecasts and climate risk information accessible to farmers, policymakers, and disaster managers in actionable formats.

17. The web-based Decision Support System (DSS) sets out to revolutionize early warning capabilities by integrating flood and drought monitoring, water allocation models, and risk assessments into a single platform. Combined with automated SMS/email alerts and cloud

⁴ See funding proposal section B.2.

computing for real-time data processing, GMET is poised to deliver timely, location-specific warnings directly to at-risk communities. Key digital components which may include centralized web portals, AI-driven analytics, and geospatial visualization tools are designed to work together to convert complex climate data into practical, decision-ready information.

18. This potential digital transformation aligns GMET with global best practices in CSA and disaster risk reduction, while addressing critical gaps in last-mile information delivery. By transitioning from traditional methods to an integrated, technology-driven approach, GMET can position itself as a modern meteorological service capable of meeting 21st century climate challenges turning raw data into life-saving and livelihood-protecting actions for Ghana's most vulnerable populations. The result will hopefully be a more responsive, precise, and user-centered national meteorological agency that leverages cutting-edge technology to fulfil its mandate more effectively than ever before.

19. Activities 1.4 and 1.5 plan for information dissemination and a national plan for coordinating drought and flood hazard management. There is however no mention about how to develop preparedness and response capabilities. It would be advisable to ensure that all elements of an Early Warning System be in place and well-aligned with the needs of farmers and other groups to which the system is to cater for.

20. **CIEWS scaling up.** The project outlines a strong potential for scaling up climate-informed early warning systems in northern Ghana by addressing key gaps in coverage, data accuracy, and information dissemination. Existing drought and flood warnings are limited in reach and effectiveness particularly in vulnerable rural areas. The project proposes expanding hydrometeorological monitoring, enhancing radar coverage, and training national and local agencies in advanced forecasting and response especially GMet. A national framework will hopefully guide the generation and community-level dissemination of climate risk information, and its planned integration into district planning and budgeting systems will support long-term sustainability. The plan to embed climate adaptation into local development plans and establishing financing mechanisms enhances the project's potential for a replicable CIEWS model worthy of a nationwide scale-up.

21. **Accelerating a paradigm shift towards climate-resilient agricultural economy in Northern Ghana.** The proposed project presents a transformative opportunity to shift northern Ghana's agriculture from a vulnerable, rainfed, and unsustainable system toward a low-carbon, climate-resilient economy. The integrated approach combining climate-resilient agriculture (CRA), ecosystem-based adaptation (EbA), and locally led investments in rural infrastructure addresses both environmental and socio-economic vulnerabilities. CRA practices such as drought-resistant crops, sustainable livestock systems, and water-efficient irrigation will stabilize yields under shifting climatic conditions. Parallel EbA interventions (i.e., reforestation, riverbank restoration, agroforestry, and fire management) will restore ecosystem services, reduce land degradation, and buffer communities against floods and droughts.

22. The introduction of alternative, climate-resilient livelihoods and strengthened local climate advisory services enables farmers to move away from environmentally harmful practices like slash-and-burn farming. Investments in meteorological infrastructure and early warning systems will improve climate risk forecasting and inform adaptive decisions at the farm level. Institutional strengthening through district-level technical assistance, integration of adaptation into development plans, and climate finance access will also enable greening of the public financial management and the sustainability and scalability of these interventions. These strategies create the conditions for a paradigm shift; empowering smallholder communities to adopt resilient practices, increase productivity, and build a more sustainable rural economy aligned with national climate goals.

1.3.1. Risk to paradigm shift potential

23. **Gaps in digital innovation integration.** While the project has plans to upgrade the GMet meteorological services using the latest technologies, it has not fully explored the integration of digital finance infrastructure for the agriculture and finance sector due to the limited study done in support of this proposal.

24. Overall, the paradigm shift potential is medium to high.

1.4 Sustainable development potential

Scale: High

1.4.1. Sustainable Development Goals (SDGs)

25. The proposal aims to contribute to the achievements of the following Sustainable Development Goals (SDGs), i.e., No Poverty (SDG 1); Zero Hunger (SDG 2); Good Health and Well-being (SDG3); Gender Equality (SDG 5); Clean Water and Sanitation (SDG 6); Decent Work and Economic Growth (SDG 8); Industry, Innovation and Infrastructure (SDG 9); Reduced Inequalities (SDG 10); Sustainable Cities and Communities (SDG 11); Responsible Consumption and Production (SDG 12); Climate Action (SDG 13); Life on Land (SDG 15); and Partnerships for the Goals (SDG 17).

26. **Transformative Gender Empowerment.** The project's Gender Assessment and Action Plan (GAAP) empowers women not merely as participants, but as active agents of change in climate-resilient development. It promotes women's leadership in FBOs, Village Savings and Loan Associations (VSLAs), and Community Climate Action Groups, positioning them in roles traditionally dominated by men. Women are equipped with practical skills in climate-resilient agriculture, land restoration, and ecosystem-based adaptation, and are supported to access financial services including credit and insurance through tailored financial literacy and enterprise development training. The GAAP also facilitates women's engagement in value chains, linking them to markets and building their capacity to manage climate-resilient livelihoods. By addressing structural barriers such as limited access to land, finance, and decision-making spaces, the GAAP ensures that women are not only included but also empowered to drive long-term climate adaptation and economic resilience in their communities.⁵

27. **Socio-Economic Development.** The project has strong potential to drive inclusive socio-economic development, particularly among Indigenous Peoples, women, youth, and rural communities in northern Ghana. Guided by the GCF Indigenous Peoples Policy, an Indigenous Peoples Planning Framework (IPPF) has been developed to ensure meaningful engagement throughout the project lifecycle. This includes mapping indigenous communities, respecting customary decision-making processes, and preparing site-specific Indigenous Peoples Plans (IPPs) to align project interventions with community priorities. These processes ensure that indigenous people's rights, perspectives, and development opportunities are fully integrated and safeguarded.

28. The project will strengthen livelihoods for approximately 142,000 farming households by increasing agricultural productivity, improving post-harvest processing, and promoting value-added activities. This will enhance rural incomes, reduce food losses, and diversify market opportunities. Women, who make up at least 50% of beneficiaries, will be empowered through targeted support for alternative livelihoods such as shea butter production, beekeeping, and small livestock rearing; helping to shift gender roles and improve household resilience. Youth and women will also be involved in the dissemination of early warning information, reinforcing their leadership in climate adaptation. Collectively, these interventions contribute to inclusive

⁵ See funding proposal annex 8.

economic growth, poverty reduction, and more equitable participation in climate-resilient development.

29. Overall, the sustainable development potential is assessed as high.

1.5 Needs of the recipient

Scale: Medium to high

30. **Need for access to finance.** Financial constraints present a major obstacle to climate resilience for smallholder farmers in northern Ghana⁶. Adaptation technologies like drought-resistant seeds and irrigation systems exist are costly from a farmer's perspective thus leading to low adoption. Agricultural financing could bridge this gap, but access remains severely constrained. Local financial institutions (LFIs) typically avoid lending to individual smallholder farmers due to four interrelated challenges, i.e., (i) low financial literacy which limits farmers' ability to engage with formal banking systems; (ii) absence of credit histories which creates uncertainty for risk-averse lenders; (iii) collateral shortages which prevent farmers from meeting traditional loan requirements; and (iv) climate sensitive activities such as rain-fed agriculture increase perceived repayment risks. This financial exclusion perpetuates a cycle of vulnerability. Output 4 specifically addresses these barriers by developing innovative financing mechanisms tailored to smallholder farmers' unique circumstances, thereby enabling investment in critical climate adaptation measures. The initiative recognizes that without financial solutions that account for these realities, even the most effective adaptation technologies will remain out of reach for those who need them most. Project component 2 output 4 will address this. (See footnote 2).

31. **Urgent need to reduce climate impacts on farmers' livelihoods in Northern Ghana.** Climate change is significantly undermining the livelihoods of smallholder farmers in Northern Ghana, where most households depend on traditional, rainfed agriculture. Erratic and shorter rainy seasons, rising temperatures, and prolonged dry spells have reduced crop productivity, increased post-harvest losses, and worsened food insecurity. Limited irrigation options hinder dry-season farming, while extreme weather events such as floods and droughts damage crops, livestock, and infrastructure. Degraded ecosystems, frequent bushfires, and dwindling water sources further reduce land productivity. With approximately 37% of Ghana's labour force in agriculture, the declining yields directly translate to income instability, food shortages, and rising malnutrition. Climate-vulnerable regions like Upper West, Upper East, and North East face poverty rates exceeding 70%, reinforcing regional inequalities. Crop failure and livestock loss trap households in cycles of debt and deepen rural poverty, highlighting the urgent need for targeted climate adaptation interventions. Project component 2 on climate resilient livelihoods will address these. (See footnote 2)

32. **Need for strengthened CIEWS to reduce vulnerability.** The increasing frequency and intensity of climate-related hazards such as droughts, floods, and extreme rainfall in northern Ghana highlight the urgent need for CIEWS. Traditional knowledge is no longer sufficient to anticipate these rapidly changing climate risks, leaving smallholder farmers and vulnerable communities unprepared. Many rural areas lack access to timely, accurate, and locally relevant climate information, limiting their ability to make informed decisions on crop planning, livestock management, and disaster response. Strengthening early warning systems by integrating meteorological and hydrological data with community-based dissemination channels will enhance preparedness, enable early action, and reduce losses. Ensuring that warnings are accessible to women, youth, and marginalized groups is critical to building inclusive resilience and reducing long-term vulnerability across the region. Project component 1 will address these. (See footnote 2)

⁶ See funding proposal section B.2

33. **Need for ecosystem-based adaptation (EbA) as a critical response to climate risks.**

The rising frequency and intensity of climate linked hazards (e.g., droughts, floods, and erratic rainfall) in Northern Ghana calls for urgent and integrated responses that go beyond conventional risk management. Smallholder farmers and vulnerable communities struggling with limited access to reliable climate information face mounting challenges in anticipating and responding to these climate shocks. Strengthening CIEWS is essential, but it must be paired with EbA to address the root causes of vulnerability. EbA offers nature-based solutions that restore degraded landscapes, enhance water retention, improve soil health, and provide natural buffers against climate extremes. EbA interventions not only mitigate the impacts of hazards but also sustain agricultural productivity and livelihoods. Combined with accessible, community-based early warning systems especially those inclusive of women, youth, and marginalized groups, EbA options becomes a powerful tool for enhancing preparedness, reducing long-term vulnerability, and building climate-resilient rural economies. Among the EbA options considered are as follows, river bank restoration, agroforestry, reforestation, flood risk management and fire risk reduction. Project component 2 output 3 will address this.⁷

34. **Needs of displaced populations due to impacts of climate.** Individuals and households affected by flood events typically migrate to urban areas in search of alternate livelihoods, usually in non-climate sensitive sectors such as informal trading. Specifically, people from the north migrate to the south to stay with close relations only for a period and then move back. During this period some may change their livelihoods from farming to the service sector and then move back. Project components 1 and 2 collectively address this.

1.5.1. **Risk to needs of recipients**

35. **Gaps in digital finance integration.** Although access to climate finance issues are addressed under Project Component 2, Output 4, the potential impact remains constrained by the project's limited consideration of digital advancements in agricultural finance.

36. Overall, the needs of the recipients is assessed as medium to high.

1.6 Country ownership

Scale: Medium to high

37. **NOL secured.** UNEP has secured from the National Designated Authority on 8th of April 2025, signed by the duly authorized representative of the Ministry of Finance, the National Designated Authority/ Focal Point of the Republic of Ghana.⁸

38. **Alignment with Ghana development and climate priorities.** The project is well-aligned with Ghana's national climate and development priorities, particularly the National Climate Change Policy (NCCP), the National Climate Change Adaptation Strategy (NCCAS), the Nationally Determined Contributions (NDCs), and the forthcoming National Adaptation Plan (NAP). The funding proposal supports key adaptation priorities identified in these frameworks, including resilient livelihoods, sustainable agriculture, land and water resource management, early warning systems, and decentralized climate governance. The project contributes directly to Ghana's 2021 NDC adaptation actions by enhancing climate services, strengthening early warning and disaster risk management systems, and promoting climate-resilient livelihoods for women and youth in vulnerable agro-ecological zones. These priorities are expected to remain central in the upcoming NDC 3.0 (2025). The project complements nature-based adaptation measures under the agriculture sector outlined in the draft NAP, being finalized under the GCF Readiness programme led by the Environment Protection Agency (EPA) and UNEP. The project reflects strong government ownership and coherence with national planning processes that

⁷ See funding proposal section B.3

⁸ See funding proposal annex 1.

have been developed through extensive consultations across national, regional, district, and community levels.⁹

39. **Demonstrated track record and core expertise of the implementing partners in Ghana.** As Executing Entity, the EPA will implement the project in partnership with the Ministry of Food and Agriculture (MoFA) and the GMet both of which have established records in delivering large-scale climate and environmental programs in their respective fields of expertise. The EPA, under the Ministry of Environment, Science, Technology and Innovation (MESTI), serves as the national focal point for the UNFCCC and has led landmark climate and environmental initiatives, including the Ghana Environmental Management Project (GEMP), the Sustainable Land and Water Management Project (SLWMP), and the GCF-financed National Adaptation Plan (NAP). EPA brings expertise in land restoration, climate-resilient agriculture, and EbA, and will coordinate implementation in collaboration with MoFA, the GMet, and the Water Resources Commission (WRC). MoFA will provide technical support to FBOs, promote access to finance, and train extension officers in climate-smart agriculture. GMet, with its technical expertise in climate forecasting and drought monitoring, will lead improvements in early warning systems under Output 1, while WRC will integrate hydrological data to strengthen climate risk assessments. These agencies have established effective coordination mechanisms and governance structures through prior multi-sectoral projects, ensuring their readiness to deliver targeted interventions and achieve project outcomes at scale. MESTI will chair the Project Steering Committee (PSC), ensuring strategic oversight and alignment with national climate and development priorities.¹⁰

1.6.1. Risk to county ownership

40. **Complex implementation.** The implementation arrangements between and amongst the Executing Entities and Implementing Entities show a complex coordination set up which risks potential delays, miscommunication, inefficiencies, and ineffective execution of project activities and achievement of project deliverables and outcomes. Absent copies of the Memorandum of Understanding (MoU), or agreements, iTAP raised these issues with the AE that the MoU appears as a “best efforts” arrangement compared to the “agreements” (i.e., Project Cooperation Agreement, grant agreements/ contracts)¹¹ which are likely to be more solid legal documentations. In response, the AE stated that “As part of updating the FP (Figures 5 and 6 and associated text) and the term Sheet (Annex 14), and following consultations with EPA Head of Legal Department, and whilst MOUs can be stated as legally binding, MOUs will be replaced with legal binding Implementation Agreements between EPA and GMET, EPA and MOFA, and GMET and WRC. The legally binding Implementation Agreements will specify roles, responsibilities, timelines, deliverables and financial commitments.”¹²

41. **Limited Capacity of EPA to Implement Output 4 and Recommended Measures.** While the EPA has strong credentials in environmental governance and climate policy implementation, it has limited expertise in financial systems, and this presents a significant risk to the effective delivery of Output 4: “Increased access of smallholder farmers to financial resources and engagement with the private sector.” EPA lacks institutional experience in key areas such as financial product design, blended finance structuring, credit risk assessment, and financial literacy programming. Specific gaps include developing credit scoring systems and actuarial models, supporting Business Advisory Centres (BACs) and LFIs, and facilitating the establishment and regulation of FBOs and VSLAs. These capacity constraints may compromise the technical soundness and financial viability of interventions and limit EPA’s ability to

⁹ See funding proposal section B.4.

¹⁰ See funding proposal section B.4 and AE responses to iTAP questions submitted on 12 May 2025.

¹¹ See funding proposal figure 5.

¹² See AE responses to iTAP question # 27 dated 12 May 2025.

coordinate complex, multi-stakeholder engagement across rural areas where institutional presence is often weak. EPA's limited experience in data governance and digital infrastructure may affect the creation and management of secure, reliable databases needed for credit profiling, risk-based pricing, and loan processing. Lastly, EPA needs support for the technical capability to deliver financial literacy training and facilitate access to markets functions that are critical for Output 4's success. As mentioned above, MoFA will provide technical support to FBOs, promote access to finance, and train extension officers in climate-smart agriculture; however, the agreement between EPA and MOFA is based on an MOU; MOFA capacity for output 4 also remains to be seen given finance is outside their core expertise. Other organizations mentioned expressing interest, collaboration, support, etc. should be secured

42. **Recommendations.** To address these gaps, the following measures are recommended:
- (a) **Establish strategic partnerships with financial sector institutions.** EPA should partner with fit-for-need organizations (e.g., Ecobank¹³, ARB Apex Bank, GIRSA, microfinance networks, VSLA-experienced NGOs (e.g., CARE), and development finance institutions (e.g., AfDB). These partnerships must be properly formalized prior to implementation through technical cooperation agreements focused on product development, capacity building, and finance delivery mechanisms.
 - (b) **Delegate financial functions as appropriate.** EPA should identify strategic partners or explore subcontracting fit-for-need experts, or co-implement technical aspects of Output 4 with relevant financial/ sectoral specialists, particularly for designing credit scoring and actuarial tools, conducting portfolio risk assessments, and delivering targeted technical assistance to BACs and LFIs.
 - (c) **Embed financial experts within EPA's project team.** Recruit financial inclusion experts and a dedicated blended finance advisor to enhance internal capacity to manage Output 4, oversee coordination, and ensure financial product design is contextually appropriate and operationally viable.
 - (d) **Invest in data and IT infrastructure.** Secure, cloud-based or mobile-accessible platforms should be developed to support real-time data management and credit profiling, while ensuring adherence to data privacy and security standards.
 - (e) **Pilot test before scaling.** A pilot phase in select districts should be conducted to validate training content, financial delivery models, and institutional coordination mechanisms before rolling out to all 120 communities. This will allow for iterative learning and adaptive management.
43. Country ownership is assessed as medium to high.

1.7 Efficiency and effectiveness

Scale: Medium

44. The proposed project enhances climate resilience through Nature-Based Infrastructure (NBI) and Hybrid (combined nature-based and conventional) interventions. These include reforestation, climate-smart agriculture, ecosystem restoration, and improved water storage/irrigation systems. The project's effectiveness is evaluated under three Shared Socioeconomic Pathways (SSPs)¹⁴ (i.e., SSP1, SSP3, and SSP5) to assess economic feasibility and resilience benefits.

¹³ See footnote 8

¹⁴ The Shared Socioeconomic Pathways (SSPs) framework outlines five global development trajectories used in climate modeling. Three key scenarios include SSP1 which is characterized by equitable growth, rapid

45. **Impact of Climate Scenarios.** Economic analysis demonstrates that NBI and Hybrid interventions significantly reduce climate risks. For example, in Lambussie, flood damages to agriculture under SSP3 decline from USD 67.2 million to USD 60.7 million by 2050, while generating additional income (USD 0.27 million from tree planting and USD 2 million from new agricultural practices). The interventions also mitigate costs from floods (buildings and crops) and droughts, underscoring their dual role in risk reduction and livelihood improvement.
46. **Economic and Financial Viability.** Benefit-Cost Ratios (BCR). Discounted BCRs range from 1.55–2.95 across districts (Lambussie: 1.55–2.36; Lawra: 1.63–2.59; Jirapa: 1.65–2.62; WA West: 2.05–2.95), confirming strong economic viability, cost-effectiveness, better resource allocation, and resilience to uncertainties (e.g., climate shocks), therefore outweighs the costs.
47. **Net Present Value (NPV).** The NBI scenario yields NPVs of USD 22.4 million (under SSP1), USD 18.4 million (SSP3), and USD 23.5 million (SSP5). The Hybrid scenario performs even better, with NPVs of USD 68.8 million (SSP1), USD 58.8 million (SSP3), and USD 68.5 million (SSP5). The results demonstrate consistent positive returns confirming the projects' economic viability regardless of climate future. The Hybrid approach generates 2-3 times greater economic value than NBI alone, highlighting the benefits of integrated solutions. While SSP3 shows slightly reduced NPVs, the returns remain robust and demonstrates that the interventions' effectiveness even under adverse conditions. The Hybrid scenario under SSP1 yields the highest NPV (USD 68.8M), suggesting optimal outcomes when sustainable development aligns with integrated infrastructure.
48. **Internal Rate of Return (IRR).** The IRR ranges overall between 17.4% and 37.9%; this range highlights the project's potential to deliver robust financial and socio-economic benefits, especially in climate-vulnerable communities.
49. **Financing and leverage.** With a GCF grant of USD 63.2 million, the Government of Ghana commits USD 6.99 million in co-financing (USD 250,000 cash; USD 6.74 million in-kind). The resulting leverage ratio of 0.11 reflects Ghana's financing constraints while demonstrating stakeholder engagement.
50. The project is economically viable across all climate scenarios, with Hybrid solutions offering the highest returns. By integrating nature-based and conventional infrastructure, it not only mitigates climate impacts but also generates co-benefits for communities and ecosystems.

1.7.1. Risk to effectiveness and efficiency

51. **Limitations of the exit strategy and the need for strengthened financial access and institutional support.** The project's exit strategy presents a comprehensive framework for sustaining climate adaptation through capacity building, community engagement, financial inclusion, and knowledge sharing.¹⁵ However, several risks could undermine its long-term effectiveness if not proactively addressed. District Authorities are expected to lead future implementation but the risk of limited fiscal resources and competing local priorities may hinder the mainstreaming of climate actions into development planning. Similarly, training and alternative livelihoods will be introduced but their sustainability depends on access to markets and continuous financing. Community engagement among women and youth may vary in depth and consistency thus risking weak ownership of interventions. Efforts to improve financial access through FBOs and VSLAs are crucial but may be insufficient without reliable long-term

decarbonization, and strong international cooperation, this pathway aligns with low-emission futures (e.g., RCP1.9 or RCP2.6), limiting warming to 1.5–2°C by 2100. SSP3 features geopolitical fragmentation, weak institutions, and slow technological progress, leading to high emissions (typically RCP7.0) and ~4°C warming by 2100. SSP5 assumes rapid growth reliant on fossil fuels, with delayed climate action resulting in very high emissions (RCP8.5) and 4–5°C warming. See IPCC AR6 (2021) and Riahi et al. (*Global Environmental Change*, 2017) for deeper analysis.

¹⁵ See funding proposal section B.6.

funding sources. The knowledge-sharing component also faces limitations in reach, participation, and the ability to drive autonomous replication. Strengthening access to finance both during and beyond the project period is essential to sustain and scale these interventions. This requires building stronger linkages with financial institutions, de-risking private sector participation, and establishing mechanisms to mobilize additional public and private capital for continued investment in climate-resilient livelihoods and systems. The identified risk to country ownership (see paragraph 31 i and ii) will compound these risks.

52. **Outdated baseline and limited integration of latest technologies in agriculture.**

The project's effectiveness and efficiency are at risk due to a dated baseline assessment that fails to reflect significant recent developments in Ghana's agricultural landscape particularly the rapid adoption of digital technologies in the agriculture and agri finance sector. While use of latest technologies is considered under Component 1, and under the monitoring and evaluation plan¹⁶, project component 2 of the funding proposal does not adequately capture the growing role of digital tools such as AI and emerging technologies for national development; among them are the digital agriculture tools such as mobile-based extension services (e.g., Esoko, Farmerline), digital financial platforms (Growforme), digital insurance, GIS-based monitoring systems. Despite Ghana's progress as highlighted in the Ghana Digital Agriculture Strategy (2021) and multiple donor-supported initiatives, the project design does not systematically these advancements. This oversight weakens Component 2's optimal design.

53. **Recommendation.** The AE should revisit and update the feasibility study to align with Ghana's rapidly evolving landscape in climate-resilient agriculture and digital finance infrastructure. Given the accelerated adoption of AI, mobile-based financial tools, and digital extension services, leveraging the most current data and technological developments is crucial to ensuring the project's relevance and long-term impact. Without considering these advancements such as predictive analytics for climate adaptation, mobile money interoperability, and smart farming solutions, the project risks being outdated at inception, lagging behind national progress, and most of all, potentially weakening Component 2 output 4 - Increased access of smallholder farmers to financial resources and engagement with the private sector. This update will ensure interventions are fit-for-purpose forward-looking, context responsive, and positioned to drive transformational change in Ghana's agri-finance sector.

54. Given the above, efficiency and effectiveness are rated as medium.

II. Overall remarks from the independent Technical Advisory Panel

55. The iTAP recommends this funding proposal for approval by the Board subject to the following recommendations and covenant.

56. While supporting the proposal, the iTAP notes several risks that may affect the efficiency, effectiveness and long-term sustainability of the project, as mentioned in paragraphs 23 (Gaps in digital innovation integration), 35 (Gaps in digital finance integration), 40 (Complex implementation), 41 (Limited capacity of EPA to implement output 4 and recommended measures), 51 (Limitations of the exit strategy and the need for strengthened financial access and institutional support) and 52 (Outdated baseline and limited integration of latest technologies in agriculture) of this assessment.

¹⁶ Use of Digital Technologies in M&E. The monitoring and evaluation plan includes the use of digital photos and videos for visual inspections, as well as the most recent satellite imagery at a reasonable pixel resolution of the project area overlaid on topographical base maps. The project will integrate satellite data (Landsat/Sentinel) and drone imagery into Ghana's GIS-based monitoring system developed by the Technical Partner. This GIS mapping of land cover change will be used for independent change analysis reports. See funding proposal annex 11 monitoring and evaluation plan.

57. **Recommendation 1: strengthen EPA's capacity and execution strategy for Output 4.** To mitigate risks associated with complex implementation arrangements and limited in-house technical expertise, the iTAP recommends the following measures be undertaken prior to or during early implementation:

- (a) **Establish formal partnerships** with financial and development institutions (e.g. , African Development Bank ARB Apex Bank, Ecobank, GIRSAL) through technical cooperation agreements focused on product development, delivery mechanisms and capacity-building;
- (b) **Delegate technical functions** where appropriate by partnering with or subcontracting qualified sector specialists to co-implement financial tools such as credit scoring systems and portfolio risk assessment methodologies;
- (c) **Embed financial expertise** within the project team, including recruitment of a financial inclusion expert and a blended finance adviser to support implementation of output 4;
- (d) **Develop secure digital infrastructure** to support data-driven credit profiling, real-time performance tracking and institutional reporting; and
- (e) **Pilot the approach** in selected districts to test delivery models and coordination mechanisms before national scale-up to all 120 target communities.

58. **Recommendation 2: update feasibility and baseline assessments.** To ensure that project interventions are evidence based, context responsive and aligned with recent developments in Ghana's agricultural and financial sectors, the iTAP recommends that the AE:

- (a) **Update the project feasibility study and baseline assessment** to reflect post-COVID-19 economic dynamics, climate-resilient agricultural practices, digital finance infrastructure and community-level vulnerability data; and
- (b) **Ensure that revised baseline data incorporates disaggregated indicators** for components 2 and 4 (particularly in relation to financial inclusion) and local adaptation capacity.

59. The measures in recommendation 2 are critical to ensure that the interventions remain relevant and fit for purpose at inception, particularly as technological and market conditions have significantly evolved since the proposal's initial design.

60. **Covenant:** In respect of the Implementation Agreements, the Accredited Entity shall:

61. Contractually require the Executing Entities that the Implementation Agreements with the Implementation Partners contain the following:

- (a) Clearly defined institutional roles, performance milestones, deliverables, and financial commitments; and
- (b) Enforcement clauses covering accountability, reporting obligations, and remedial action in case of underperformance.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP265)

Proposal name:	Climate-resilient landscapes for sustainable livelihoods in northern Ghana
Accredited entity:	UNEP
Country/(ies):	Ghana
Project/programme size:	Medium

Impact potential

We value and affirm iTAP's assessment of the project's potential impact as high. We note that no specific recommendations are made.

Paradigm shift potential

We note iTAP's conclusion that the paradigm shift potential is medium to high owing to further potential to integrate digital finance infrastructure for the agriculture and finance sector. We reaffirm the intention to support expansion of Ghana's fast developing digital infrastructure in the project's 8 northern districts. We sought guidance from Finance Access, a social enterprise specialized in rural financial services with headquarters in the Netherlands and an office in Ghana, in the project design. The intention is to support development of a digital tool for credit scoring systems which we will explore extending to include portfolio risk assessment.

Sustainable development potential

We value and affirm iTAP's assessment of the project's sustainable development potential as high. We note that no specific recommendations are made.

Needs of the recipient

We note iTAP's conclusion that the need of the recipients is assessed as medium to high owing to the project's limited consideration of digital advancements in agricultural finance. The project will address this risk as explained in our response concerning paradigm shift potential and in addressing the iTAP recommendations in the section "overall remarks".

Country ownership

We note iTAP's conclusion that country ownership is assessed as medium to high owing to the complex implementation arrangements and limited capacity of EPA to implement Output 4. We accept the recommendations made to strengthening legally binding agreements, establishing strategic partnership with financial sector institutions as envisaged, ensuring

delegated financial functions under output 4, and to secure financial experts within the project team, and invest in data and IT infrastructure, and pilot test before scaling up in the 8 districts.

Efficiency and effectiveness

We note iTAP's conclusion that the effectiveness and efficiency assessment is medium due to lack of details on financial expertise to deliver on access to credit markets by smallholder farmers. We have added clarifications in the Funding Proposal, that were in Annex 2, that will engage two partners to support the implementation of the workplan: GIRSA (Ghana Incentive-Based Risk-Sharing System for Agricultural Lending) and an entity specialized in building credit scores that can bring to bear the latest in AI-driven analytical capability. Similarly, the project will update the project's feasibility study when the baseline study is developed in the inception stage, factoring in the recent development in Ghana's agricultural landscape including adoption of digital technologies for agriculture and agriculture finance.

Overall remarks from the independent Technical Advisory Panel:

We take note of iTAP's recommendations and commit to undertake the following within the parameters of the submitted FP package:

Recommendation 1: Strengthen EPA's capacity and execution strategy for Output 4:

We have added clarifications in the FP that will engage two partners to support the implementation of the workplan: GIRSA (Ghana Incentive-Based Risk-Sharing System for Agricultural Lending) and an entity specialized in building credit scores.

The AE will also explore with EPA and MOFA to second financial inclusion and blended finance expertise from a government institution (e.g. MOFA, Min. of Finance, GIRSA, Development Bank Ghana) potentially as part of Government cofinancing and to establish a Working Group on Finance with members from FIs, agricultural finance programmes, Government institutions (e.g. MOFA, Min of Finance, GIRSA, Development bank Ghana, etc.) and independent experts to exchange knowledge and experience related to financial inclusion, digitalization and blended finance.

Recommendation 2: Update feasibility and baseline assessments and ensure that the revised baseline data incorporates disaggregated indicators.

This will be undertaken as part of the baseline survey planned during the project's six-month inception period (See E3 and budget line M&E1).

We agree to the proposed Covenant in respect of the Implementation Agreements, where the Accredited Entity shall a) clearly define institutional roles, performance milestones, deliverables and financial commitments, and b) enforcement clauses covering accountability, reporting obligations, and remedial action in case of underperformance.

CLIMATE-RESILIENT LANDSCAPES FOR SUSTAINABLE LIVELIHOODS IN NORTHERN GHANA

ANNEX 8

GENDER ASSESSMENT AND ACTION PLAN (GAAP)

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Part one: Gender assessment

1. Introduction

The overall objective of the 'Climate Resilient Landscapes for Sustainable Livelihoods' project is to contribute to improved resilience of local small-holder farmers and vulnerable communities in the Upper East, Upper West and Northern regions of Ghana.

Experience and evidence indicate that women, youth, children and disabled individuals are among the most vulnerable to the impacts of climate change, and that a large proportion of vulnerable people who are highly dependent on local natural resources for their survival are women and children. The Government of Ghana (GOG) has therefore established that gender-responsive planning, resource-allocation and implementation in the context of the proposed project will be a priority. Supporting transformational change both in the lives of female beneficiaries and in prompting gender mainstreaming in national climate change policymaking and implementation was therefore considered a priority during stakeholder consultations.

To fulfil the project's design requirements, this Gender Assessment (GA) was undertaken in accordance with GCF guidelines to inform the project design on gender barriers, constraints and opportunities for women to benefit from the project's activities. In addition, the Gender Action Plan (GAP) will ensure women, Indigenous Peoples (IPs) and other marginalised groups are included in the project's activities. The GAP is based on the proposed project's logical framework and is focused on avoiding and minimising the negative impacts of the project's activities on the local communities and maximising the positive impacts and social benefits, particularly on women, IPs and marginalised groups.

2. Methodology

This GA primarily used qualitative data collection methods, including informant interviews with multiple stakeholders and partners (See Annex 7h: stakeholder engagement plan) as well as mixed community consultations in seven local communities across four districts, namely Jirapa, Lawra, East Mamprusi and Yunyoo-Nasuan (Figure 1). The method involving open community meetings proved particularly effective as it allowed for the provision of information by individuals through open discussions, accompanied by immediate expressions of approval or disapproval and corroborated by testimonies or cases from other group members. This facilitated an accurate representation of the gender-related challenges faced by their respective communities. The GA has also made use of: i) a desktop review of available literature; ii) gender equality studies conducted by research institutions and donor agencies; and iii) details made available by the GOG.

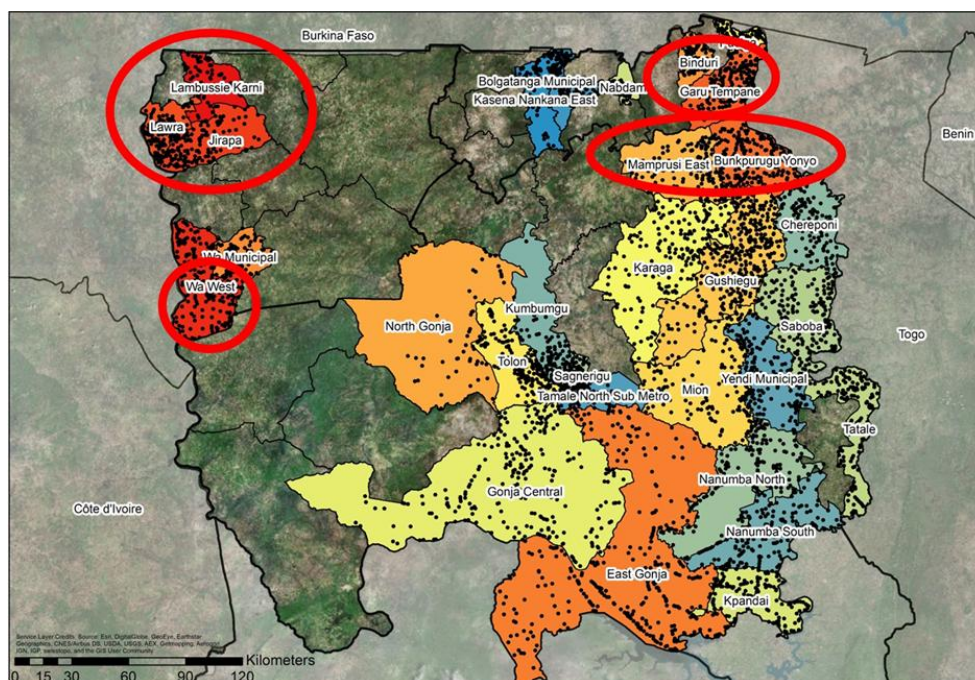


Figure 1. Map of the project sites and consultation locations.

To elicit informed and relevant responses from community members, a brief overview of the project's components, objectives and anticipated outcomes was shared with participants prior to engaging in discussions. The questions posed to the community members focused on the lived experiences of individuals and communities regarding the effects of climate change and their gendered responses to these challenges. This data collection method was preceded and supplemented by a review of existing literature on gender dynamics and equality in Ghana — particularly within the rural and agricultural sectors — to establish a framework aligned with the project's objectives and anticipated impact. The findings of the GA form the basis of the GAP (part two).

3. The Northern Ghana Context

The three northern regions of Ghana — the Upper East, Upper West and Northern regions — are threatened by harsh agro-climatic conditions, resulting in limited agricultural outputs per capita. Communities in these regions have limited livelihood options besides smallholder farming, with less service provision and urbanisation than in the south of the country. Approximately 70% of the ~4 million people living in northern Ghana depend on traditional, small-scale¹, rainfed agricultural systems to generate household incomes and maintain food security². These agricultural systems are vulnerable to changes in rainfall patterns and extreme climate events.

Furthermore, a large proportion of northern Ghanaians rely on climate-dependent ecosystem goods produced in the agro-ecological landscapes in which their livelihoods are embedded. This reliance on climate-sensitive livelihoods makes rural communities in Ghana vulnerable to the effects of current and future climate change. Ghanaian women are considered particularly vulnerable to climate change because of their limited access to education, land resources and

¹ Farms < 2 hectares in size.

² Ministry of Food and Agriculture – 2012 – Agriculture in Ghana: Facts and Figures.

credit³. However, given that they account for approximately half the population, women play a substantial role in socio-economic development. Therefore, by building the capacity of women to adapt, the project will substantially improve the overall resilience of rural communities in northern Ghana.

This GA examines the gender situation in Ghana, focusing on women from rural communities within the Upper East, Upper West and Northern regions. The objectives of the GA are to: i) identify relevant cultural and policy matters that contribute to gender inequality in Ghana; and ii) determine potential opportunities for addressing gender inequality through project interventions. Addressing gender inequality is necessary to addressing climate change, because of the substantial role women play in the adoption of climate change adaptation practices. Lessons learned from the implementation and evaluation of similar projects in the region have also been incorporated into the GA. The findings have been integrated into a GAP to ensure adequate consideration of gender issues in the design of project activities, targets, indicators, monitoring and evaluation protocols.

3.1. *Existing gender inequality in northern Ghana*

Ghana has a long history of commitment to gender equality, dating back over 50 years to the establishment of the Affirmative Action Act — shortly after achieving independence in 1957. Gender equality is further embedded in Ghana's constitution, which states that “all persons shall be equal before the law” and “a person shall not be discriminated against on the grounds of gender, race, colour, ethnic origin, religion, creed, social or economic status”. More recently, Ghana has highlighted the importance of empowering women and promoting gender equality as part of their commitment to the Sustainable Development Goals (SDGs). Despite these efforts to ensure gender equality, women in the northern regions of Ghana still face several gender-specific barriers limiting the capacity of communities to adapt to climate change.

The ability of men and women to adapt to climate change is dependent on their: i) involvement in policy development and decision-making; ii) integration into local committees that are focused on climate change adaptation; and iii) capacity to implement Ecosystem-based Adaptation (EbA) practices⁴. Building the capacity of both men and women will lead to increased adaptive capacity at the household and community levels.

Barriers to achieving gender equality and building the adaptive capacity of women in Ghana include, *inter alia*: i) strong patriarchal family structures; ii) limited influence in decision making; iii) increased out-migration of men leaving women with an increased labour burden⁵; iv) greater difficulty translating labour into secure incomes⁶; and v) limited financial independence. Approximately 68% of women in rural Ghana are employed in the agricultural sector⁷. However, less than 20% of northern households are headed by women⁸ and, therefore, the role of women in farming decision-making is limited. These challenges are compounded by a lack of policy

³ FAO. 2012. Gender Inequalities in Rural Employment in Ghana – Policy and Legislation. Gender, Equality and Rural Employment Division of FAO.

⁴ Gaye I. 2009. Gender and climate change: Women matter. UN Economic Commission for Africa.

⁵ Baden S, Green C, Otoo-Oyortey N & Peasgood T. 1994. Background paper on gender issues in Ghana. Bridge Development – Gender Report No 19.

⁶ FAO. 2012. Gender Inequalities in Rural Employment in Ghana – Policy and Legislation. Gender, Equality and Rural Employment Division of FAO.

⁷ Ghana Statistical Service. 2014. Ghana Living Standards Survey Round 6.

⁸ Ibid.

frameworks directed at integrating social and gender equality in rural labour markets⁹. This makes women particularly vulnerable to the impacts of climate change. Addressing these barriers to gender equality would lead to numerous advantages to rural communities, building their resilience. For example, through stakeholder consultations (see Annex 7h: Stakeholder Engagement Plan) it was noted that if women are provided access to fertile land, improved technologies and financial services, they are likely to invest in the land, leading to increased production — and potential increased income — which benefits not only themselves, but their families and communities too. Furthermore, women are most likely to invest in health and education which will improve the overall situation within households and communities.

The following sections detail the gender dynamics currently experienced in Ghana and their implications for building the adaptive capacity of women — and their communities — to the effects of climate change.

3.1.1. Gender Inequality Index

The Global Gender Gap Report¹⁰ published in 2024 ranks Ghana 88th out of 149 countries in terms of gender equality, with an overall parity score of 0.701 (Table 1). This is based on four criteria including: i) economic participation and opportunity; ii) education attainment; iii) health and survival; and iv) political empowerment. Of these criteria, Ghana ranks 30th in terms of economic participation and opportunity, with a parity score of 0.750. Political empowerment scores the lowest at only 0.101, ranking 119th.

Table 1. Gender Gap Report for Ghana¹¹

Focus Area	Parity Score	Ranking (out of 144)
Economic participation and opportunity	0.805	30
Education attainment	0.972	103
Health and survival	0.978	36
Political empowerment	0.101	119
Overall	0.701	88

3.2. *Poverty and employment*

There is a considerable disparity in depth and severity of poverty across Ghana, with greater poverty rates in the northern than southern regions. In addition to lower economic activity and development^{12,13}, northern Ghana underperforms relative to the south in terms of social development. Child mortality is relatively high in the north^{14,15}, while data on indicators relating to education, sanitation, water, health, security and governance suggest that the northern regions have experienced less development compared with the southern regions¹⁶. In addition to the north-south development gap, there is a notable gap in poverty levels between rural and urban

⁹ FAO. 2012. Gender Inequalities in Rural Employment in Ghana – Policy and Legislation. Gender, Equality and Rural Employment Division of FAO.

¹⁰ WEF. 2024. Global Gender Gap Report. Available at: https://www3.weforum.org/docs/WEF_GGGR_2024.pdf

¹¹ Ibid.

¹² Indicated by night light intensity.

¹³ Mellander C, Lobo J, Stolarick K, Matheson Z. 2015. Night-time light data: a good proxy measure for economic activity? Available at: <http://dx.doi.org/10.1371/journal.pone.0139779>.

¹⁴ Burke M, Heft-Neal S and Bendavid E. 2016. Understanding variation in child mortality across Sub-Saharan Africa: A spatial analysis. The Lancet Global Health, 2016, Volume 4, Issue 12, e936-e945.

¹⁵ ICF International (2004–2015) Demographic and Health Surveys (various) [Datasets]. Calverton, Maryland: ICF International [Distributor], 2015.

¹⁶ UNICEF. 2015. Ghana's District League Table 2015.

areas. Poverty levels in rural areas (38%) are far higher than those in urban areas (11%), with the gap growing over the past two decades¹⁷. There is also a gendered pattern between poverty levels and employment in Ghana. Although the unemployment rates in Ghana are only marginally higher for women than men (~6% compared to ~5%)¹⁸, women's employment is often concentrated in low-income activities¹⁹, including unpaid family labour and small household farms. As a result, women's control over their income is precarious and poverty rates among working women are generally higher than those of working men. This challenge is compounded by the gender wage gap of 0.75, with women earning 75 cents to the dollar compared with men²⁰. This ranks Ghana at 26th out of 144 countries in terms of the gender wage gap. However, in terms of overall economic participation and opportunity, Ghana ranks much higher at 30th place, with a gender gap index of 0.805.

3.3. *Access to and control over resources*

A major factor influencing the economic and social status of a group of people is their access to and control over assets. In rural communities such as those in northern Ghana, this often centres on the land tenure. Approximately 70% of the northern population of Ghana are reliant on small-scale subsistence farming for their livelihoods. Therefore, control over the natural resources and land-use practices strongly influences income potential, social status and power²¹. In Ghana, ~80% of the total land is controlled under a customary land tenure system, meaning that land is largely vested in male-centred kinship institutions and customary authorities²². The customary authority then grants community members access to customary lands. In the patriarchal culture, which dominates most of northern Ghana, the land-use rights are passed through the male lineage. Women's customary rights to access land are therefore restricted in favour of men's, despite women's rights to own and inherit land being protected under The Constitution²³. As of 2014, ~80 of women aged 15-49 did not own land²⁴. Furthermore, the social status of women and the stability of their marriage can influence their access to agricultural and natural resources from family-owned land, with divorcees and widows often facing discrimination when trying to access land²⁵.

3.4. *Education*

Access to education is fundamental to social and economic development. Ghana's Education Sector Strategic Plan (ESP 2010–2020) aims to provide equitable access to quality basic

¹⁷ Cooke E., Hague. & McKay A. 2016. The Ghana Poverty and Inequality Report. Available at: [https://www.unicef.org/ghana/Ghana_Poverty_and_Inequality_Analysis_FINAL_Match_2016\(1\).pdf](https://www.unicef.org/ghana/Ghana_Poverty_and_Inequality_Analysis_FINAL_Match_2016(1).pdf)

¹⁸ World Bank. Ghana – Gender Data Portal. Available at: <http://datatopics.worldbank.org/gender/country/ghana>

¹⁹ Heintz J. 2005. Employment, Poverty and Gender in Ghana. Practical Economy Research Institute. Working Paper Series 92.

²⁰ Schwab K, Samans R, Zahidi S, Leopold T.A, Ratcheva V, Hausmann R & Tyson L.D. 2016. The Global Gender Gap Report. World Economic Forum, Geneva.

²¹ Kuusaana E.D, Kidido J.K & Halidu-Adam E. 2013. Customary Land ownership and Gender Disparity – Evidence from the Wa Municipality of Ghana. GJDS 10(1&2). Pg 63-80.

²² USAID 2013. Ghana Commercial Agriculture Project. Available at: <https://www.land-links.org/country-profile/ghana/>

²³ Kuusaana E.D, Kidido J.K & Halidu-Adam E. 2013. Customary Land ownership and Gender Disparity – Evidence from the Wa Municipality of Ghana. GJDS 10(1&2). Pg 63-80.

²⁴ The World Bank Group. 2024. Ghana gender landscape. Available at: <https://documents1.worldbank.org/curated/en/099024506302242863/pdf/IDU05017dff90e8da047a70964d0d0b553ab2f02.pdf>

²⁵ Kuusaana E.D, Kidido J.K & Halidu-Adam E. 2013. Customary Land ownership and Gender Disparity – Evidence from the Wa Municipality of Ghana. GJDS 10(1&2). Pg 63-80.

education²⁶. Access to primary education is relatively high, with a 92% net enrolment ratio in 2016 and an equal gender ratio²⁷. Overall enrolment begins to decrease at the junior high school level — dropping to 50%²⁸ — and continues to decrease at the tertiary level, with the progression to senior high school declining to only 25%. Gender disparity is, however, still great among secondary education, with a ratio of 0.98. The Northern region has the highest level of gender disparity at the senior high school level, with 42% female enrolment. It is only at the tertiary education level that gender disparity is notable, dropping to a ratio of 0.67, ranking 122nd out of 144 countries²⁹.

3.5. Political participation

Women's political participation is considered a catalyst for gender equality and sustainable development in Ghana³⁰. Since achieving independence in 1957, Ghanaian women have been afforded equal rights to participate in political activities and decision-making at every level. Despite this, cultural structures and traditional practices involving male dominance and patriarchy still impede women's participation in decision making^{31,32}. Cultural perceptions of women being inferior to men have long been used to rationalise the differential roles between men and women in society³³. Consequently, women are underrepresented across all government structures. Currently, only 14.5% of parliamentary seats are held by women, with less than 30% of those holding ministerial or chief executive roles³⁴. Ghana, therefore, ranks 95th out of 144 countries in terms of political empowerment, with a parity score of 0.112³⁵. However, despite women's political representation following a positive trajectory — increasing by 0.4 over the past decade — women remain underrepresented at the highest levels of political participation³⁶.

As part of their commitment to the Sustainable Development Goals Agenda 2030, Ghanaian gender advocates have set forward a gender checklist to measure their progress towards achieving gender equality in decision-making. The checklist sets targets for women's representation across various sectors, including *inter alia*: i) 60% of parliamentary seats; ii) 60% of ministerial portfolios; iii) 60% of metropolitan, municipal and district Chief Executive posts; iv) 50% of vice chancellorships and university professorships; and v) 60% of state corporations Chief

²⁶ Ghana Ministry of Education. 2016. Education Sector Performance Report. Available at: http://www.moe.gov.gh/assets/media/docs/ESPR2016_Final_Version_Final.pdf

²⁷ Schwab K, Samans R, Zahidi S, Leopold T.A, Ratcheva V, Hausmann R, & Tyson L.D. 2016. The Global Gender Gap Report. World Economic Forum, Geneva.

²⁸ Ghana Ministry of Education. 2016. Education Sector Performance Report.

²⁹ Schwab K, Samans R, Zahidi S, Leopold T.A, Ratcheva V, Hausmann R, & Tyson L.D. 2016. The Global Gender Gap Report. World Economic Forum, Geneva.

³⁰ Asuako J. 2017. Women's political participation – a catalyst for gender equality and women empowerment in Ghana. UNDP Ghana.

³¹ Ibid.

³² FAO. 2012. Gender Inequalities in Rural Employment in Ghana – Policy and Legislation. Gender, Equality and Rural Employment Division of FAO.

³³ Asuako J. 2017. Women's political participation – a catalyst for gender equality and women empowerment in Ghana. UNDP Ghana.

³⁴ The World Bank Group. 2024. Ghana gender landscape. Available at: <https://documents1.worldbank.org/curated/en/099024506302242863/pdf/IDU05017dff90e8da047a70964d0d0b553ab2f02.pdf>

³⁵ Schwab K, Samans R, Zahidi S, Leopold T.A, Ratcheva V, Hausmann R, & Tyson L.D. 2016. The Global Gender Gap Report. World Economic Forum, Geneva.

³⁶ Ameyibor F. Agenda 2030: Women in Power and Decision-Making. Government of Ghana. Available at: www.ghana.gov.gh

Executive Officer positions^{37,38}. The Minister of Gender, Children and Social Protection has officially called on all state institutions and non-governmental organisations to support Agenda 2030.

3.6. *Gender and climate change*

The gender inequalities described above disproportionately affect the adaptive capacity of women to current and future climate change, as opposed to men. The effects of climate change on households and communities will be exacerbated if the adaptive capacity of both men and women are not strengthened in parallel. Overlooking either gender will affect production and income within a household, having knock-on effects on families, children and the community. Although traditional knowledge has established coping mechanisms within communities to the harsh climatic conditions experienced in northern Ghana, it is unlikely to adapt quickly enough to overcome the effects of future climate change³⁹.

Women — along with children and the elderly — are among the most vulnerable groups to the effects of climate change. This is largely because women and female-headed households have fewer resources to cope with reductions in food and water security and the effects of extreme weather events such as droughts and floods. This is particularly pertinent in rural communities, where people are often reliant on climate-sensitive natural resources for their livelihoods. Several factors contribute to the vulnerability of women to climate change including, *inter alia*, the effects of: i) migration; ii) water availability; iii) decreased agricultural yields; and iv) reliance on climate-sensitive livelihoods. These are described in more detail below.

- When climatic conditions become less favourable for agricultural practices, men often migrate out of rural areas in search of work in other sectors. Women are left behind to not only continue with their domestic duties but also to continue the agricultural work on their land. Much of this agricultural work comes in the form of subsistence farming, with little or no pay, leaving women without the financial resources to adapt to climate change.
- Women have limited rights to own or control land. Without ownership of the land they are working on, women are unable to make the necessary decisions or access the necessary funds to improve agricultural practices, leading to reduced agricultural productivity and food security. This affects not only the women themselves but also their households and potentially the wider community.
- Water scarcity associated with shifting rainfall patterns and drought can have a substantial impact on women, most of whom are responsible for the collection of water for household uses such as drinking, cooking and cleaning. When water security is threatened, extra effort is required for women to collect water — with women having to travel further to collect water. Similar pressures are observed with regards to the collection of fuelwood which will become scarce as climate conditions worsen. This compounds the additional workload placed on women maintaining both domestic duties and agricultural responsibilities, with little capacity to adapt to growing climatic pressures.
- Many rural Ghanaian women are reliant on natural resource-based livelihoods that are climate-sensitive. As the availability of natural resources becomes threatened by climate change, so do income generation and food security.

³⁷ Government of Ghana. Ghana women set dramatic 2030 gender equality target. Available at: www.ghana.gov.gh

³⁸ Ameyibor F. Agenda 2030: Women in Power and Decision-Making. Government of Ghana. Available at: www.ghana.gov.gh

³⁹ Dampney T.M. & Essel A.K. 2012. Gender perspectives of climate change coping and adaptive strategies in Ghana. UN Women.

The intersection of the vulnerabilities highlighted above further exacerbates gender inequality in Ghana. Rural women with disabilities face physical, social and institutional barriers — such as limited mobility, unstable income and stigma — that restrict their economic participation. Young women in informal labour are more likely to have lower levels of financial literacy, making them more likely to be under-paid, while gender norms combined with household hierarchy may constrain their ability to participate in training or capacity-building initiatives. These overlapping forms of marginalisation exacerbate existing gender inequalities, further restricting access to land, finance and decision-making opportunities. As climate change increasingly threatens natural resource-based livelihoods, the ability of women with intersecting vulnerabilities to maintain food security and support household well-being becomes increasingly strained.

3.7. Gender-based violence (GBV)

Despite being considerably underreported, rape and domestic violence are considered to be a widespread problem in Ghana⁴⁰, particularly when married women are thought to be unfaithful. As many as 27% of Ghanaian women experienced some form of domestic violence in 2016⁴¹. There is also evidence of women being targeted for not conforming to expected submissive roles. In cases of domestic violence and rape, Ghanaian women are often faced with obstacles when reporting and receiving justice for abuse. This is largely rooted in the cultural belief that domestic violence is a private matter that should be settled outside of the criminal justice system⁴². Magistrates may block women's attempts to pursue cases of domestic violence through the courts, advocating instead for family unity and encouraging settlements between the parties⁴³. Women face similar hurdles when reporting cases of sexual abuse, often being unable to complete formal complaints because of fees associated with physicians' documentation needed for police medical forms. Any instances of GBV or sexual exploitation and harassment (SEAH) during the project's implementation will be reported through the Domestic Violence and Victim Support Unit (DOVVSU) as elaborated in the project's grievance redress mechanism (GRM). For further details on this process, see Annex 6c: ESAMF.

4. Legal and policy framework for gender equality

Ghana has a long history of commitment to gender equality, dating back to its independence in 1957. The basic rights of women are outlined in the Constitution and have been incorporated into legislature and policy across all levels of government. The legal framework for gender equality is, however, complicated by the inclusion of customary laws which vary between regions and ethnicities.

Matters of gender equality currently fall under the portfolio of the Ministry of Gender, Children and Social Protection (MoGCSP), established in 2013 as a successor to the Ministry of Women and Children's Affairs. The mission of the MoGCSP is to “contribute to the development of the nation by achieving gender equality and equity, facilitate the enforcement of the rights of children and promote the integration and protection of the vulnerable, excluded and persons with disabilities in the development process through appropriate policies and strategies with adequate resources.” It is the ministry's responsibility to guide the formulation of gender, child and social protection-related policies — taking the lead role in the coordination, monitoring and evaluation of matters

⁴⁰ US DoS. 2016. Country Reports on Human Rights Practices for 2016 - Ghana.

⁴¹ Ibid.

⁴² Campbell J. 2016. Violence against women in Ghana: Unsafe in the second safest country in Africa. Council on Foreign Relations – Africa in Transition.

⁴³ Annex 7h: stakeholder engagement plan.

relating to these target areas. The primary gender-related policies and development frameworks are summarised below.

4.1. *National Gender Policy*

The National Gender Policy⁴⁴ (NGP) established in 2015 is central to Ghana's efforts to promote the empowerment of women and gender equality. Through the NGP, the Government of Ghana seeks to emphasise the objectives set out in their commitment to the SDGs and Ghana's National Development Frameworks (see Section 4.2). The overarching goal for the NGP is to mainstream gender equality and women's empowerment into Ghana's development efforts by improving social, legal, civic, political, economic and socio-cultural conditions for the people of Ghana. Through the empowerment of women, Ghana aims to: i) reduce poverty levels and social injustice; ii) improve health standards; and iii) enhance the efficiency of public and private sector investments. This will be achieved by focusing on five policy commitments, namely:

- women's empowerment and livelihoods;
- women's rights and access to justice;
- women's leadership and accountable governance;
- economic opportunities for women; and
- gender roles and relations

4.2. *Ghana's National Development Frameworks and the Sustainable Development Goals*

Social development forms one of the four pillars of national development in Ghana, along with economic, environmental and institutional development. The long-term National Development Frameworks (NDFs) guide the preparation and implementation of medium- and short-term development plans, setting targets and indicators for national development. The long-term framework is binding to all successive governments but flexible to allow for each government to prepare their own medium-term plans that align with the overarching vision, goals and objectives of the framework⁴⁵. Among the focal areas of the NDFs are the targets committed to by Ghana through the SDGs. These targets, as defined under Goal 5⁴⁶ of the SDGs, include:

- ending all forms of discrimination against women and girls;
- eliminating all forms of violence against women and girls;
- eliminating all harmful practices, including early forced marriage and female genital mutilation;
- recognising and valuing unpaid domestic work and care;
- promoting shared responsibility within the household;
- ensuring all women's full and effective participation and equal opportunities for leadership at all levels of decision-making in political, economic and public life;
- ensuring universal access to sexual and reproductive health and reproductive rights;
- undertaking reforms to give women equal rights to economic resources and control over land, financial services, inheritance and natural resources;
- enhancing the use of enabling technologies to promote empowerment of women; and
- adopting and strengthening sound policies and enforceable legislation for the promotion of gender equality and the empowerment of all women and girls at all levels.

⁴⁴ MoGCSP. 2015. National Gender Policy – Mainstreaming Gender Equality and Women's Empowerment into Ghana's Development Efforts.

⁴⁵ National Development Planning Commission. 2015. The basis for a long-term National Development Plan for Ghana. Available at: <https://s3.amazonaws.com/ndpc-static/CACHES/PUBLICATIONS/2016/03/24/TheBasisforaLong.pdf>

⁴⁶ Sustainable Development Goals. 2016. Goal 5: Achieve gender equality and empower all women and girls. Available at: <http://www.un.org/sustainabledevelopment/gender-equality/>

4.3. Shared Growth and Development Agenda.

Gender equality and women empowerment are listed as a focus area for the medium-term priorities in the Shared Growth and Development Agenda (GSGDA) II⁴⁷. The objectives of the policy are to:

- promote gender equality in political, social and economic development systems;
- safeguard the rights, safety and security of vulnerable groups including girls and women;
- promote women's equal access to economic opportunities and resources; and
- integrate gender considerations at all stages and in all dimensions of data production.

Through these objectives, the GSGDA II will address the slow progress made in solving gender inequality and increase the recognition of gender equality in the public sector.

4.4. International commitments

Ghana is a signatory to several regional, continental and international commitments aimed specifically at empowering women and promoting gender equality and equity. These commitments include the Protocol on the African Charter on Human and Peoples' (ACHPR) on the Rights of Women in Africa and the United Nations (UN) Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW). Further details on international and regional commitments on gender are provided in Table 2 below.

Table 2. International and regional agreements Ghana has ratified.

Commitment	Overview
2030 Agenda for Sustainable Development (2015) ⁴⁸	The agenda provides a shared framework aimed at eradicating poverty, combating inequalities and promoting prosperity by 2030. Of the 17 SDGs and targets, SDG 5 speaks specifically to achieving gender equality and empowering all women and girls by 2030. SDG 5 is further supported by 37 gender-related targets across nine other goals.
Beijing Declaration and Platform for Action (1995) ⁴⁹	The Beijing Declaration and Platform for Action serves as an agenda plan for the empowerment of women and provides appropriate actions required by, <i>inter alia</i> , multilateral development institutions, governments, intergovernmental organisations and public and private institutions, as well as communities and women's and youth organisations. The Platform for Action provides several strategic objectives targeted towards addressing, <i>inter alia</i> , gender inequalities in the access to education and healthcare, insufficient mechanisms to promote the advancement of women and gender inequalities in the management and protection of the natural environment and resources.
United Nations (UN) Convention on the Elimination of All Forms of Discrimination Against Women (CEDAW, 1979) ⁵⁰	CEDAW emphasises that discrimination against women: i) undermines the principles of equality of rights and respect for human dignity; ii) hinders the participation of women as equal to their male counterparts in the social, political, economic and cultural spheres of their countries; iii) inhibits the development and prosperity of societies; and iv) challenges the empowerment of women in the service of their countries and humanity. Therefore, CEDAW provides measures required by state parties to enable and ensure the elimination of all forms of

⁴⁷ Government of Ghana. 2015. Ghana Shared Growth and Development Agenda II. National Development Planning Commission.

⁴⁸ UN. 2022 History of the Sustainable Development Goals. Available at: <https://sdgs.un.org/goals#history>

⁴⁹ UN Women. 1995. Beijing Declaration and Platform for Action. Available at:

<https://www.icsspe.org/system/files/Beijing%20Declaration%20and%20Platform%20for%20Action.pdf>

⁵⁰ UN General Assembly. 1981. Convention on the Elimination of All Forms of Discrimination against Women. Available at: <https://www.ohchr.org/sites/default/files/Documents/ProfessionalInterest/cedaw.pdf>

Convention on the Rights of Persons with Disabilities ⁵¹ (CRPD, 2006)	discrimination against women, such as developing appropriate legislation to ensure equality of men and women, promoting and supporting women empowerment and modifying social and cultural patterns that perpetuate gender inequality. Ghana ratified the CRPD in 2012, a treaty aimed at promoting, protecting, and ensuring the full and equal enjoyment of all human rights by persons with disabilities. It emphasises principles such as dignity, non-discrimination, equality, accessibility and full participation in society. The convention covers various rights, including access to education, healthcare, employment and the justice system, while also promoting awareness, independent living and social inclusion. It calls on governments to implement legislative, administrative, and other measures to uphold these rights.
Protocol to the African Charter on Human and Peoples' (ACHPR) on the Rights of Women in Africa (The Maputo Protocol, 2003) ⁵²	This protocol was developed based on the recommendation of the African Commission on Human and Peoples' Rights to provide a protocol on women's rights in Africa. The objective of the protocol addresses, <i>inter alia</i> , the elimination of discrimination against women and harmful practices, women's rights to participate in political and decision-making processes, as well as their rights to education, training, economic and social welfare. The protocol ensures that the rights of African women are promoted, realised and protected, and further includes provisions.

5. Situational analysis — Northern Ghana target areas

5.1. Gender and agriculture

In the agricultural sector of Ghana, gender dynamics shape the responsibilities and contributions of men and women across numerous communities in northern Ghana. These dynamics are evident in the distinct division of labour, access to resources and the economic roles that men and women occupy within their respective communities. For example, in the Upper West Region, agricultural work in the Tuggoh community is divided along gender lines, with men taking on physically demanding tasks such as land preparation, while women focus on planting, sowing and compost production. Women also support men's work by collecting water and preparing food during land preparation activities. However, their involvement in certain types of crop cultivation, such as yams, is limited due to the physical demands of the task. Additionally, women are often responsible for selling agricultural produce at local markets, but they must first seek permission from their husbands or elders, highlighting the influence of gender norms on economic activities⁵³.

Similarly, in the Ping community, gender roles are clearly defined, with men leading in the cultivation of major crops such as maize and groundnuts and engaging in tasks such as pesticide application, which is reserved exclusively for them. This is primarily because of the belief that pesticides can bring harm to women's reproductive capacity. Nevertheless, both men and women participate equally in livestock farming, and women extend their involvement beyond agriculture into petty trading and shea butter processing. Despite these contributions, women face challenges

⁵¹ UN General Assembly. 2006. Convention on the Rights of Persons with Disabilities. Available at: <https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-rights-persons-disabilities>

⁵² African Union. 2003. Protocol to the African Charter on Human and People's Rights on the Rights of women in Africa. Available at: <https://www.ohchr.org/sites/default/files/Documents/Issues/Women/WG/ProtocolontheRightsofWomen.pdf>

⁵³ Annex 7h: stakeholder engagement plan.

in sustaining dry season gardening due to unreliable water sources, demonstrating how environmental factors can exacerbate existing gender inequalities in agriculture⁵⁴.

The challenge of land ownership further complicates gender dynamics in agriculture, as seen in the Bagri community. Historically, women have had limited ownership rights over farmland, restricting their ability to fully engage in agricultural activities. However, initiatives such as the Women in Agriculture Programme (WAP) have begun to address these disparities by enabling women to form farmer-based organisations (FBOs) and purchase land. While this has empowered women to take greater control over their agricultural activities, challenges remain, such as the reallocation of land that women have maintained and improved to others male smallholder farmers, which discourages long-term investment. These obstacles underscore the ongoing struggle for gender equality in the agricultural sector⁵⁵.

In communities like Sumniboma 1 in the Upper East region of Ghana, the division of labour between men and women reflects both collaboration and the persistence of gendered responsibilities in agriculture. While some tasks, such as ploughing, are performed by men, women take on cooking and planting, often working alongside men in the fields. Specific tasks, such as groundnut farming, are divided, with women overseeing the plucking and men digging up the plants. Despite these shared responsibilities, certain roles, such as pesticide application, remain reserved for men, although some women also participate, depending on their physical condition⁵⁶.

In Fulani herding communities, gendered responsibilities are defined by specific roles. For example, men primarily manage herding, migration in search for grazing land, and cattle trade, along with ensuring the protection of livestock. Women are responsible for processing dairy products, caring for young livestock and performing household tasks such as cooking and gathering firewood. Additionally, Fulani women often sell milk and other dairy products in local markets, contributing economically to the household. This division of labour aligns with cultural norms, where men focus on external tasks, while women manage domestic and economic activities within the household⁵⁷.

5.2. *Alternative livelihoods*

In rural communities across northern Ghana, alternative livelihoods are necessary for diversifying income sources and sustaining families, particularly during the dry-season when farming activities are limited. For example, the Bogni community relies on dry season farming as an alternative livelihood strategy. However, limited reliable water sources and electricity presents considerable challenges, forcing some residents to migrate to southern regions, cities, or mining sites in search of additional income. Despite these difficulties, women's traditional practices such as beekeeping, wild honey harvesting, and brewing Poitou⁵⁸ remain viable alternative livelihoods that supplement household incomes during the dry season⁵⁹.

Similarly, in the Sumniboma 1 community, a variety of alternative livelihoods support both men and women. For example, women participate in shea butter production, while men and women

⁵⁴ Ibid.

⁵⁵ Annex 7h: stakeholder engagement plan.

⁵⁶ Annex 7h: stakeholder engagement plan.

⁵⁷ Setrana, M. 2021. Citizenship, Indigeneity, and the Experiences of 1.5- and Second-Generation Fulani Herders in Ghana. *African Spectrum*.

⁵⁸ Poitou is a traditional beer brewed by the local populations in Ghana.

⁵⁹ Annex 7h: stakeholder engagement plan.

participate in livestock rearing. Charcoal production and firewood sales also provide income, with the community ensuring wood is sourced from private farming land rather than the community forests, which is reserved for gathering medicinal plants. Women also engage in food preparation, trading, and local fowl rearing, though they face barriers to expanding these activities due to limited capital. Dry season gardening is also practiced, but water access challenges makes it less viable⁶⁰.

In the Ping and Bagri communities, alternative livelihoods are similarly diverse. The proximity of the Volta River to the Bagri community makes fishing a predominant activity, while in the Ping community, women engage in petty trading and shea butter processing. Similar to other communities, dry season gardening is practiced, but it is hindered by unreliable water sources, leading to a reliance on boreholes and wells. Additionally, some young men in the Ping community participate in informal gold mining during the dry season. While this provides temporary income, it poses considerable safety risks and contributes to environmental degradation⁶¹.

5.3. *Access to and control over land*

Within this context, various communities in the Upper West, Upper East and Northern regions of Ghana exhibit differing levels of access to land for women, often shaped by local customs and initiatives aimed at improving equity. For example, in the Lawra district, women have begun to form farmer-based organisations (FBOs) through the Women in Agriculture Programme (WAP) and Rural Women in Agriculture (RWA) facilitated by each District Agricultural Office. These FBOs can purchase and manage land collectively, offering women greater control over their agricultural activities, although challenges remain, such as the reallocation of improved land to other community members. In the Zambo Baagangn community, women have access to farming land without specific restrictions on their agricultural activities but are still denied formal ownership as a result of the Customary Land tenure favouring patriarchal ownership (see Section 3.3: Access to and control over resources). Similarly, in the East Mamprusi district, women are heavily involved in the labour-intensive aspects of farming but lack control over land ownership, underscoring ongoing gender disparities associated with the Customary Land tenure systems. The Nyagu community presents another dimension to the land access challenge, where women, although not given formal ownership, have access to land because their farming activities benefit the household. Men in this community have recognised the higher productivity of women's farms and are therefore supportive of women's access to land. These examples highlight the complexities of land access in rural northern Ghana and the need for more equitable practices to ensure that women, who are substantial contributors to agriculture, have secure and fair access to land and its resources⁶².

5.4. *Access to finance*

Access to finance in the northern region of Ghana plays a substantial role in supporting household livelihoods, farming activities, and overall community well-being. Although only ~30% of women and ~50% of men in Ghana have bank accounts⁶³, Village Savings and Loan Associations (VSLAs) are a common financial mechanism for local communities in the Upper West, Upper East and Northern regions of Ghana that enables community members to save, borrow, and invest in

⁶⁰ Ibid.

⁶¹ Ibid.

⁶² Annex 7h: stakeholder engagement plan.

⁶³ The World Bank Group. 2024. Ghana gender landscape. Available at: <https://documents1.worldbank.org/curated/en/099024506302242863/pdf/IDU05017dff90e8da047a70964d0d0b553ab2f02.pdf>

a way that supports both individual and collective needs⁶⁴. These community-driven savings groups are necessary in areas where formal financial services are limited or inaccessible. By pooling resources and offering loans, VSLAs empower individuals, particularly women, to manage their finances, invest in agriculture, and contribute to household expenses⁶⁵.

For example, in the Tuggoh community, VSLAs provide a platform for farmers, traders, and students to access necessary funds for farming inputs and other needs. Profits from market sales are divided, with portions allocated to household expenses and farming reinvestments. Women receive a share of the proceeds for personal use, but much of the money is reinvested into sustaining household operations. In contrast, the Ping community has tailored its VSLAs to cater to different demographics and needs, with groups that are women-led or mixed. These VSLAs not only support farming and health-related expenses but also foster social unity through co-management and collaboration. Meanwhile, in the Sumniboma 1 community, financial management is more independent, with men and women managing their own incomes based on their roles in farming and household responsibilities. These examples highlight the diverse ways in which VSLAs and access to finance can be adapted to meet the unique needs of local communities, ultimately contributing to their economic resilience and social cohesion⁶⁶.

6. Recommendations

6.1. *Project design and implementation*

Climate change affects men and women differently, with cultural practices making women more vulnerable than men. Mainstreaming gender into the climate resilient and sustainable livelihoods project design – including conceptualisation, implementation, monitoring and evaluation – is therefore necessary to build the adaptive capacity of households and communities in the Upper West, Upper East and Northern regions. The results from this gender assessment will guide the design of project activities, ensuring that women (including divorcees and widows), the youth, elderly and other vulnerable groups are not only included in the project, but that the relationships between the cultural roles of men and women are considered. The proposed project interventions will, therefore, address gender equality through several pathways, including:

- incorporating a gender action plan into the development of the development plans and community climate action plans, based on the specific gaps identified within each community;
- integrating women into the project specific institutional structures and involving them in the project's decision-making process;
- ensuring women's participation in all climate change adaptation training;
- ensuring equal opportunity and access to women farmers for on-the-ground adaptation interventions;
- targeting women for specific adaptation interventions, with a particular focus on climate-resilient additional livelihoods;
- providing women with business and financial training;
- raising awareness of the importance of equal opportunity for all people and the role women play in climate change adaptation across a community;

⁶⁴ The World Bank Group. 2024. Ghana gender landscape. Available at: <https://documents1.worldbank.org/curated/en/099024506302242863/pdf/IDU05017dff90e8da047a70964d0d0b553ab2f02.pdf>

⁶⁵ Ibid.

⁶⁶ Ibid.

- establishing long-term monitoring and evaluation programs to assess the effectiveness of interventions for achieving gender equality; and
- sharing lessons learned through project interventions to help create a paradigm shift in the approach to gender equality in climate change adaptation projects.

It is also essential to avoid exposing women to greater risks of sexual exploitation and harassment (SEAH), GBV and intimate partner violence as a result of their empowerment through project activities that increase their income accessibility. Project interventions should be sensitive to these dynamics, ensuring that women can exercise their agency and empowerment, while also engaging male community members in women empowerment.

The gender assessment and problem analysis identified gaps in access and control over resources and opportunities. Mainstreaming gender in the project design is achieved by using sex-disaggregated data, facilitating awareness raising on grievance redress mechanisms (GRM) for SEAH and GBV, developing strategies and actions to close those gaps, allocating resources and expertise for implementing equality strategies, and monitoring results using gender-sensitive indicators. A SEAH risk assessment will be undertaken during the project's inception phase of implementation (see Annex 6B: ESMF for additional details).

6.2. *Monitoring and evaluation*

Gender will be considered throughout the planning, implementation, monitoring and evaluation of the project. Gender-specific indicators defined in the gender action plan (GAP) below will be used to quantitatively assess the gender-specific benefits of project interventions on an ongoing basis. These indicators will ensure that the monitoring and evaluation process is gender-responsive and that project interventions are effectively promoting gender equality in climate change adaptation. The results of these assessments will be incorporated into the annual Project Implementation Reports, Mid-term Report and Terminal Evaluation and will serve as information communication products produced by the project.

7. Part two: Gender Action Plan

7.1. Introduction

The Environmental Protection Agency (EPA) and the Ministry of Food and Agriculture (MoFA) are leading the preparation of a GCF funding proposal titled *Climate-resilient landscapes for sustainable livelihoods in Northern Ghana* with United Nations Environment Programme (UNEP) as its international accredited entity. Agroecological systems in northern Ghana, on which the livelihoods of smallholder farmers rely, are under considerable stress because of among other things: i) unfavourable climate conditions; ii) extensive environmental degradation; and iii) outdated and unsustainable farming methods. Therefore, this proposed project will be implemented in eight districts in northern Ghana that have been specifically chosen because of their high vulnerability to climate change impacts. The project will directly support at least 120,000 males and females of different age groups. These direct beneficiaries represent ~15% of the total population of the eight districts and 1.5% of the people in Ghana.

The Gender Action Plan (GAP) has two primary objectives: i) to guide the project in understanding and addressing the differential impacts of climate change on women and men; and ii) identify gender entry points for actions that ensure equitable benefits for women, men, and vulnerable populations in the target areas. The GAP serves as the main strategy for integrating gender metrics into the proposed project's implementation. Building on the gender assessment (Part One), the GAP ensures targeted actions to address existing gender gaps and inequalities faced by women, people with disabilities and the youth.

7.2. Impact Statement

Research and anecdotal evidence indicate that women, children and youth are among the groups most vulnerable to the impacts of climate change. In addition, a large number of vulnerable people in communities that are highly dependent on local natural resources for their survival are women and children. Therefore, the Government of Ghana (GOG) has established that gender-responsive planning, resource-allocation and implementation in the context of this proposed project will be a priority. Supporting transformational change both in the lives of women beneficiaries and in prompting gender mainstreaming in national climate change policy-making and implementation was therefore seen as a priority in stakeholder consultations.

Expected total numbers of direct and indirect beneficiaries (reduced vulnerability or increased resilience) are:

- 691,125 people directly benefit from project interventions (331,743 men and 359,385 women); and
- 2,861,674 people indirectly benefit from the project interventions (1,459,474 women and 1,402,200 men).

The project will also:

- ensure that all reviewed policies, recommended policy revisions, and economic development planning frameworks and guidelines promote gender inclusion and responsiveness.;
- ensure sustainability and long-term impact of gender-responsive outcomes, outputs and actions;

- build capacity for government institutions, traditional authorities and faith-based institutions in acknowledging gender differentials in responding to climate risks;
- ensure that women and youth take leadership with regards to project components related to them, however, without the marginalisation of men with the intent to avoid their resistance and negative influence; and
- increase social accountability.

7.3. Outcome Statement

The project aims to enhance the climate resilience of at least 124,410 women directly, and approximately 2.2 million women indirectly, by improving their access to resources for agricultural practices and/or grant facilities.

The GAP outlined in Table 3 serves as a framework for integrating gender considerations into the climate resilient and sustainable livelihoods project in northern Ghana. It will be used to ensure that both men and women — including divorcees and widows — are actively involved in planning, decision-making and the implementation of climate-resilient agricultural practices, early warning systems and livelihood interventions. The EPA, MoFA, WRC and other governmental bodies will collaborate closely with the project management unit's (PMU) ESS/Gender Specialist, who holds primary responsibility for overseeing gender mainstreaming efforts. The plan will be monitored through gender-disaggregated indicators such as participation rates and consultation outcomes, with reporting scheduled at intervals throughout the project's implementation. Regular workshops, focus group discussions, and training sessions will be used to ensure continuous feedback and adaptation of the plan. During the 2024 mission, women in the target district local communities indicated that they have previously been unable to attend workshops or training sessions in the afternoon and early evening as a result of their household responsibilities. Therefore, all workshops, consultations and meetings within the local communities will take place during weekdays in the morning hours. The implementation status of the GAP will be a recurring agenda item for all project steering committee (PSC) meetings.

Table 3. Gender Action Plan

Project management activities	Gender activity	Baseline	Indicators	Targets	Indicative timeline	Responsibility	Budget
	Ensure a gender balance in establishing the Project Management Unit (PMU) and in managing its human resources.	No PMU established	Documentation of contracting process — such as vacancy announcements, shortlisting, interview notes — and proportion of women employed in unskilled, technical, management, or supervisory roles.	Demonstrable efforts to recruit at least 50% women, including in positions with project	Year 1	UNEP Task Manager	N/A

				responsibilities		
				67		
Recruit a ESS/Gender specialist in the PMU to support and monitor the effective implementation of the Gender Action Plan (50%) and oversee any grievances related to gender-based violence (GBV) or sexual exploitation and harassment (SEAH).	No Gender Specialist in the PMU	ToRs to recruit for the position	85% of their time is dedicated to supporting the gender-integrated implementation of the project. 15% of their time is allocated to support grievances for SEAH or GBV.	Year 1	UNEP Task Manager	N/A
Gender Training for the PMU. Training materials will include topics of, <i>inter alia</i> : i) SEAH and GBV; understanding how to analyse the differential impacts of projects on men and women; iii) strategies to promote equitable access to project benefits for all social groups; and iv) familiarisation with national, regional and international gender policies and conventions.	No training for the PMU	Online and in-situ training provided to team members (certificates, modules), knowledge assessed through interviews	100% of PMU team members are knowledgeable on gender challenges and able to report on gender-responsive indicators of the project such as certificates, modules and knowledge assessments.	Year 1	PMU ESS/Gender Specialist	
Develop and implement mandatory SEAH prevention training for project staff, CCAGs, FBOs and communities	No SEAH prevention training	Percentage of project staff, CCAGs, FBOs and community members who have completed the	100% of project staff, CCAGs, FBOs, and 80% of community members	Year 1	PMU ESS/Gender Specialist	N/A

⁶⁷ Measurability will be improved in the implementation stage by tracking and reporting % of female hires in the APR , including any challenges encountered.

			mandatory SEAH prevention training.	trained in SEAH prevention within the first six months of the programme's implementation.			
	Integrate SEAH prevention measures in the financial credit line development programmes for the local financial institutions (LFIs) engaged under Output 4 to adhere to SEAH free policies and to ensure women have direct access to financial services without intermediaries.	No SEAH integration	Number of LFIs engaged under Output 4 that have integrated SEAH prevention measures and adopted SEAH-free policies, ensuring direct access to financial services for women.	100% of LFIs engaged under Output 4 integrate SEAH prevention measures and SEAH-free policies, with 80% of women accessing financial services directly, without intermediaries, within the first two years of programme implementation.	Year 1–2	PMU ESS/Gender Specialist	N/A
Project output	Gender activity		Indicators	Targets	Indicative timeline	Responsibilities	Budget
Component 1: Early warning systems							
Output 1: Improved climate data and early warnings made available to facilitate proactive drought and flood management.	Ensure gender-sensitive training for technical staff at national, regional, and district levels, including GMet, HYDRO, and WRC, focusing on the unique needs and vulnerabilities of women in flood and	0% of technical staff have received training in gender-responsive flood and drought management practices; Less than 5% of women	- Percentage of technical staff trained in gender-responsive flood and drought management practices. - Percentage of women in each community reporting increased access to and understanding of	At least 40% of trained technical staff (national, regional, district) are women, and 100% of staff trained receive	Year 1	Water Resource Commission (WRC), DHI and project ESS/Gender Specialist	National consultant (gender expert) procured to undertake the FGDs, develop the report and collaborate with system developers: US\$12,000

drought management. • Integrate gender-specific data and information into the new early warning system, ensuring that women's access to flood and drought early warning information is prioritized, especially in vulnerable farming communities. • Facilitate women's active participation in the development of the national framework for disseminating DSS and climate-related hazard management information to ensure their specific concerns and priorities are addressed.	report having access to or understanding of flood and drought early warning information.	flood and drought warnings. • Number of women actively involved in consultations and decision-making processes for the development of the national framework on drought and flood hazard management.	gender-sensitive training by the end of the second year of implementation. • 80% of all women across all the communities report an increase in access to early warning messages by the third year of implementation. • Women represent at least 50%⁶⁸ of the participants in consultations for the development of the national framework by the second year	• Travel to each target district and DSA: US\$18,400. • The PMU ESS/Gender Specialist salary collaborating with system developers to integrate gender-responsive features into the early warning system.
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⁶⁸ 50% has been selected as a target for this activity based on the 2024 mission where there was considerably high female attendance. See Annex 7h: stakeholder engagement plan.

					of implementat ion.			
Component 2: Climate resilient livelihoods								
Output 2: Climate-resilient agricultural practices implemented in beneficiary communities.	- Establish mentorship and capacity-building programmes specifically designed to enhance women's entrepreneurial skills in climate-resilient livelihoods, including market access, value chain development and business management, to increase their economic independence and long-term resilience. - Ensure women-led cooperatives in the local communities lead climate-resilient agricultural practices and EbA initiatives, enhancing women's leadership, decision-making power and economic resilience.	No women entrepreneurs have received training or mentorship in climate-resilient livelihood development or business management.	- Number of women entrepreneurs trained and mentored in climate-resilient livelihood development and business management. - Number of women-led cooperatives facilitating in climate-resilient agriculture and EbA practices.	- At least 80% of women in each community report increased financial literacy and profit margins on products sold as a result of the training. - At least 50 women in each beneficiary district are mentored and supported to launch or expand climate-resilient businesses by the third year of the project, with specific strategies to improve their access to markets	Year 1–7	District Agricultural Offices and project ESS/Gender Specialist	- Costs of the mentorship and capacity building programmes are embedded in the activity costs (US\$192,000). - The PMU ESS/Gender Specialist salary	

					- and value chains. - At least one women-led cooperative or group is established in 75% of beneficiary communities by the second year of the project, with ongoing support provided for their leadership development and access to resources.			
Output 3: Restoration of landscape to reduce drought and flood risk	Facilitate the active participation of women in planning and implementing land restoration activities, including training women in nurseries mgmt, agroforestry and flood-based farming techniques.	Women are minimally involved in communal land restoration activities and have limited access to training in nursery operations, agroforestry or flood-based farming practices.	Number of women trained and actively participating in land restoration and agroforestry interventions in the 120 target communities.	At least 3,600 women (30 per community) trained and actively engaged in land restoration and agroforestry activities by the end of the project.	Year 1–7	District Agricultural Offices and project ESS/Gender Specialist	The PMU ESS/Gender Specialist salary	

Output 4: Increased access of smallholder farmers to financial resources and engagement with the private sector.	• Prioritise the formation of women-led Farmer-Based Organisations (FBOs) and Village Savings and Loan Associations (VSLAs), ensuring that women have leadership roles and decision-making authority in these groups to directly influence access to credit and insurance for both farming and non-farming livelihood activities. • Develop tailored financial products and insurance packages that address the specific needs and risks experienced by women smallholder farmers and entrepreneurs, ensuring that women have equal access to financial services and are not excluded due to collateral or credit history requirements.	0 women report having access to or using formal financial products such as credit and insurance; Women smallholder farmers and entrepreneurs have received minimal to no training in financial literacy or mentorship on engaging with private sector actors.	• Percentage of women-led FBOs and VSLAs established and actively accessing financial resources for climate-resilient farming and alternative livelihoods. • Number of women who report increased access to financial products such as credit and insurance. • Number of women smallholder farmers and entrepreneurs trained in financial literacy and mentored in engaging with private sector actors⁶⁹.	• At least 50% of all newly established FBOs and VSLAs are women-led by the second year of the project, and 100% of these groups have access to credit and insurance products by the end of the third year. • 60% of women smallholder farmers in beneficiary communities accessing these products by the third year of the project. This will be measured through surveys	Year 1–5	District Agricultural Offices and project ESS/Gender Specialist	• National consultant (gender expert) to prepare materials for and facilitate the training sessions (US\$16,000). • National consultant (gender expert) to engage with the BAC on facilitating mentorship and financial literacy programmes in each target district (\$1,600). • Travel and DSA for the national consult to travel to each target district to facilitate the training and engage with the BAC (US\$22,400). • The cost for the training workshop is covered under the training
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⁶⁹ Engagement with the private sector refers to collaboration with businesses and financial institutions to leverage their resources, expertise, and services.

Establish mentorship and financial literacy programmes for women farmers and entrepreneurs to build their capacity to manage credit, understand insurance products, and engage with private sector actors, fostering long-term financial empowerment and business resilience.	capturing the number of women who self-report improved access to financial products compared to the baseline. At least 100 women in each beneficiary district are trained in financial literacy and business skills by the second year of the project, with at least 50% of them actively engaging with private sector actors for climate-resilient agriculture by the end of the third year.	costs for sub-activity 4.1.2 (US\$286,000). The PMU ESS/Gender Specialist salary
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Output 5: Knowledge and awareness of climate threats to agricultural livelihoods and available adaptation options increased to inform the upscaling of climate change adaptation across northern Ghana.	- Establish and support women-led knowledge hubs in beneficiary communities where women farmers and entrepreneurs can lead the dissemination of climate adaptation information, share best practices, and provide peer-to-peer learning on climate-resilient agricultural practices and adaptation options. - Facilitate women's leadership in knowledge-sharing and awareness-raising events, ensuring that women climate champions and leaders from various sectors — such as farming, small businesses and community leadership — are given platforms to share their experiences, insights, and success stories in climate-resilient agriculture and adaptation.	0 women-led knowledge hubs currently exist or are operational for sharing climate adaptation information and best practices; Women are underrepresented as key speakers or facilitators, with less than 10% of events featuring women in these roles.	- Number of women-led knowledge hubs established and operational in sharing climate adaptation information and best practices. - Percentage of knowledge-sharing and awareness-raising events where women serve as key speakers or facilitators.	- At least one women-led knowledge hub is established in each target communities by the end of the fourth year of the project, with these hubs actively engaged in generating and disseminating climate-resilient agricultural knowledge. - Women are key speakers or facilitators in at least 50% of knowledge-sharing and awareness-raising events by the second year, with these events specifically targeting at least 70%	Year 1–4	District Agricultural Offices and project ESS/Gender Specialist	- National consultant (gender expert) to facilitate the establishment of women-led knowledge hubs in each target district (\$8,000) - Travel and DSA for national consultant to go to each community (\$14,400). - The PMU ESS/Gender Specialist salary
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female
attendance
to ensure
broader
outreach to
women
farmers and
entrepreneu
rs.

Table 4. Summary of GAAP budget

Output number	Gender activity description	Budget
Output 1	- Ensure gender-sensitive training for technical staff at national, regional, and district levels, including GMet, HYDRO, and WRC, focusing on the unique needs and vulnerabilities of women in flood and drought management. - Integrate gender-specific data and information into the new early warning system, ensuring that women's access to flood and drought early warning information is prioritized, especially in vulnerable farming communities. - Facilitate women's active participation in the development of the national framework for disseminating DSS and climate-related hazard management information to ensure their specific concerns and priorities are addressed - Undertake the FGDs, develop the report and collaborate with system developers	- National consultant (gender expert): US\$12,000 - Travel and DSA: US\$18,400.
Output 2	- Establish mentorship and capacity-building programmes specifically designed to enhance women's entrepreneurial skills in climate-resilient livelihoods, including market access, value chain development and business management, to increase their economic independence and long-term resilience. - Ensure women-led cooperatives in the local communities lead climate-resilient agricultural practices and EbA initiatives, enhancing women's leadership, decision-making power and economic resilience.	- National consultant (gender expert): USD 4,000 - Travel and DSA: USD 13,000
Output 3	Facilitate the active participation of women in planning and implementing land restoration activities, including training women in nurseries mgmt, agroforestry and flood-based farming techniques.	- National consultant (gender expert): US\$12,000 - Travel and DSA: US\$18,400.
Output 4	Engage with the Business Advisory Centre (BAC) on facilitating mentorship and financial literacy programmes in each target district.	- National consultant (gender expert): US\$17,600 - Travel and DSA: US\$22,400

Output 5	• Prepare materials for and facilitate the training sessions. • Facilitate the establishment of women-led knowledge hubs in each target district	• National consultant (gender expert): US\$8,000 • Travel and DSA: US\$14,400
Monitoring & Safeguards	• Environmental and Social Safeguards and Gender Specialist	• Remuneration over 7 years (USD 327,957)
GAP total budget		US\$468,157